

注意力机制与外部记忆



内容



- 注意力机制
 - 指针网络
 - 自注意力模型
- 记忆增强网络
 - 结构化外部记忆
 - 基于神经动力学的联想记忆

网络能力



全连接
前馈网络

循环网络

卷积网络

图网络

增加网络能力的另一种思路：

- 注意力机制
- 外部记忆

例子：阅读理解



Facebook bAbi tasks

1 John moved to the bedroom.

2 Mary grabbed the football there.

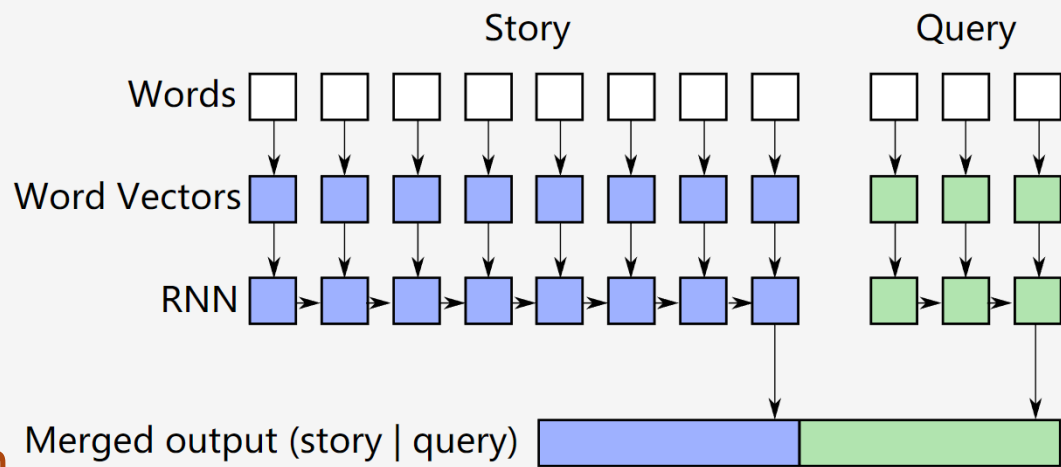
3 Sandra journeyed to the bedroom.

4 Sandra went back to the hallway.

5 Mary moved to the garden.

6 Mary journeyed to the office.

7 Where is the football? office



例子：阅读理解 SQuAD



The first recorded travels by Europeans to China and back date from this time. The most famous traveler of the period was the Venetian Marco Polo, whose account of his trip to "Cambaluc," the capital of the Great Khan, and of life there astounded the people of Europe. The account of his travels, *Il milione* (or, *The Million*, known in English as the *Travels of Marco Polo*), appeared about the year 1299. Some argue over the accuracy of Marco Polo's accounts due to the lack of mentioning the Great Wall of China, tea houses, which would have been a prominent sight since Europeans had yet to adopt a tea culture, as well the practice of foot binding by the women in capital of the Great Khan. Some suggest that Marco Polo acquired much of his knowledge **through contact with Persian traders** since many of the places he named were in Persian.

How did some suspect that Polo learned about China instead of by actually visiting it?

Answer: **through contact with Persian traders**

大脑中的注意力



- 人脑每个时刻接收的外界输入信息非常多，包括来源于视觉、听觉、触觉的各种各样的信息。
- 但就视觉来说，眼睛每秒钟都会发送千万比特的信息给视觉神经系统。
- 人脑通过**注意力**来解决**信息超载**问题。

注意力示例



注意力示例



- 鸡尾酒会效应

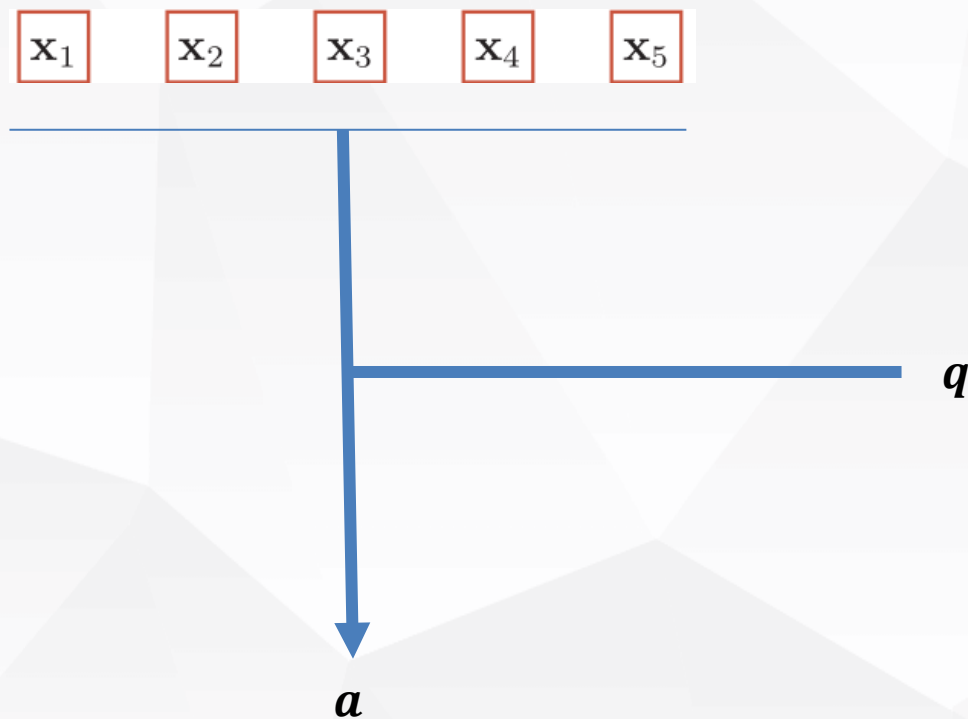
- 当一个人在吵闹的鸡尾酒会上和朋友聊天时，尽管周围噪音干扰很多，他还是可以听到朋友的谈话内容，而忽略其他人的声音。
- 同时，如果未注意到的背景声中有重要的词（比如他的名字），他会马上注意到。

注意力实验



所有黑色牌的数字之和为多少？

问题



Try to find 8 less than 8 seconds

注意力模型



软性注意力机制 (soft attention mechanism)

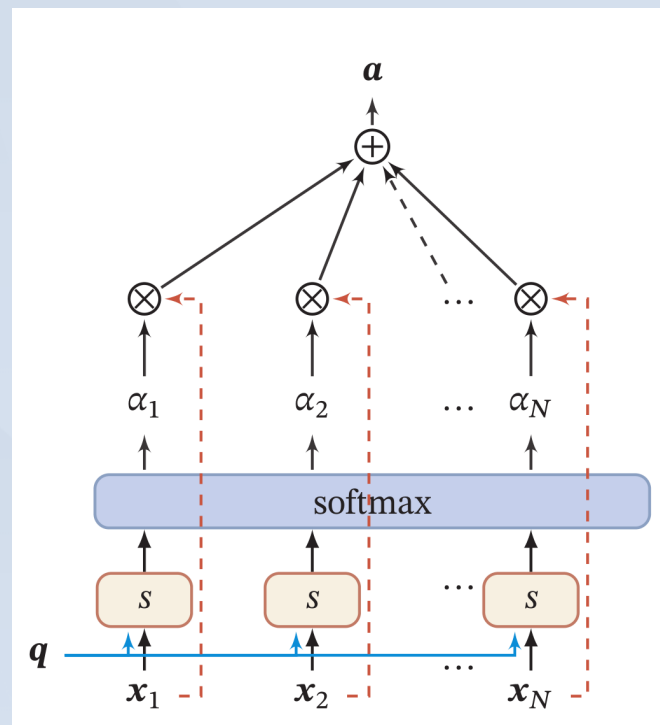
注意力机制可以分为两步：

Step1 计算注意力分布 α

$$\begin{aligned}\alpha_n &= p(z = n | \mathbf{X}, \mathbf{q}) \\ &= \text{softmax}(s(\mathbf{x}_n, \mathbf{q})) \\ &= \frac{\exp(s(\mathbf{x}_n, \mathbf{q}))}{\sum_{j=1}^N \exp(s(\mathbf{x}_j, \mathbf{q}))}\end{aligned}$$

Step2 根据 α 来计算输入信息的加权平均

$$\begin{aligned}\text{att}(\mathbf{X}, \mathbf{q}) &= \sum_{n=1}^N \alpha_n \mathbf{x}_n \\ &= \mathbb{E}_{z \sim p(z | \mathbf{X}, \mathbf{q})}[\mathbf{x}_z]\end{aligned}$$



注意力打分函数 $s(\mathbf{x}_n, \mathbf{q})$



加性模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{v}^\top \tanh(\mathbf{W}\mathbf{x} + \mathbf{U}\mathbf{q}),$$

点积模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{x}^\top \mathbf{q},$$

缩放点积模型

$$s(\mathbf{x}, \mathbf{q}) = \frac{\mathbf{x}^\top \mathbf{q}}{\sqrt{D}},$$

双线性模型

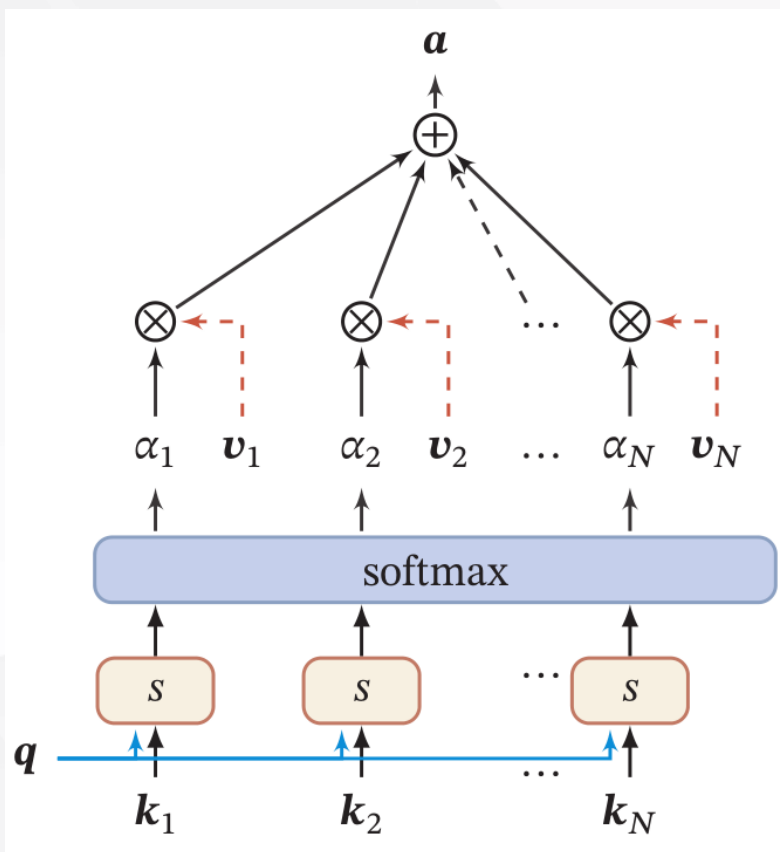
$$s(\mathbf{x}, \mathbf{q}) = \mathbf{x}^\top \mathbf{W} \mathbf{q},$$

注意力机制的变体



硬性注意力 (hard attention)

键值对注意力 (key-value pair attention)



用 $(K, V) = [(k_1, v_1), \dots, (k_N, v_N)]$ 表示 N 个输入信息

$$\begin{aligned} \text{att}((K, V), q) &= \sum_{n=1}^N \alpha_n v_n, \\ &= \sum_{n=1}^N \frac{\exp(s(k_n, q))}{\sum_j \exp(s(k_j, q))} v_n \end{aligned}$$

注意力机制的变体



➤ 多头注意力（multi-head attention）

利用多个查询 $Q=[q_1, \dots, q_M]$ ，来并行地从输入信息中选取多组信息。每个注意力关注输入信息的不同部分。

$$\text{att}((K, V), Q) = \text{att}((K, V), q_1) \oplus \dots \oplus \text{att}((K, V), q_M)$$

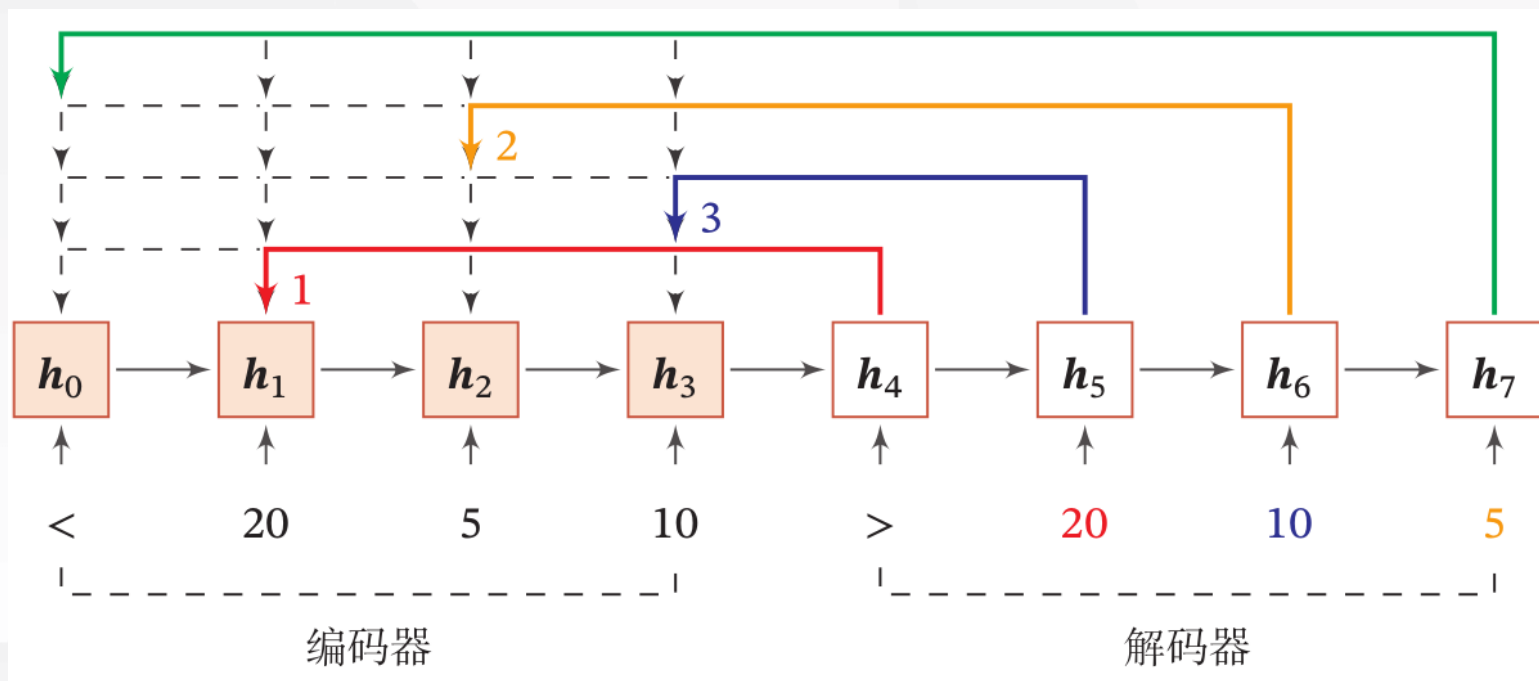
➤ 结构化注意力（structural attention）

指针网络

pointer network



- 我们可以只利用注意力机制中的第一步，将注意力分布作为一个软性的指针（pointer）来指出相关信息的位置。



一些常见的注意力



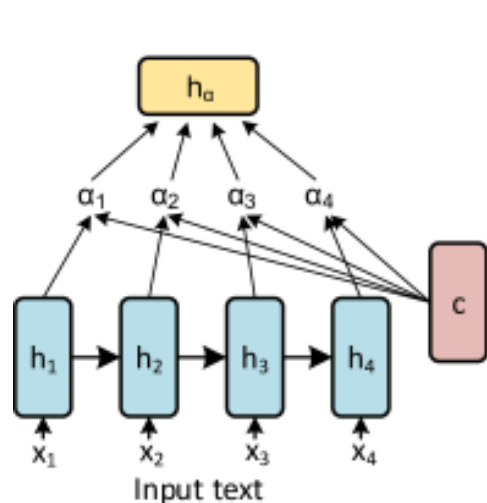
I have not read the original version of this work , but the translation lacks originality and art . A beautiful story , but the writing style lacks grace and creativity . This is the only time I have liked a movie better than the book . Do yourself a favor and skip the book . the movie is quite beautiful and moving

(a) Attention of task “Books” in SA-MTL, Output: Negative

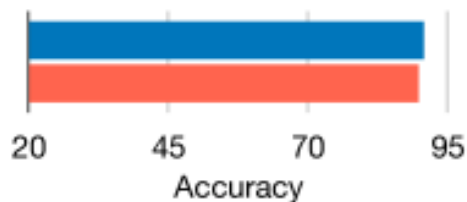
I have not read the original version of this work , but the translation lacks originality and art . A beautiful story , but the writing style lacks grace and creativity . This is the only time I have liked a movie better than the book . Do yourself a favor and skip the book . the movie is quite beautiful and moving

(b) Attention of task “DVD” in SA-MTL, Output: Positive

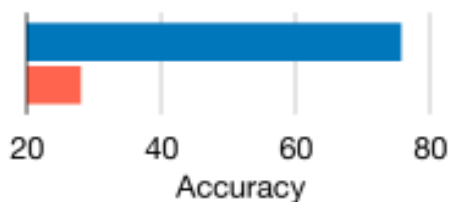
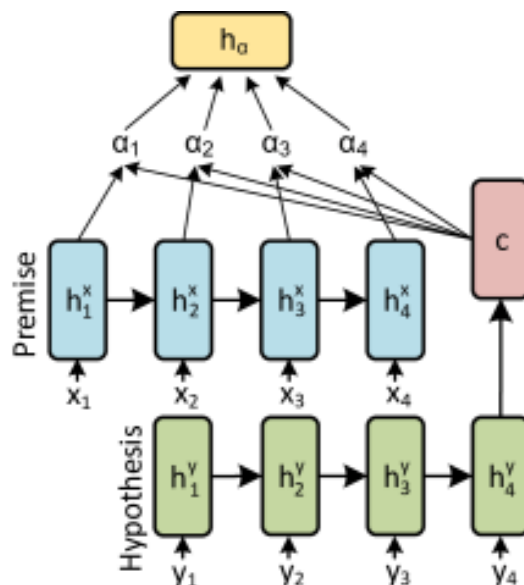
一些常见的注意力



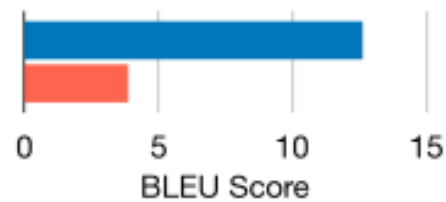
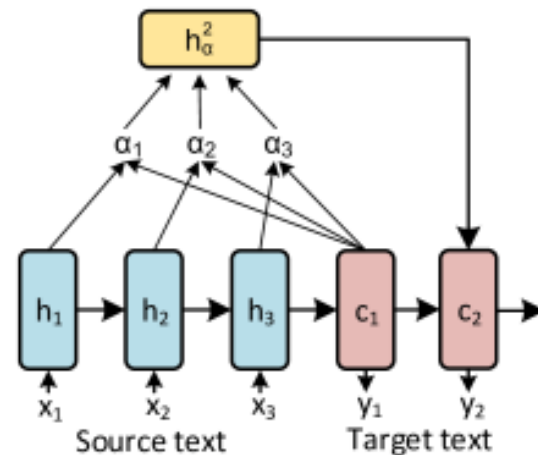
■ Attention
■ Uniform



(a) Text Classification



(b) Natural Language Inference

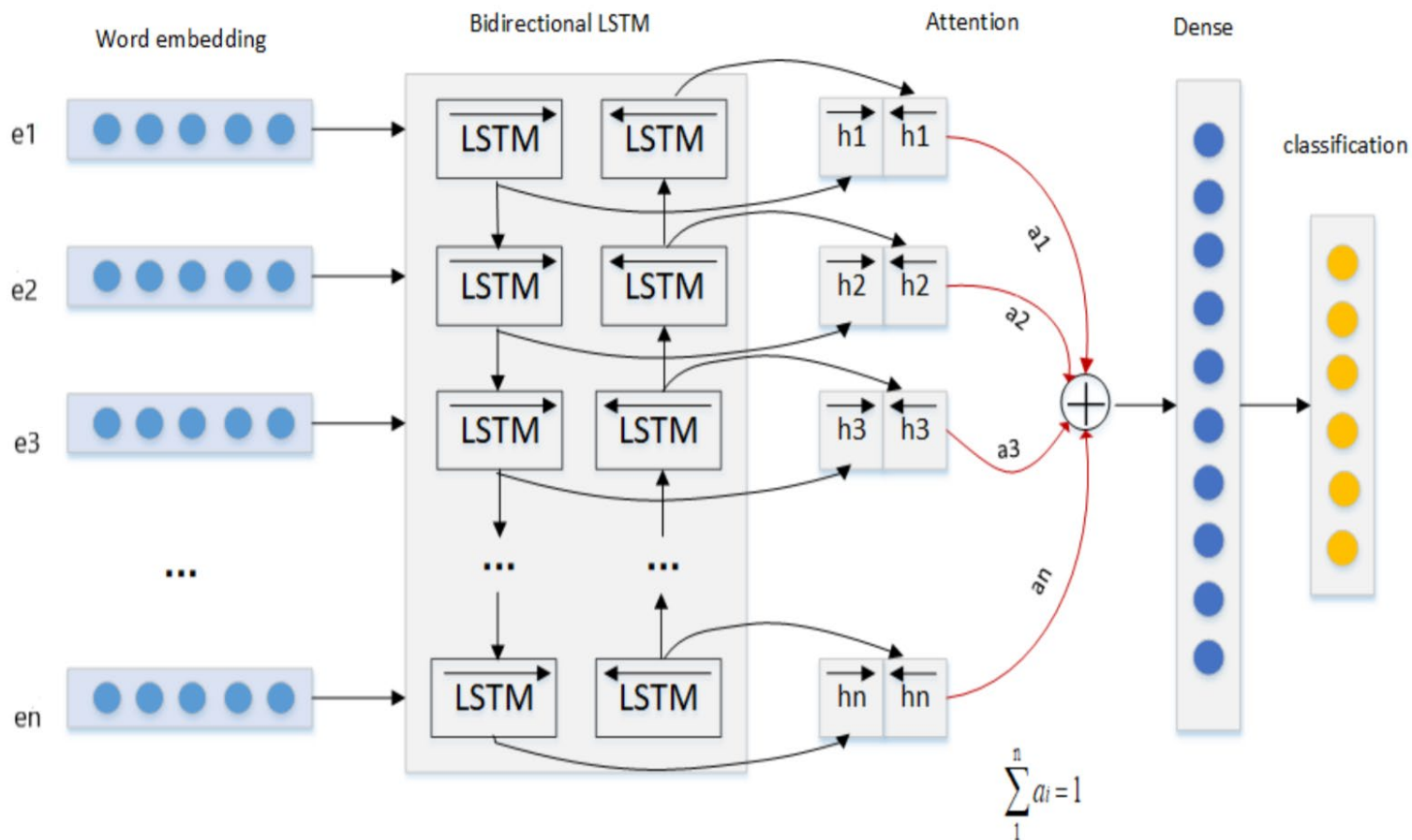


(c) Neural Machine Translation

注意力模型



- 文本分类



机器翻译

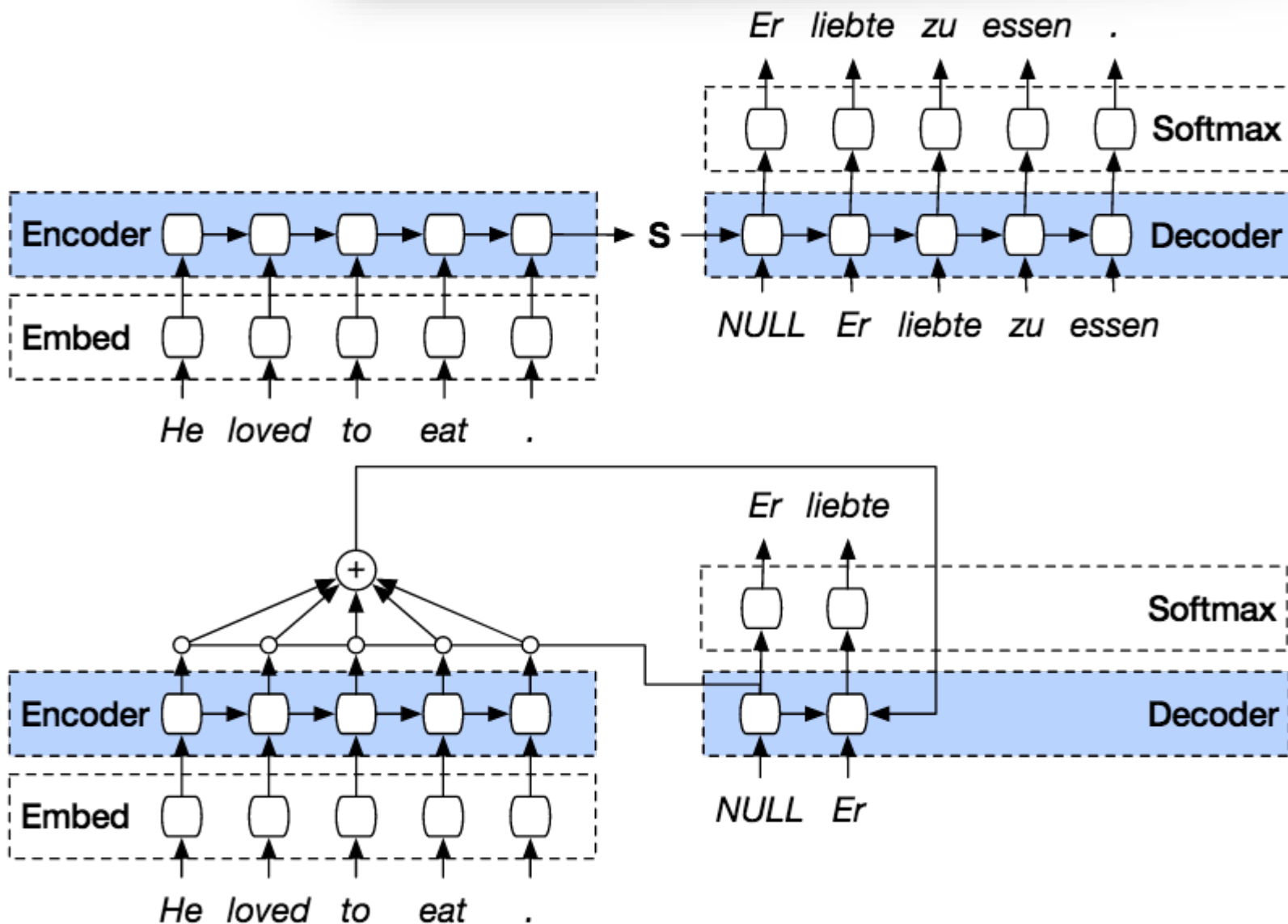


Image Caption

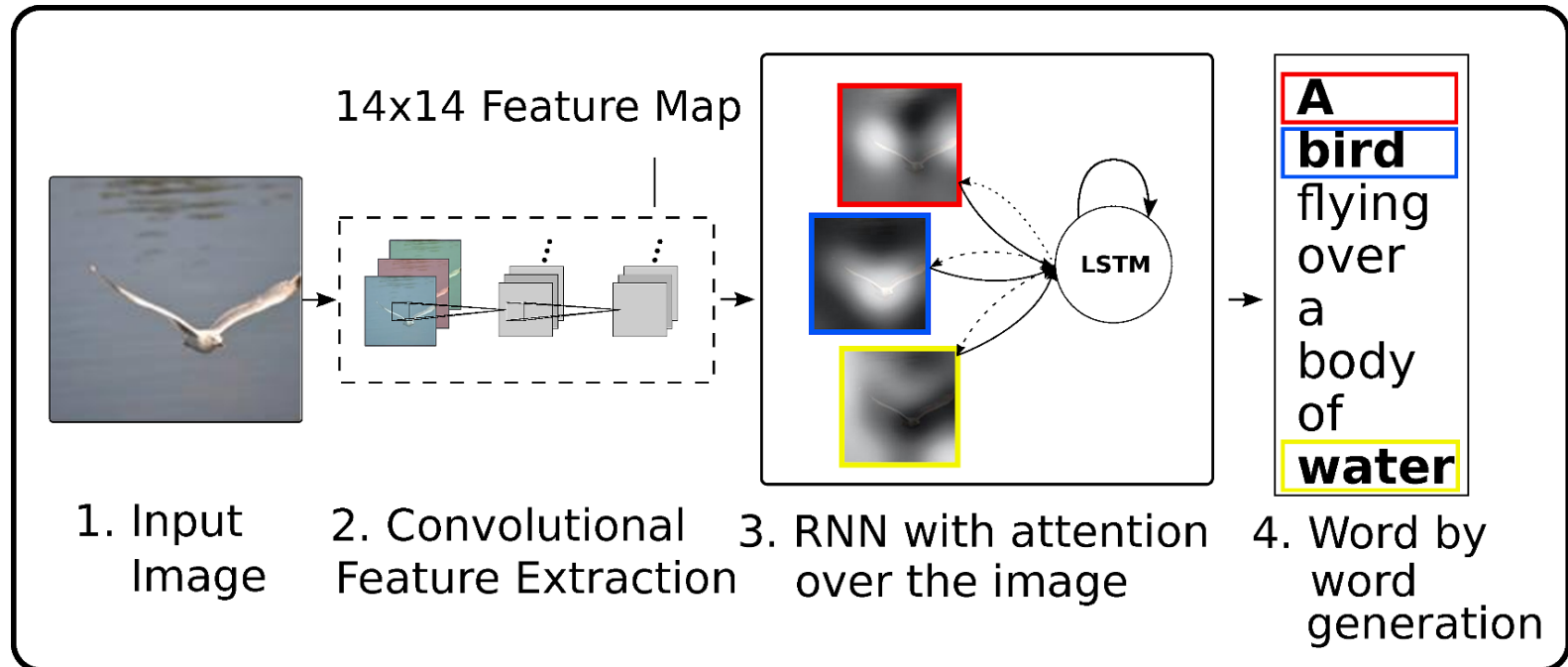


Image Caption



remote(0.23)



a(0.41)



sky(0.61)



on(0.27)



red(0.38)



woman(0.36)



of(0.19)



A(0.99)



racket(0.16)



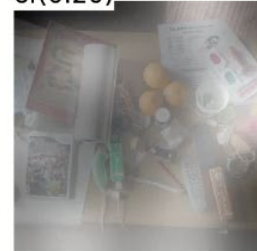
a(0.33)



sink(0.48)



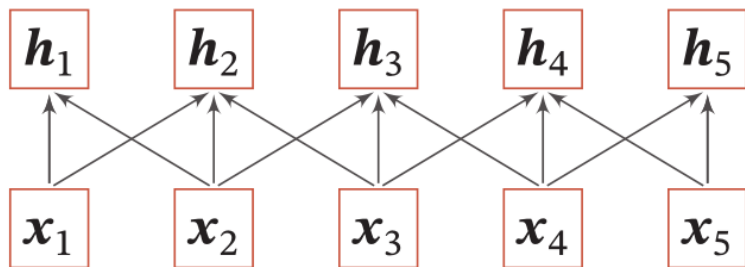
of(0.20)



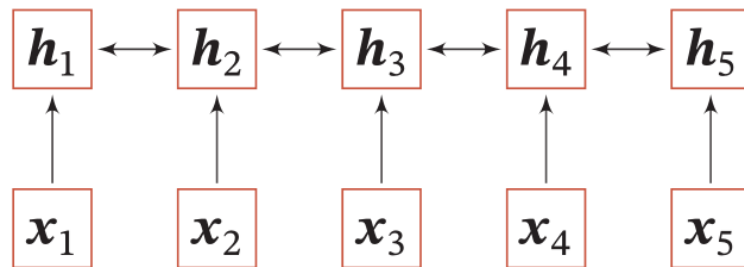
自注意力模型



- 当使用神经网络来处理一个变长的向量序列时，我们通常可以使用卷积网络或循环网络进行编码来得到一个相同长度的输出向量序列。



(a) 卷积网络



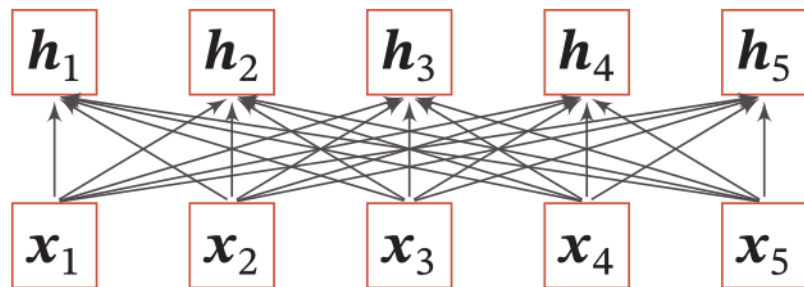
(b) 双向循环网络

只建模了输入信息的局部依赖关系

自注意力模型

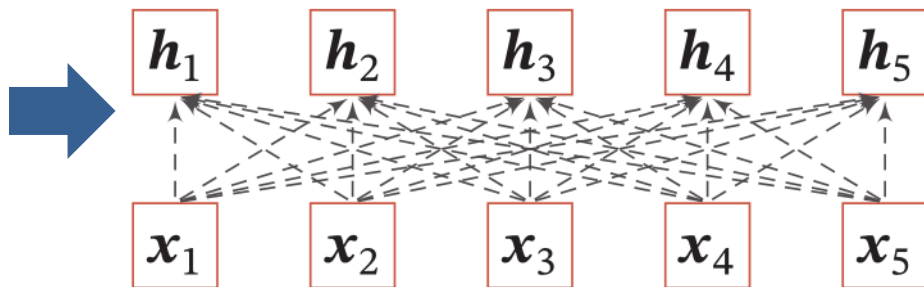


- 如何建立非局部（Non-local）的依赖关系全连接？



(a) 全连接模型

无法处理变长问题



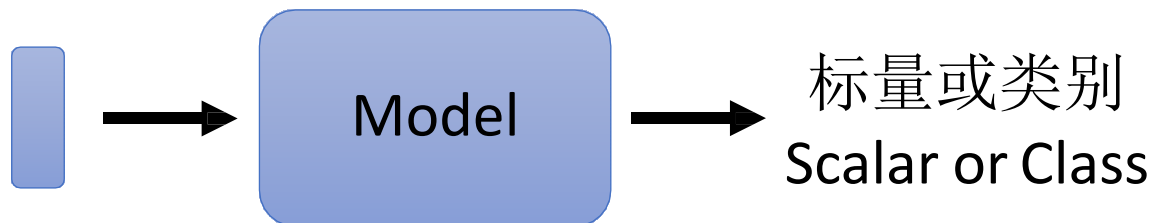
(b) 自注意力模型

连接权重 α_{ij} 由注意力机制动态生成

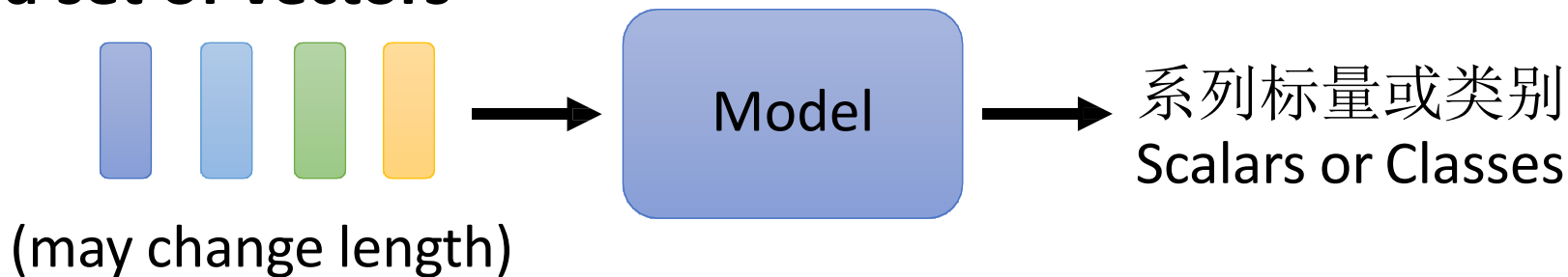
自注意力模型



- 输入向量 **a vector**



- 输入一组向量
a set of vectors

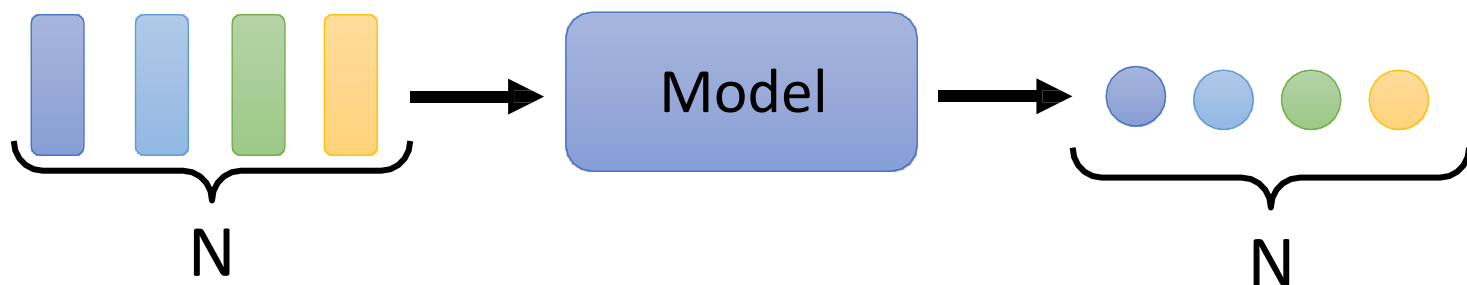


自注意力模型



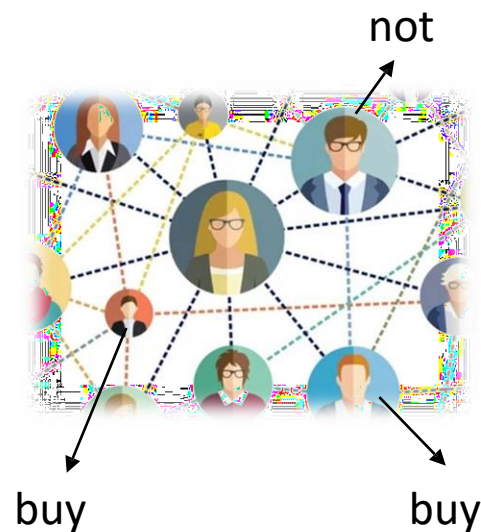
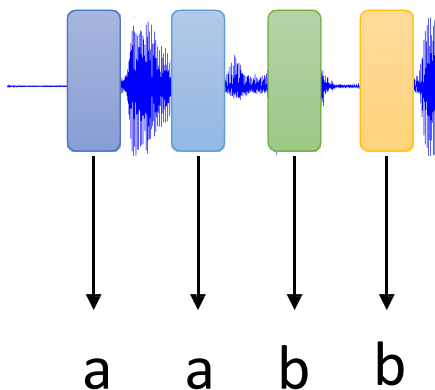
输出是什么?

- 每一个向量对应一个标签.



示例:

I saw a saw
↓ ↓ ↓ ↓
N V DET N



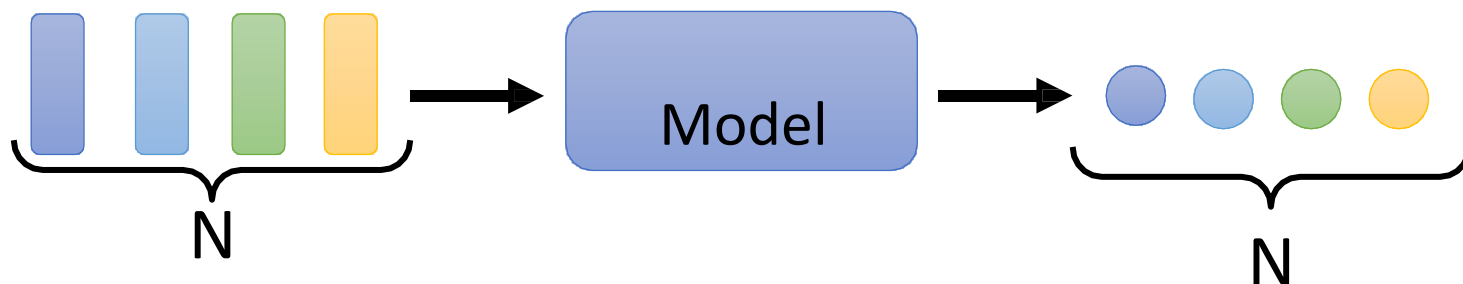
词性标注POS tagging

自注意力模型

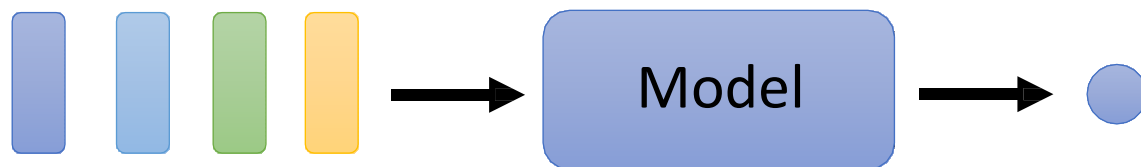


输出是什么?

- Each vector has a label.



- The whole sequence has a label.

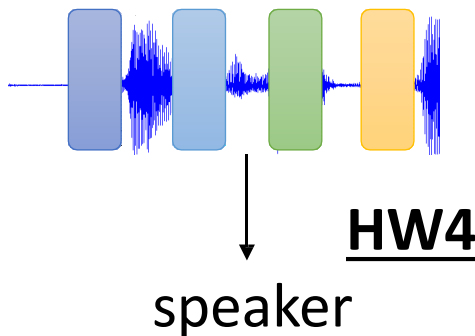


示例:

this is good

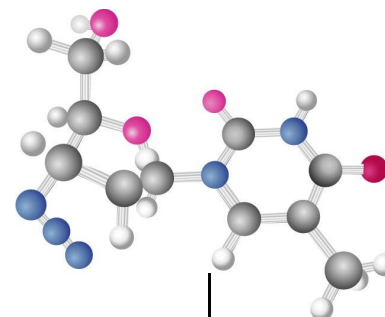
情感分析

positive



speaker

HW4



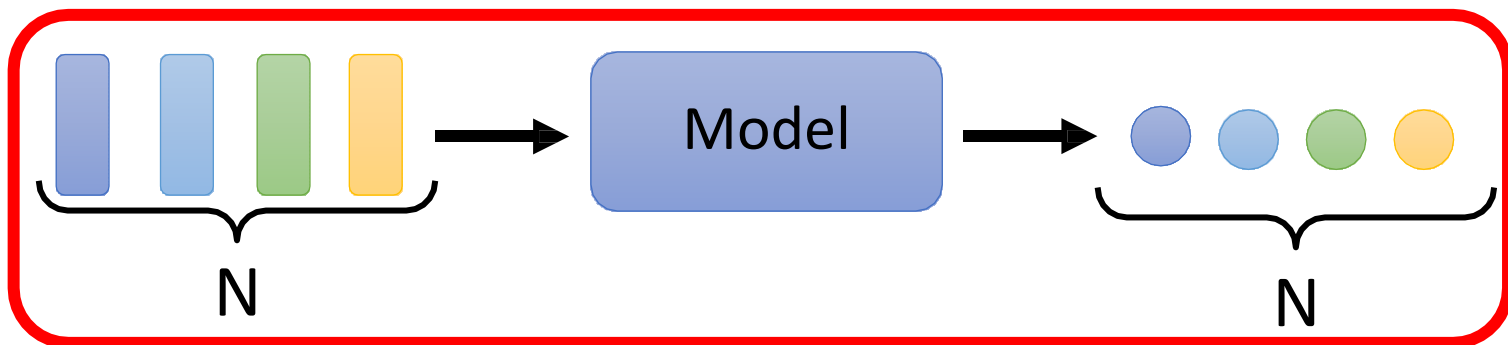
hydrophilicity₈

自注意力模型

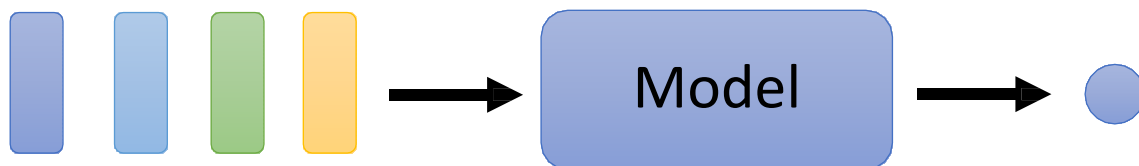


- Each vector has a label.

课程重点关注该类型

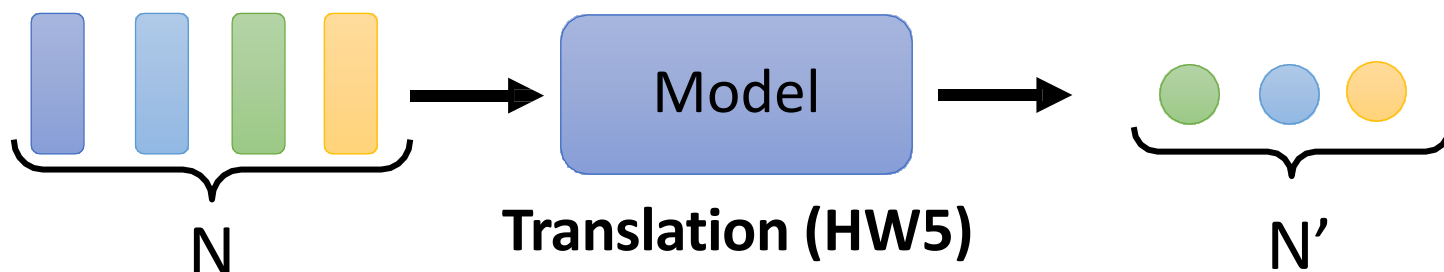


- The whole sequence has a label.



- Model decides the number of labels itself.

seq2seq



自注意力模型

序列标注



Is it possible to consider the context?

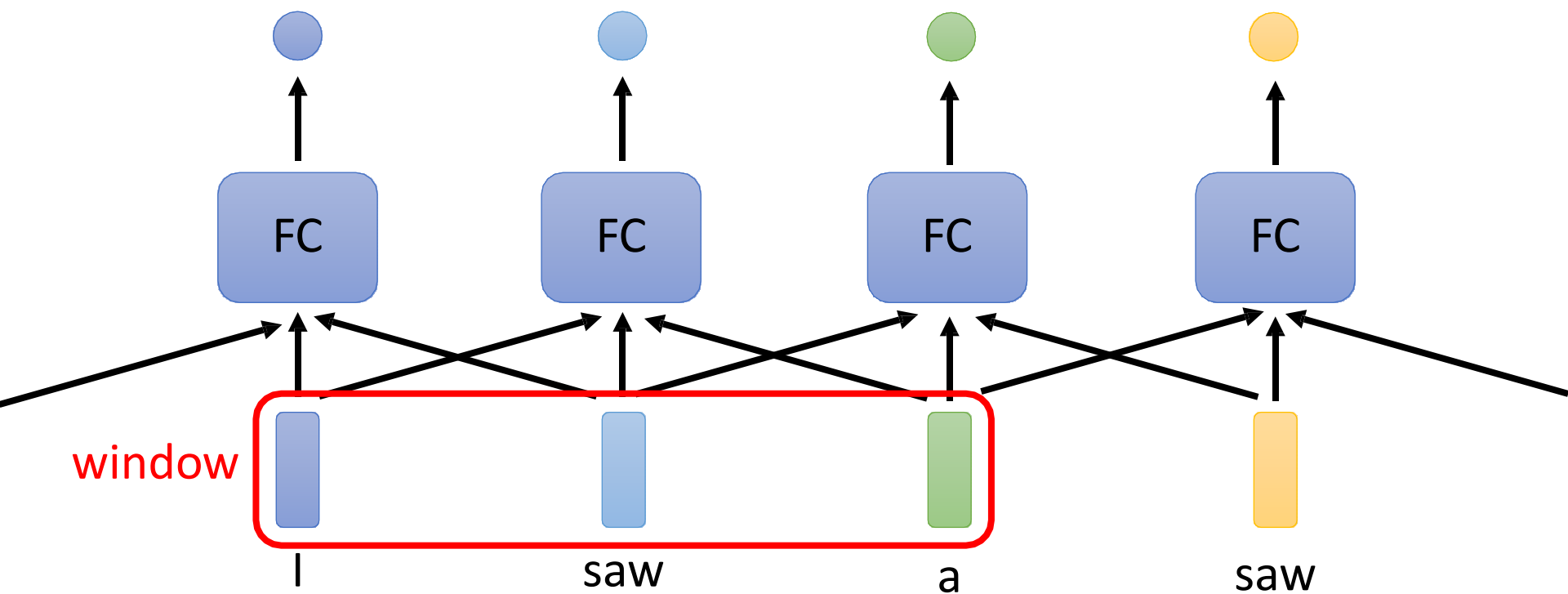
FC

Fully-
connected

FC can consider the neighbor

How to consider the whole sequence?

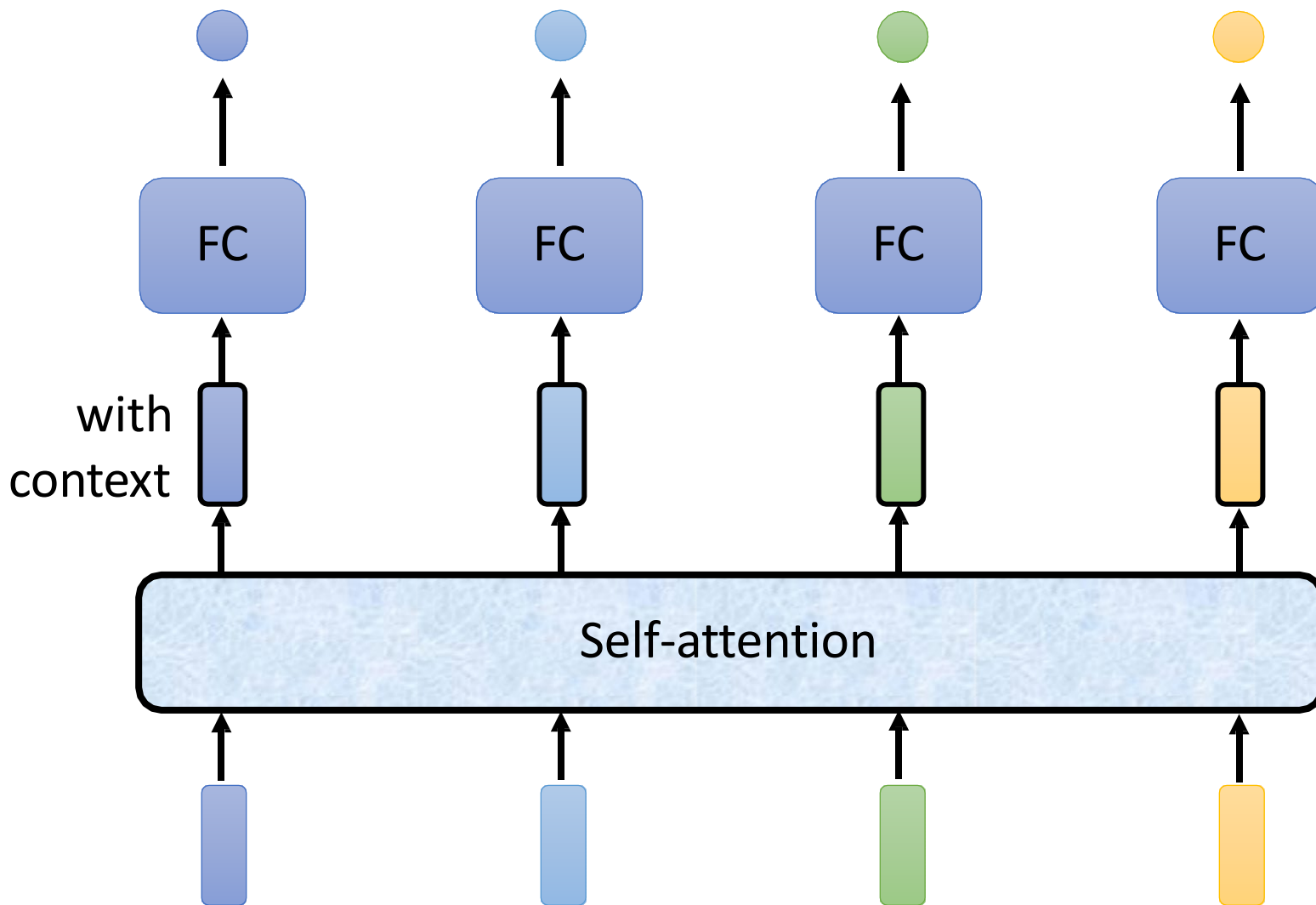
a window covers the whole sequence?



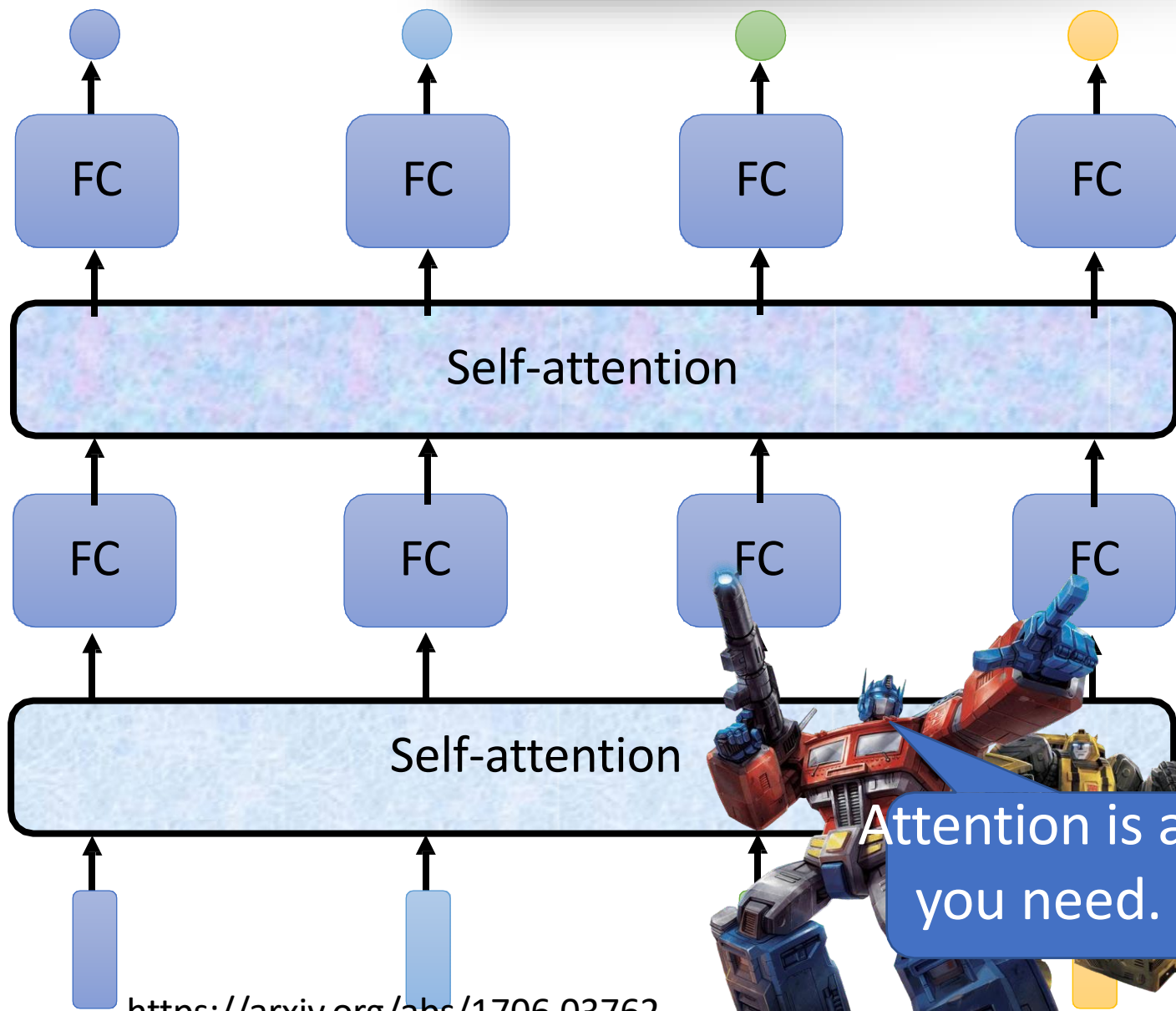
自注意力模型



Self-attention



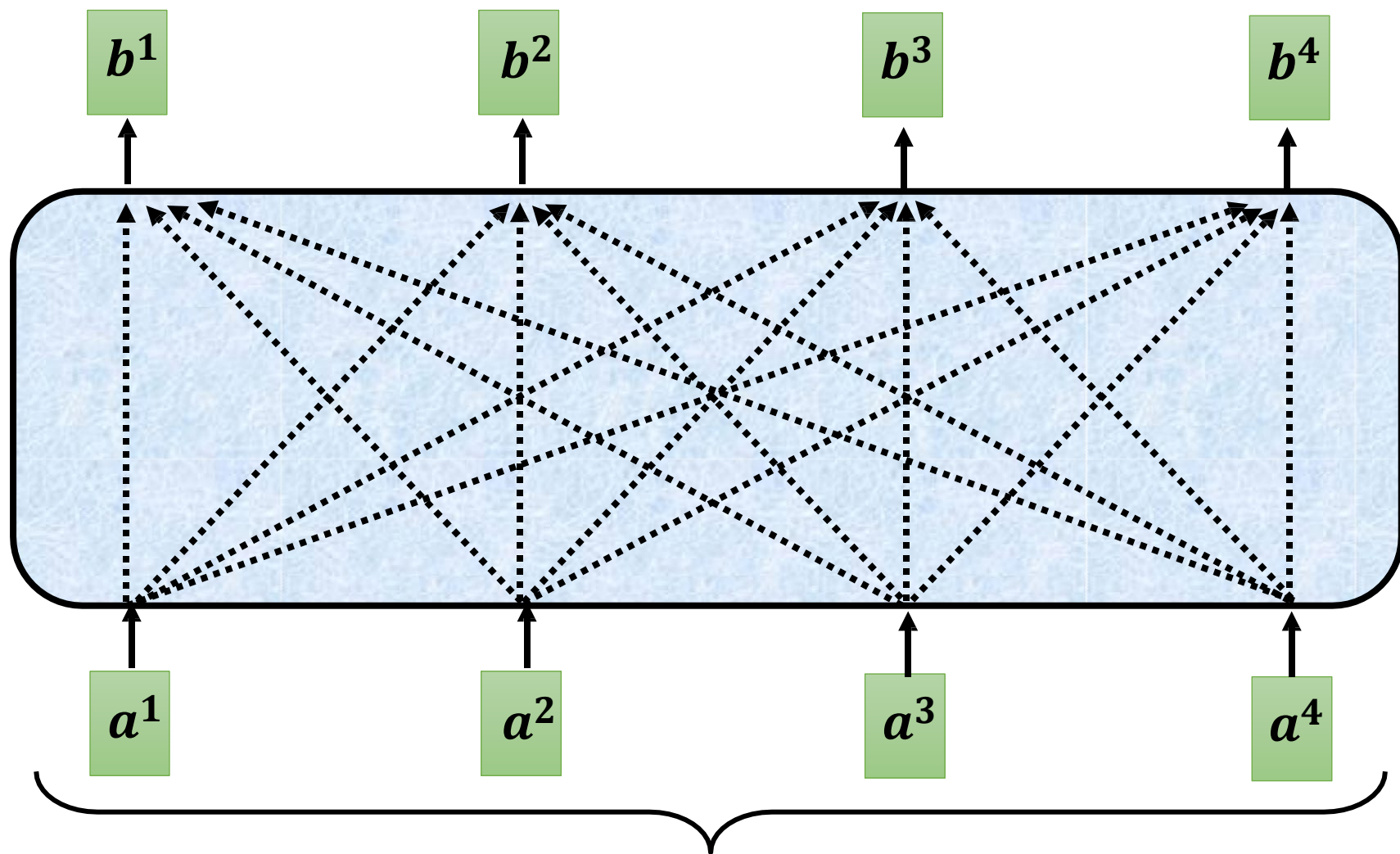
自注意力模型



Attention is all
you need.

自注意力模型

Self-attention

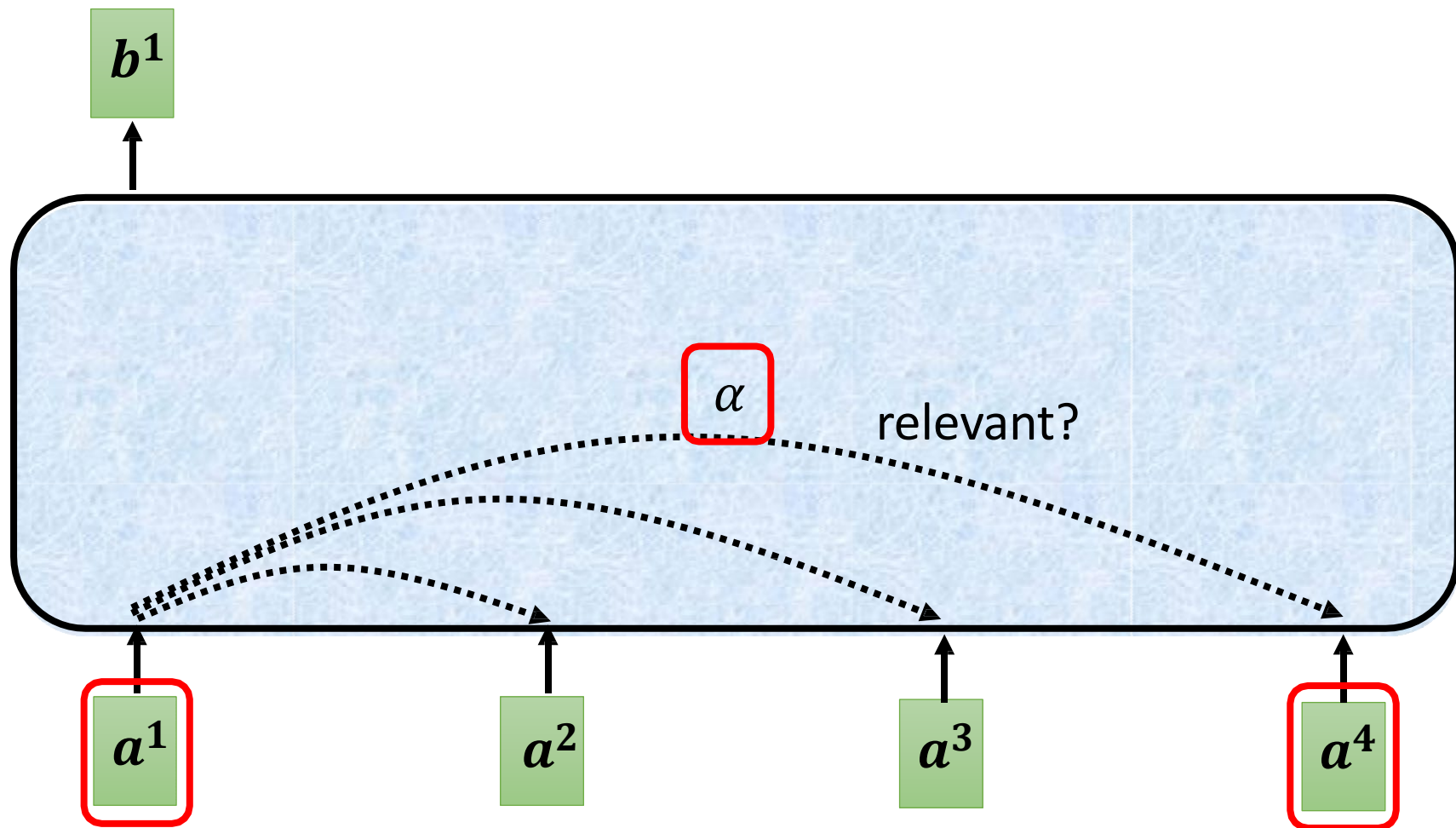


既可以是输入层，也可以是隐含层

自注意力模型



Self-attention



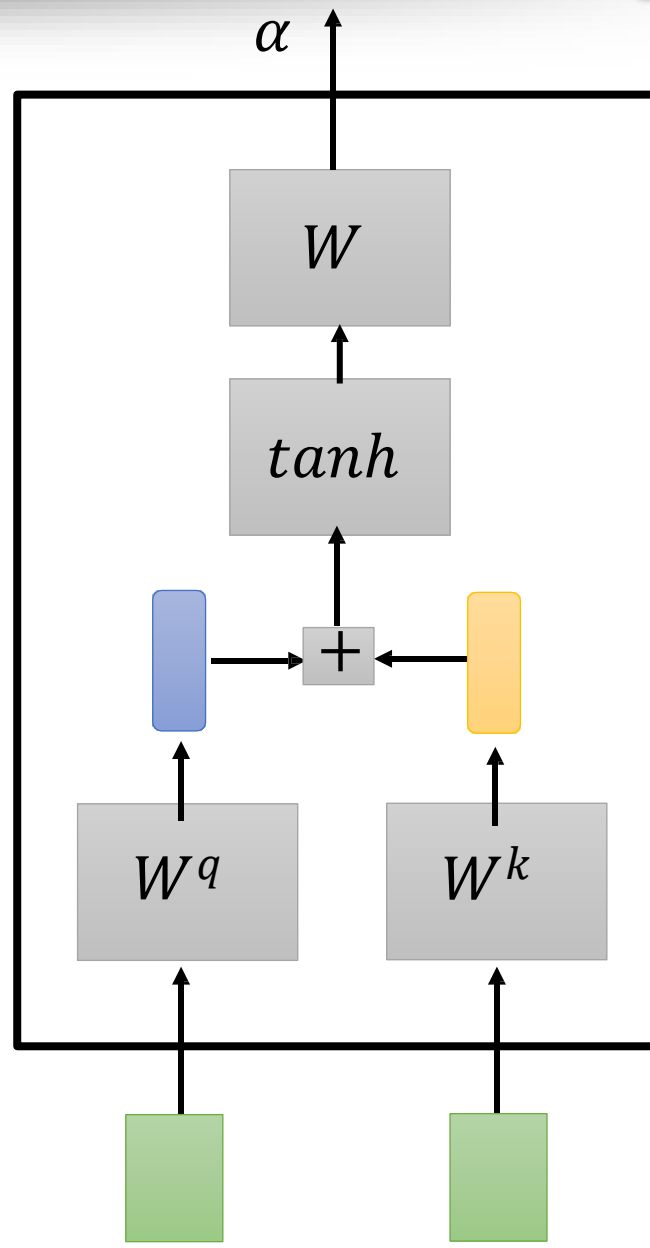
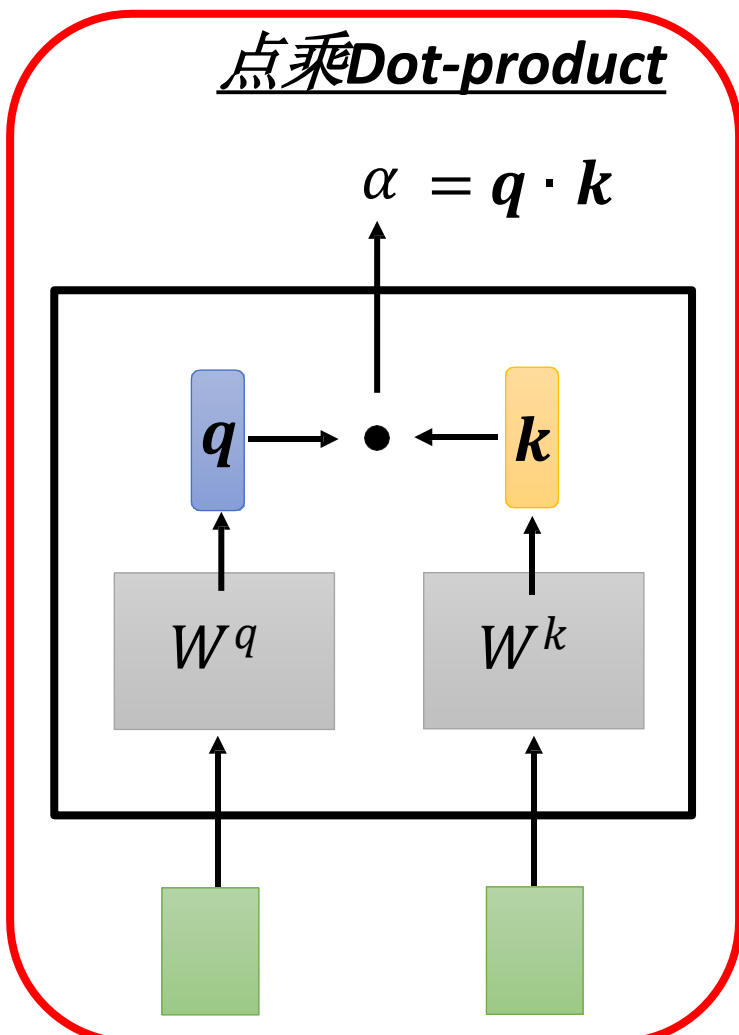
Find the relevant vectors in a sequence

自注意力模型

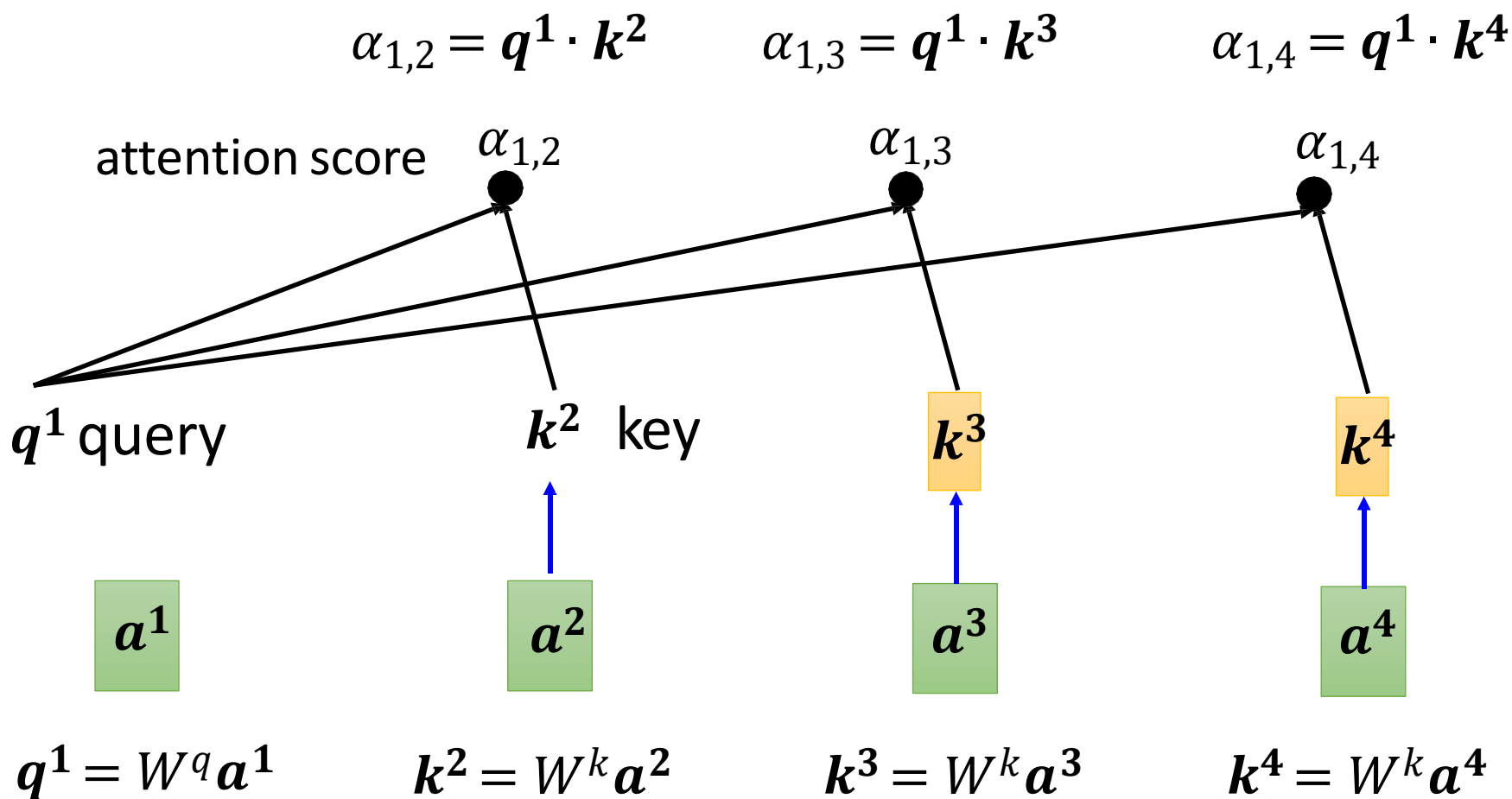


Additive

点乘 Dot-product



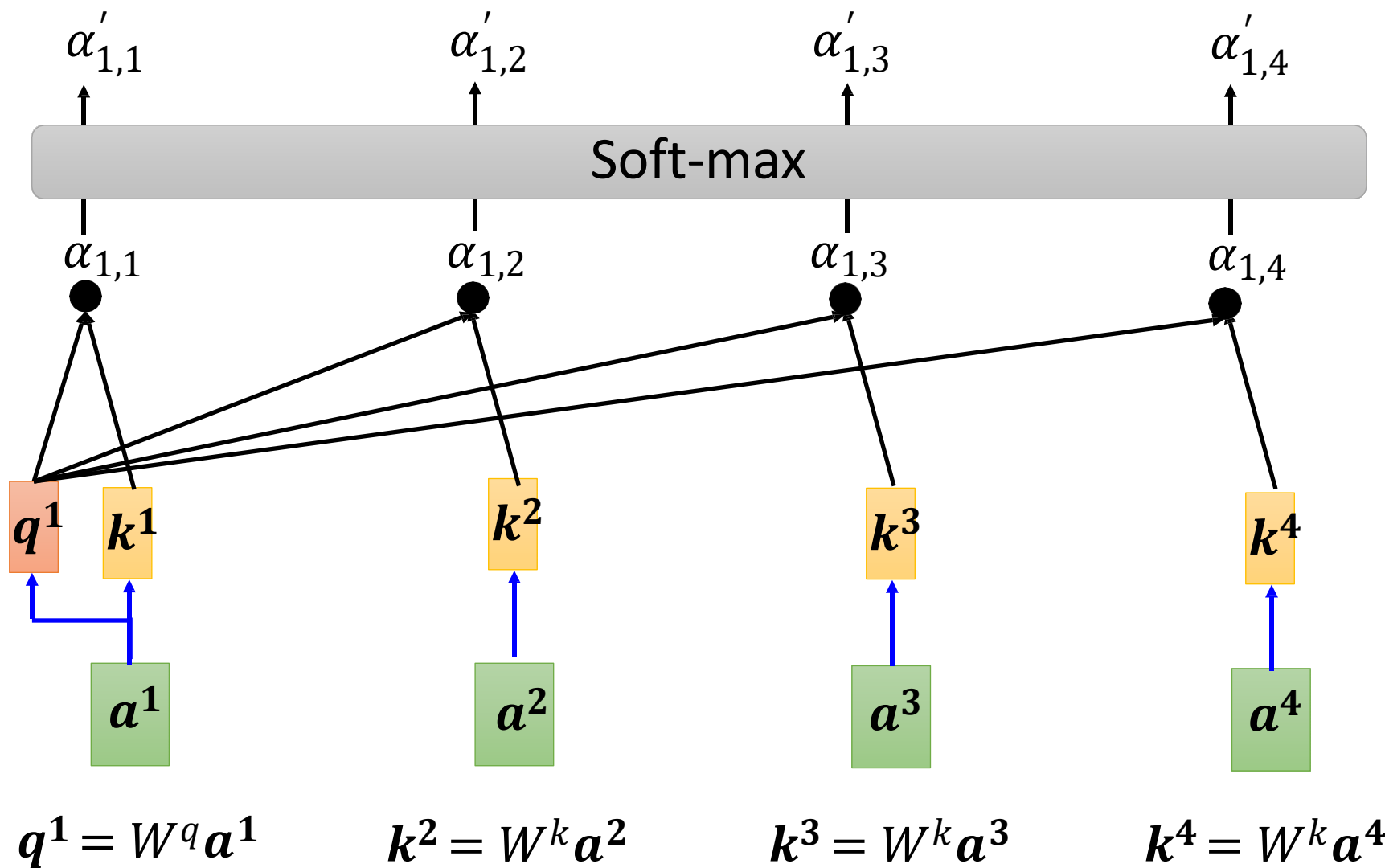
自注意力模型



自注意力模型



$$\alpha'_{1,i} = \exp(\alpha_{1,i}) / \sum_j \exp(\alpha_{1,j})$$

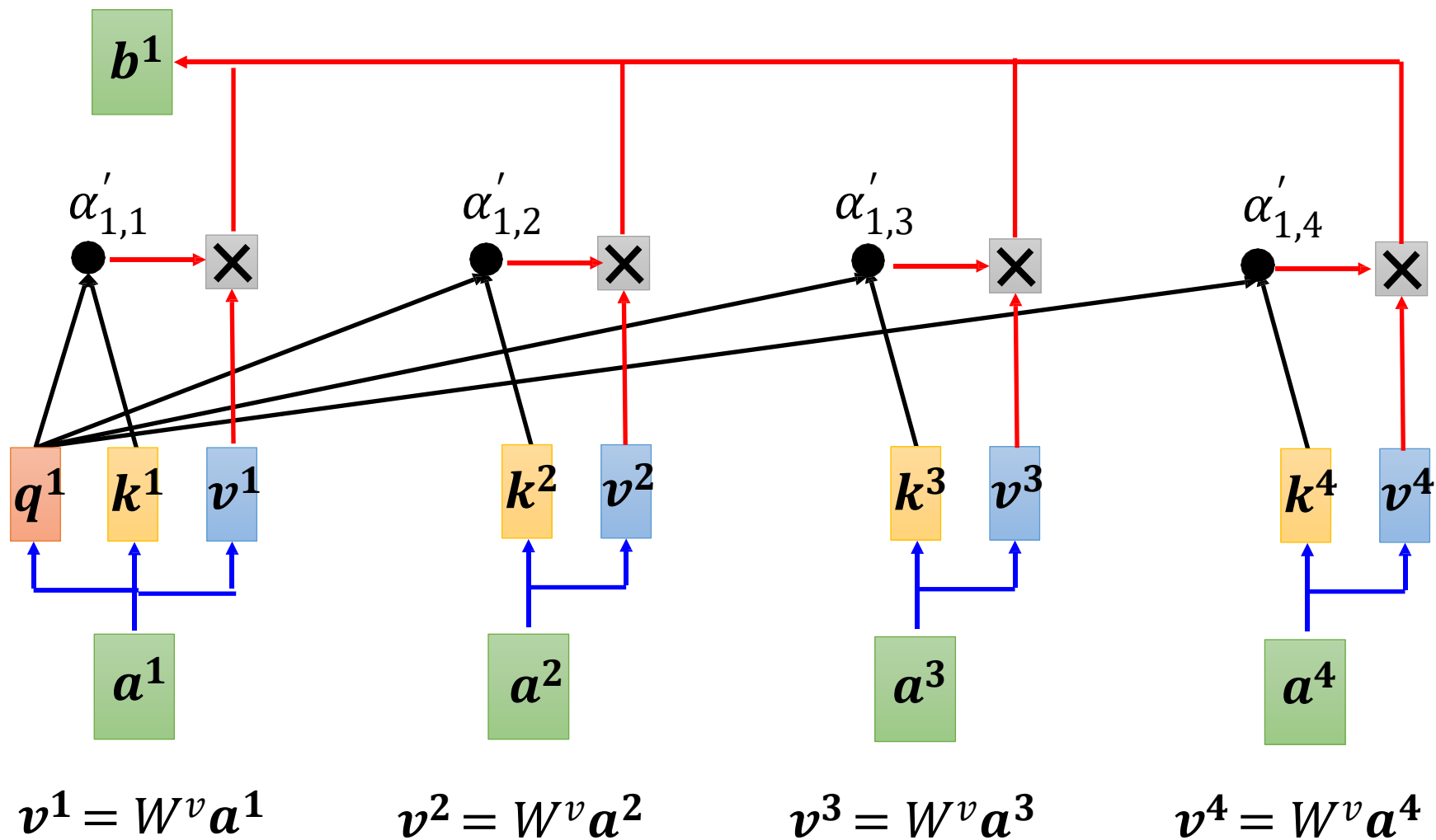


自注意力模型



基于注意力得分提取信息

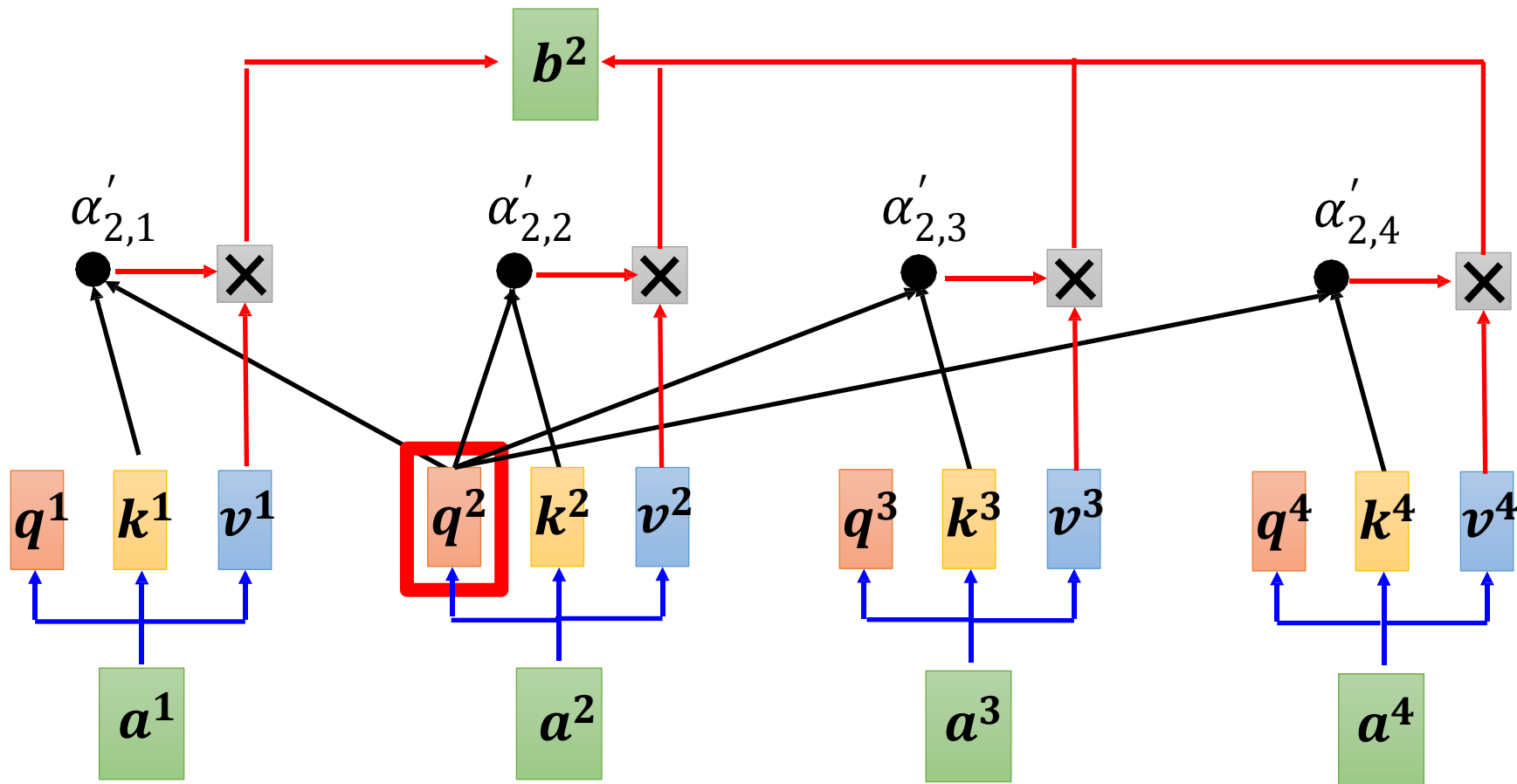
$$b^1 = \sum \alpha'_{1,i} v^i$$



自注意力模型



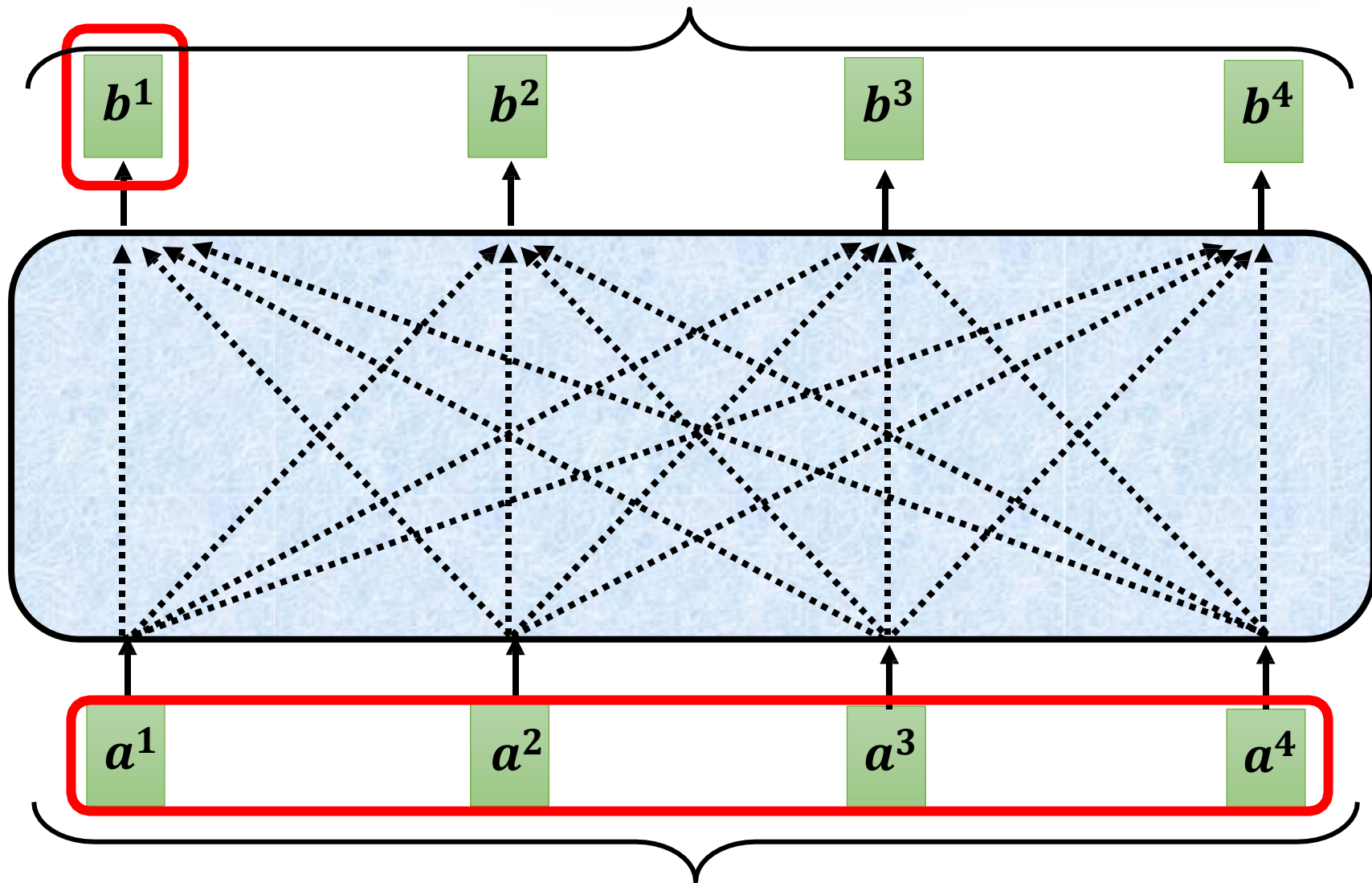
$$b^2 = \sum \alpha'_{2,i} v^i$$



自注意力模型



并行



既可以是输入层，也可以是隐含层

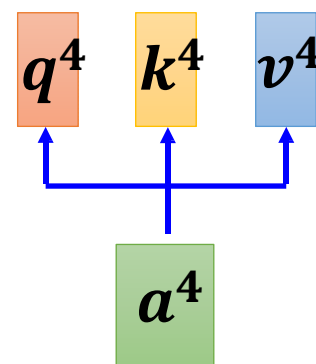
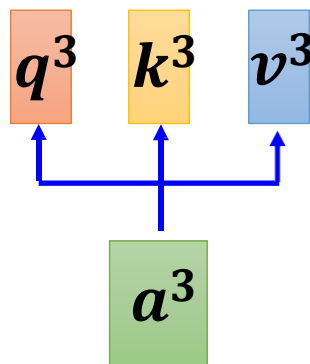
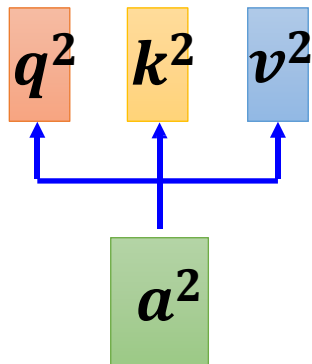
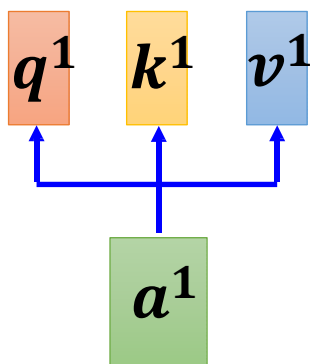
自注意力模型



$$q^i = W^q a^i \quad \begin{matrix} q^1 & q^2 & q^3 & q^4 \\ Q \end{matrix} = \begin{matrix} W^q & \begin{matrix} a^1 & a^2 & a^3 & a^4 \\ I \end{matrix} \end{matrix}$$

$$k^i = W^k a^i \quad \begin{matrix} k^1 & k^2 & k^3 & k^4 \\ K \end{matrix} = \begin{matrix} W^k & \begin{matrix} a^1 & a^2 & a^3 & a^4 \\ I \end{matrix} \end{matrix}$$

$$v^i = W^v a^i \quad \begin{matrix} v^1 & v^2 & v^3 & v^4 \\ V \end{matrix} = \begin{matrix} W^v & \begin{matrix} a^1 & a^2 & a^3 & a^4 \\ I \end{matrix} \end{matrix}$$

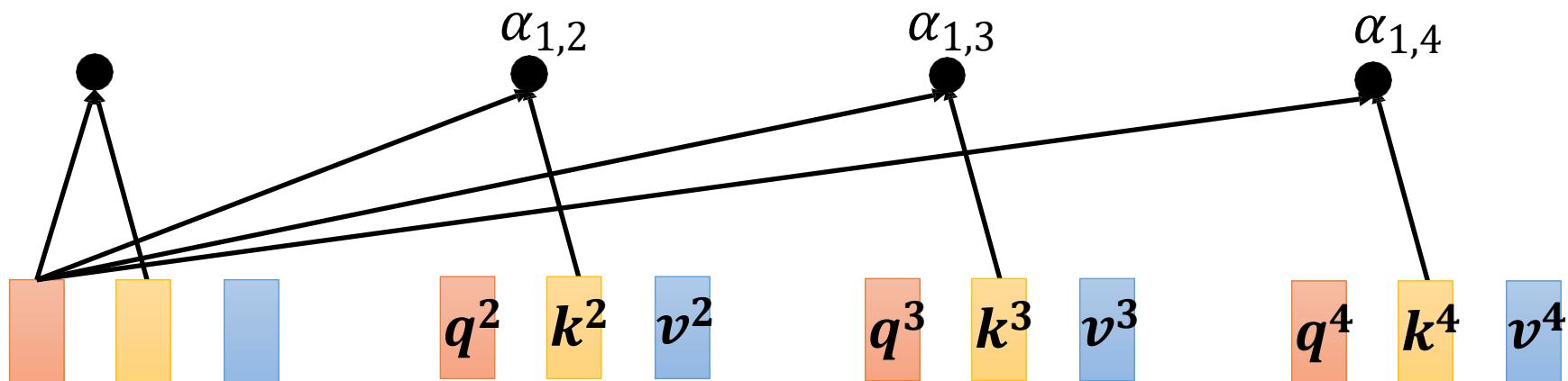


自注意力模型



$$\begin{aligned}\alpha_{1,1} &= k^1 q^1 & \alpha_{1,2} &= k^2 q^1 \\ & k^3 q^1 & \alpha_{1,4} &= k^4 q^1\end{aligned}$$

$$\begin{bmatrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \alpha_{1,3} \\ \alpha_{1,4} \end{bmatrix} = \begin{bmatrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{bmatrix} q^1$$



自注意力模型



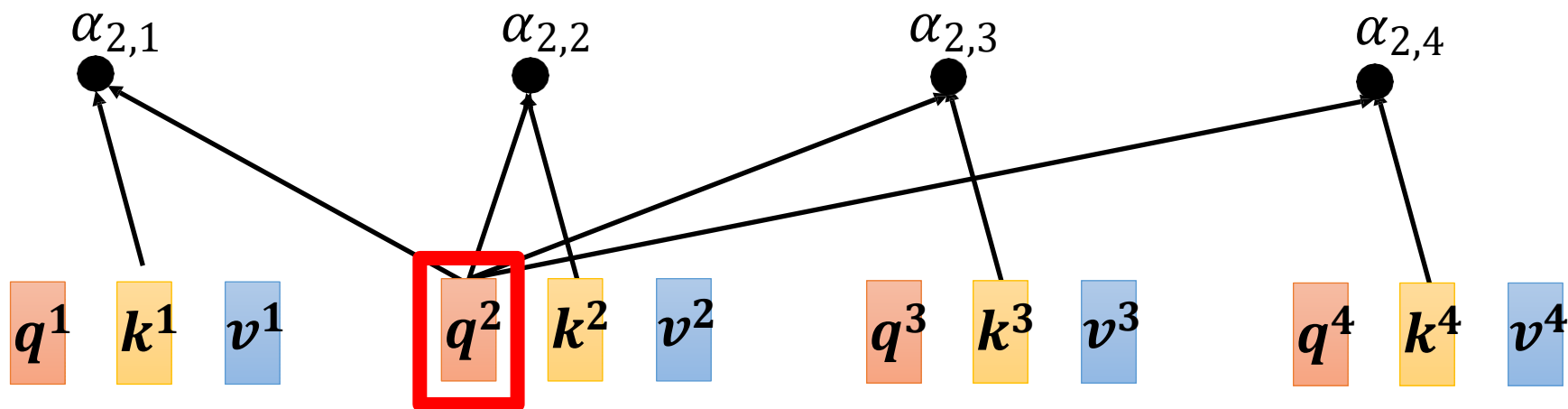
$$\alpha_{1,1} = k^1 q^1$$

$$\alpha_{1,2} = k^2 q^1$$

$$\alpha_{1,3} = k^3 q^1$$

$$\alpha_{1,4} = k^4 q^1$$

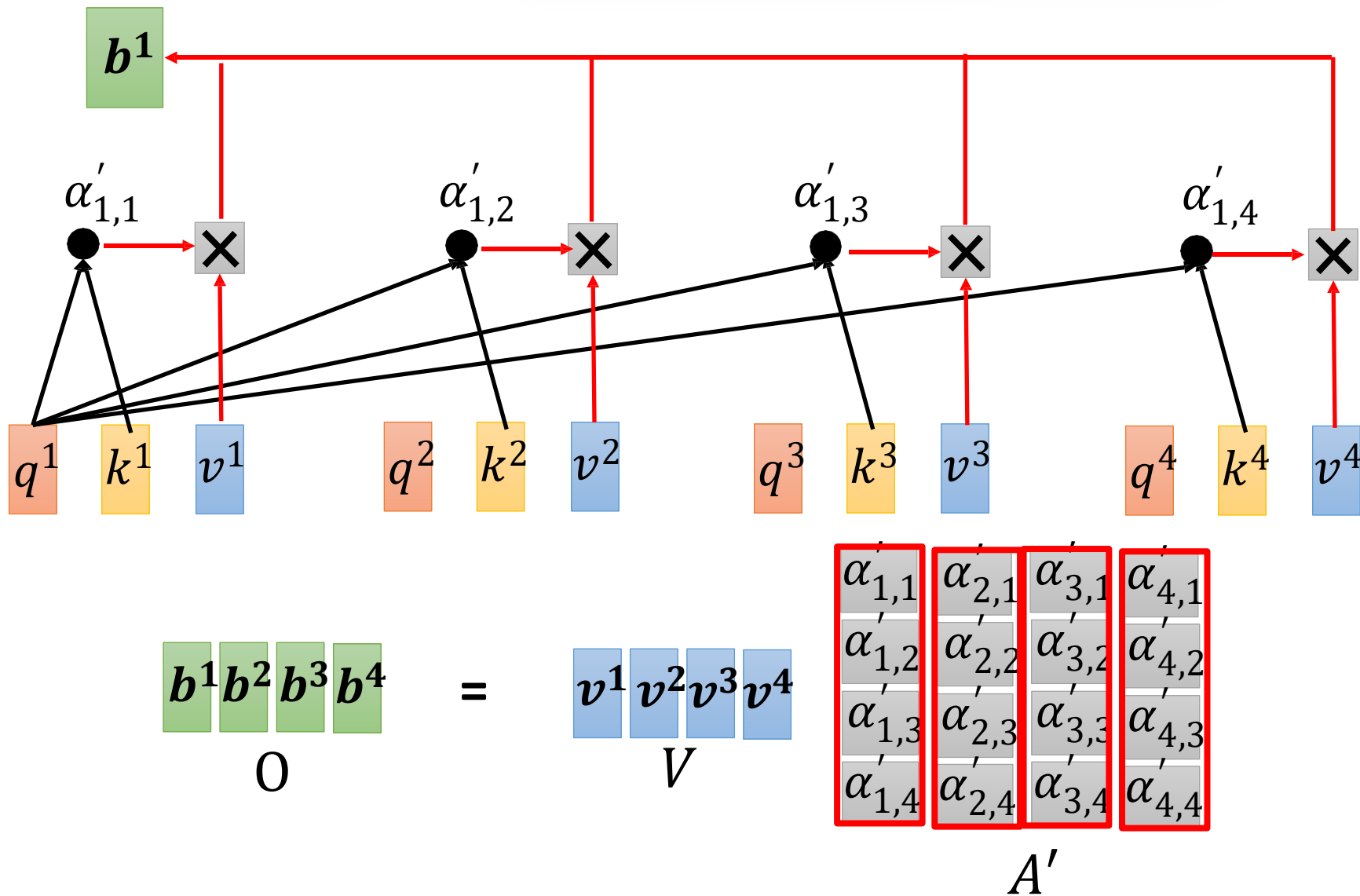
$$\begin{bmatrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \alpha_{1,3} \\ \alpha_{1,4} \end{bmatrix} = \begin{bmatrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{bmatrix} q^1$$



$$\begin{bmatrix} \alpha'_{1,1} & \alpha'_{2,1} & \alpha'_{3,1} & \alpha'_{4,1} \\ \alpha'_{1,2} & \alpha'_{2,2} & \alpha'_{3,2} & \alpha'_{4,2} \\ \alpha'_{1,3} & \alpha'_{2,3} & \alpha'_{3,3} & \alpha'_{4,3} \\ \alpha'_{1,4} & \alpha'_{2,4} & \alpha'_{3,4} & \alpha'_{4,4} \end{bmatrix} \xleftarrow{\text{softmax}} \begin{bmatrix} \alpha_{1,1} & \alpha_{2,1} & \alpha_{3,1} & \alpha_{4,1} \\ \alpha_{1,2} & \alpha_{2,2} & \alpha_{3,2} & \alpha_{4,2} \\ \alpha_{1,3} & \alpha_{2,3} & \alpha_{3,3} & \alpha_{4,3} \\ \alpha_{1,4} & \alpha_{2,4} & \alpha_{3,4} & \alpha_{4,4} \end{bmatrix} = \begin{bmatrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{bmatrix} \begin{bmatrix} q^1 & q^2 & q^3 & q^4 \end{bmatrix}$$

$K^T \quad Q$

自注意力模型

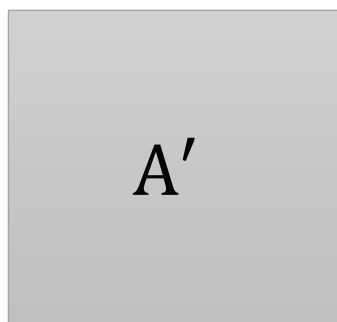


自注意力模型

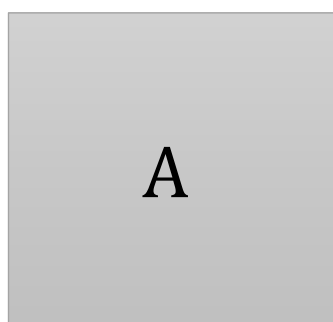


$$\begin{aligned} Q &= W^q I \\ K &= W^k I \\ V &= W^v I \end{aligned}$$

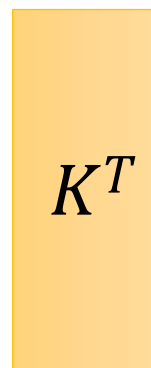
Parameters
to be learned



Attention Matrix



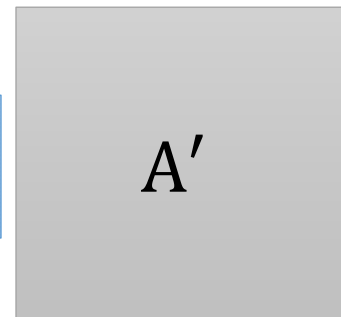
=



=



=



自注意力模型

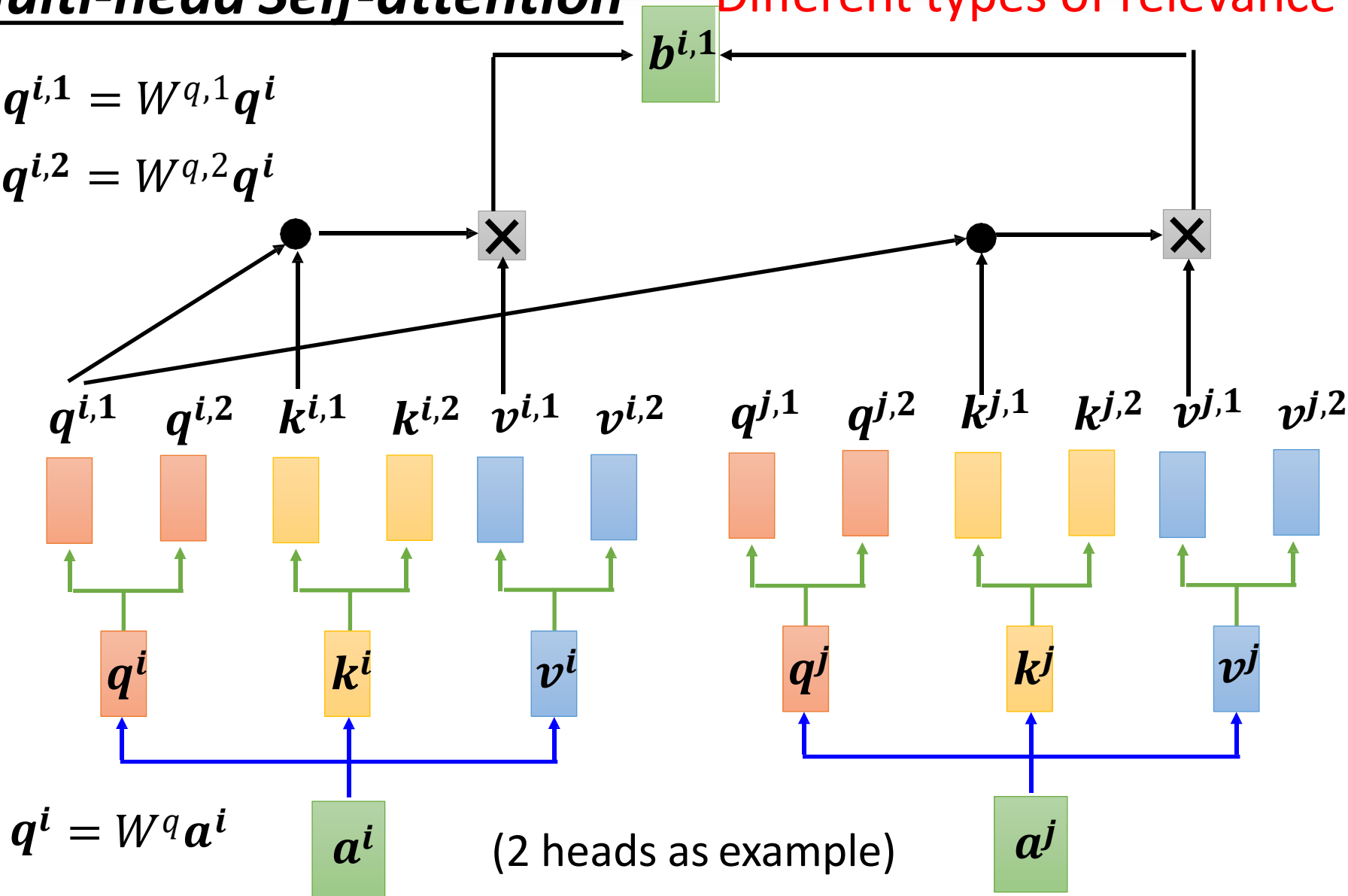


Multi-head Self-attention

Different types of relevance

$$q^{i,1} = W^{q,1} q^i$$

$$q^{i,2} = W^{q,2} q^i$$



自注意力模型



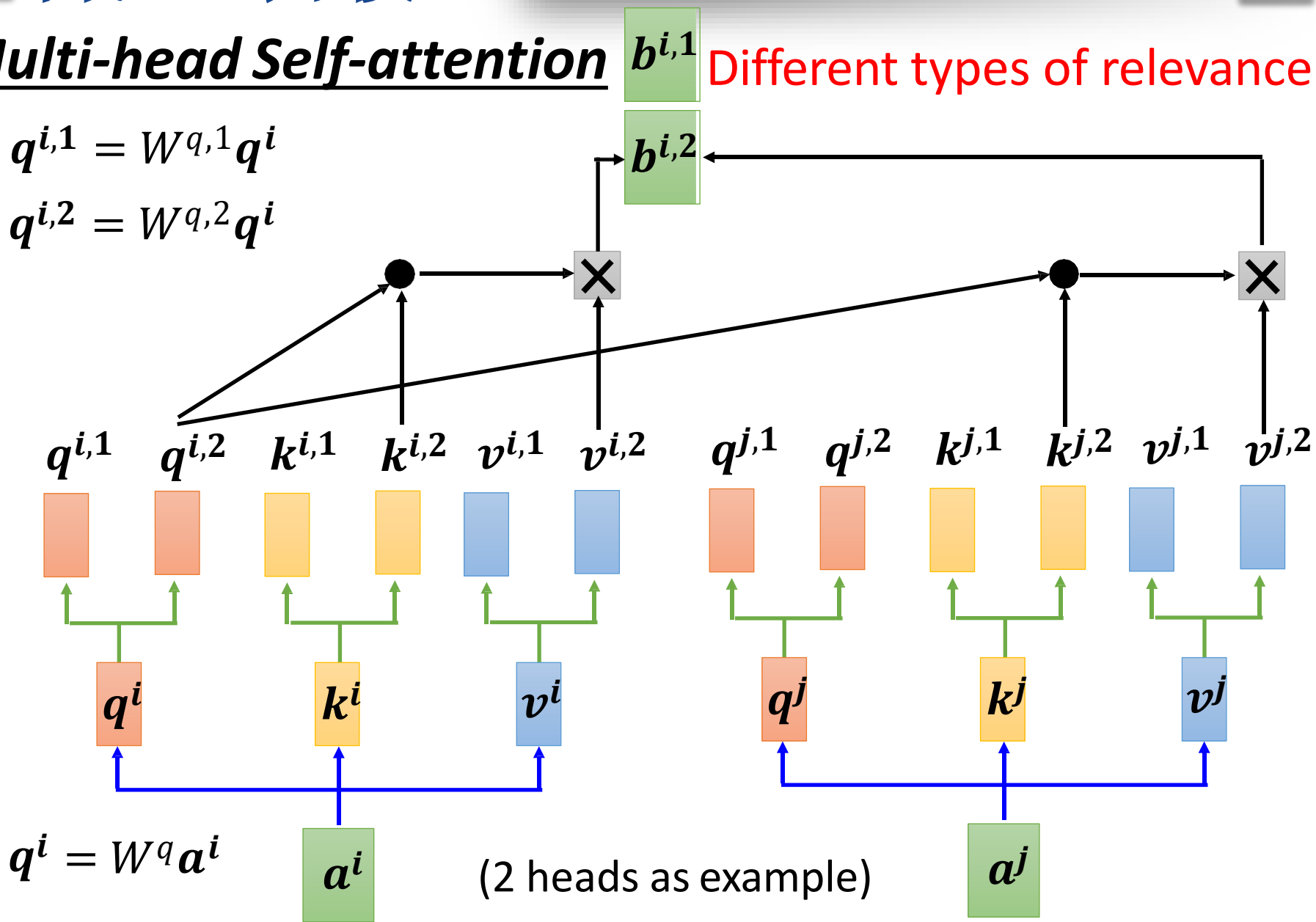
Multi-head Self-attention

 $b^{i,1}$ $b^{i,2}$

Different types of relevance

$$q^{i,1} = W^{q,1} q^i$$

$$q^{i,2} = W^{q,2} q^i$$



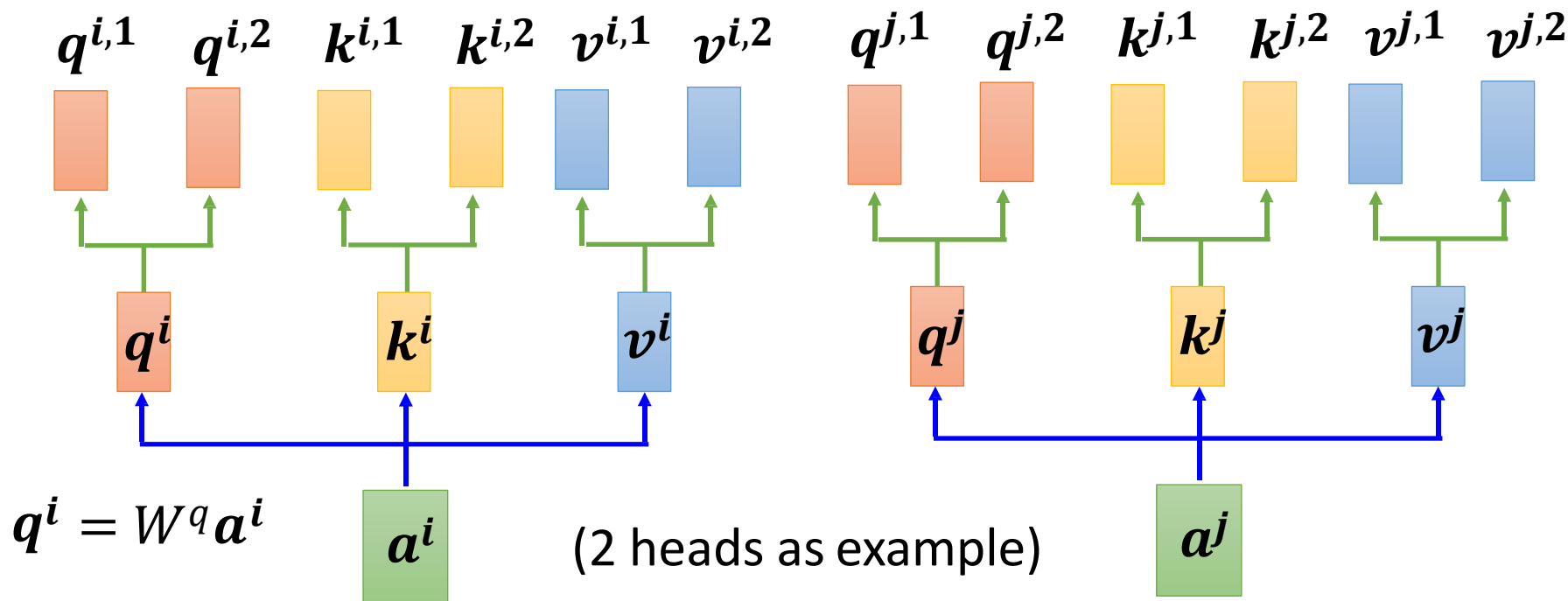
自注意力模型

Multi-head Self-attention

Different types of relevance



$$b^i = W^O \begin{bmatrix} b^{i,1} \\ b^{i,2} \end{bmatrix}$$

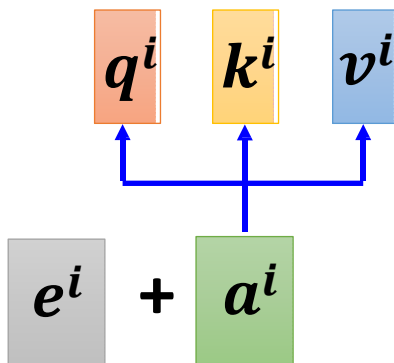


自注意力模型

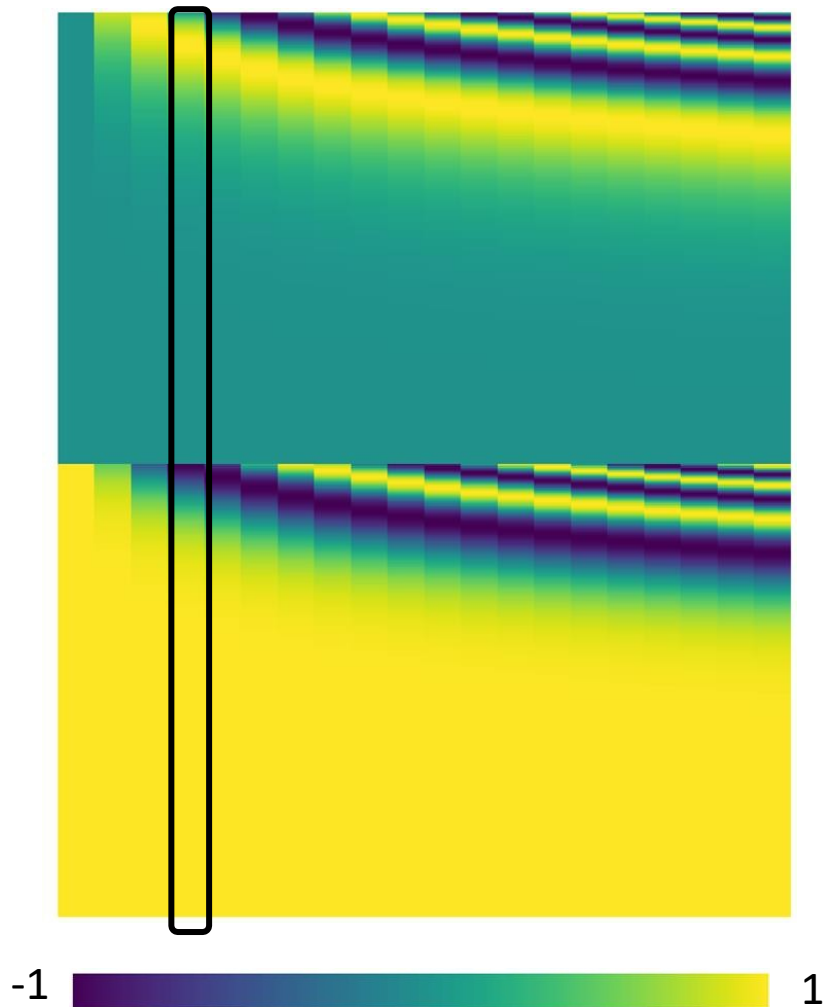


位置编码 Positional Encoding

- No position information in self-attention.
- Each position has a unique positional vector e^i
- **hand-crafted**
- **learned from data**



Each column represents a positional vector e^i



自注意力模型



Many applications



Transformer

<https://arxiv.org/abs/1706.03762>



BERT

<https://arxiv.org/abs/1810.04805>

Widely used in Natural Language Processing (NLP)!

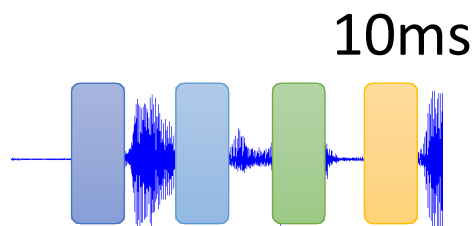
自注意力模型



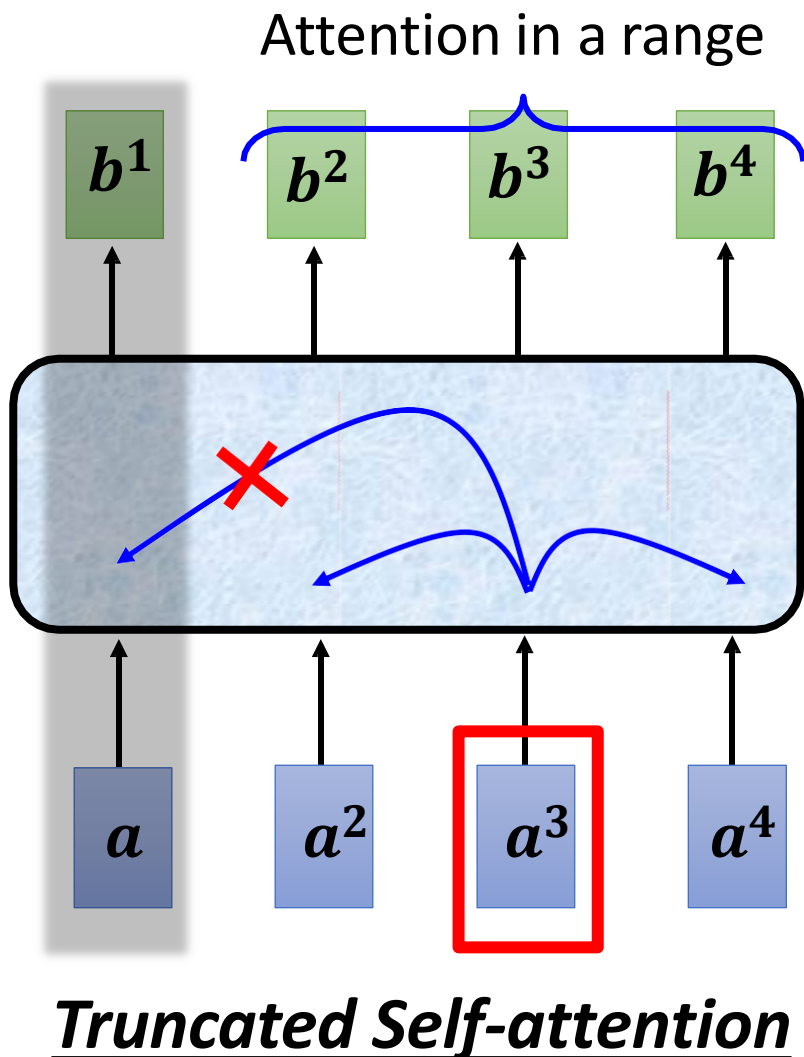
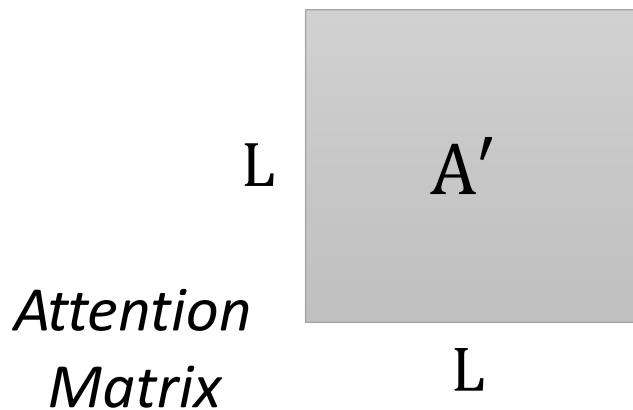
语音 Self-attention for Speech

<https://arxiv.org/abs/1910.12977>

语音是一个很长的序列。



假设输入序列长度为 L

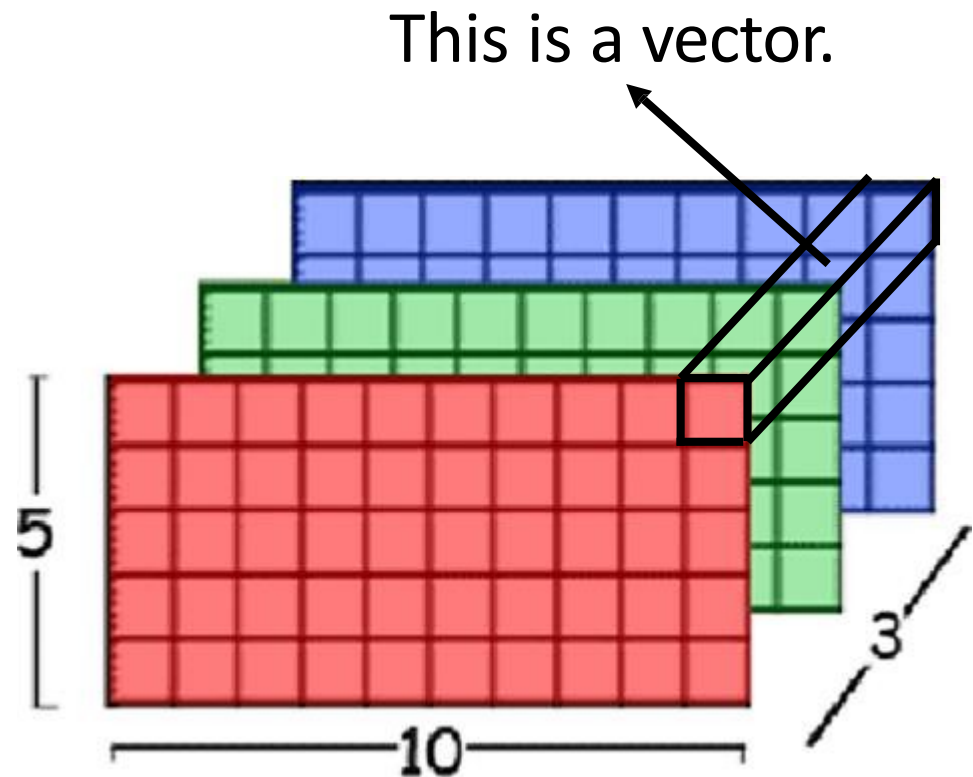
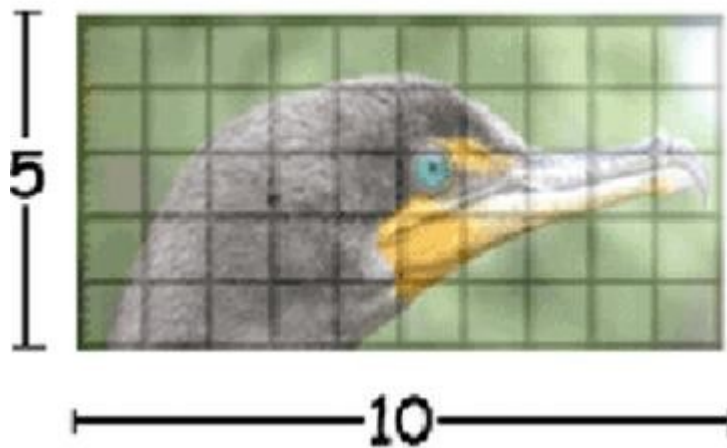


自注意力模型



图像Self-attention for Image

An **image** can also be considered as a **vector set**.

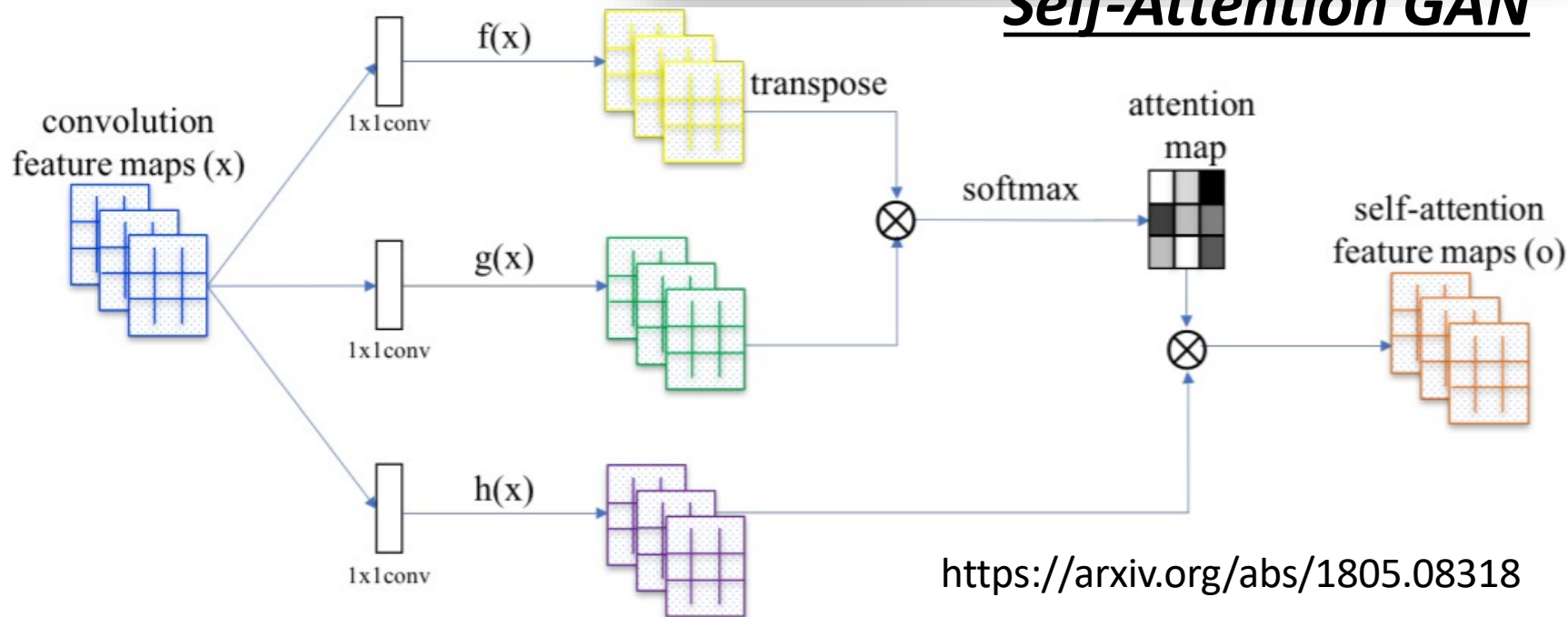


Source of image: https://www.researchgate.net/figure/Color-image-representation-and-RGB-matrix_fig15_282798184

自注意力模型

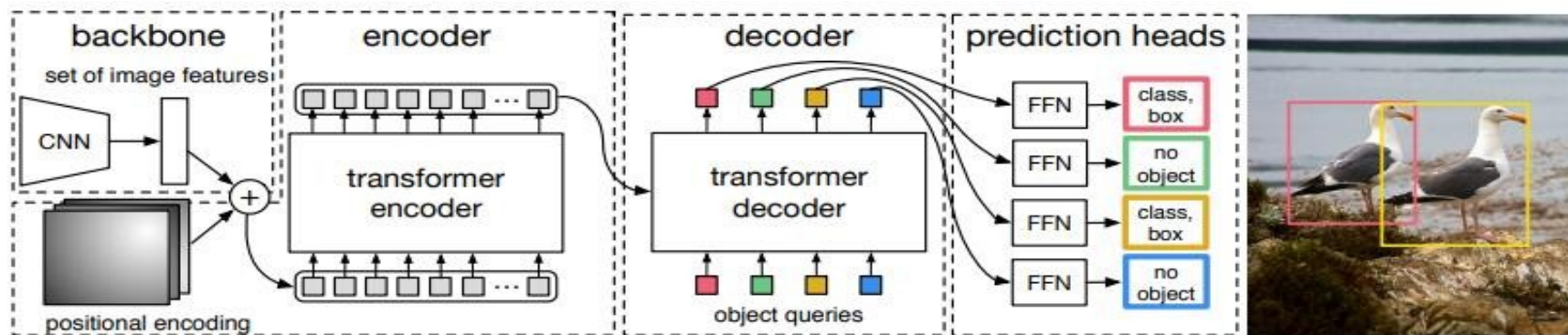


Self-Attention GAN



<https://arxiv.org/abs/1805.08318>

DEtection Transformer (DETR)



<https://arxiv.org/abs/2005.12872>



Self-attention v.s. CNN

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

- CNN: self-attention that can only attends in a receptive field
 - CNN is simplified self-attention.
- Self-attention: CNN with learnable receptive field
 - Self-attention is the complex version of CNN.

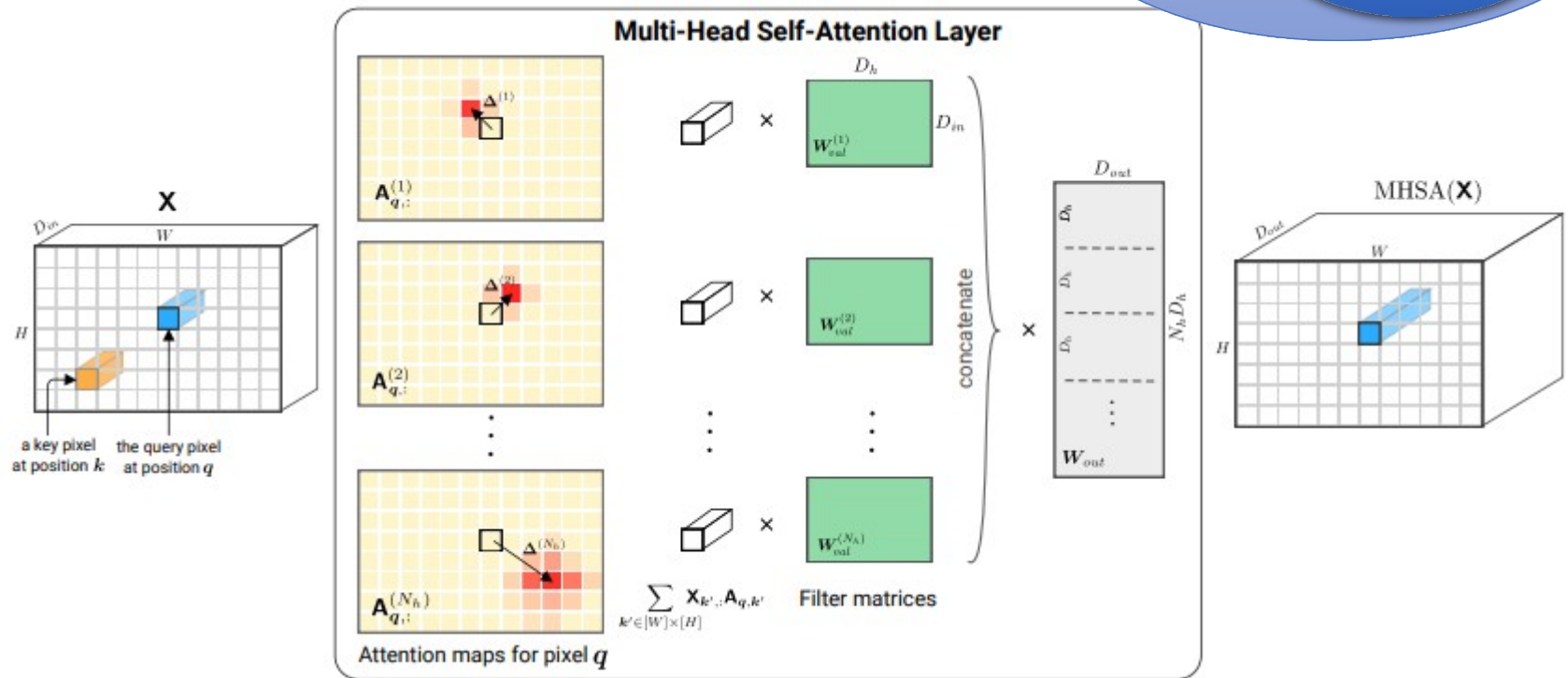
自注意力模型



Self-attention v.s. CNN

Self-attention

CNN



On the Relationship between Self-Attention and Convolutional Layers

<https://arxiv.org/abs/1911.03584>

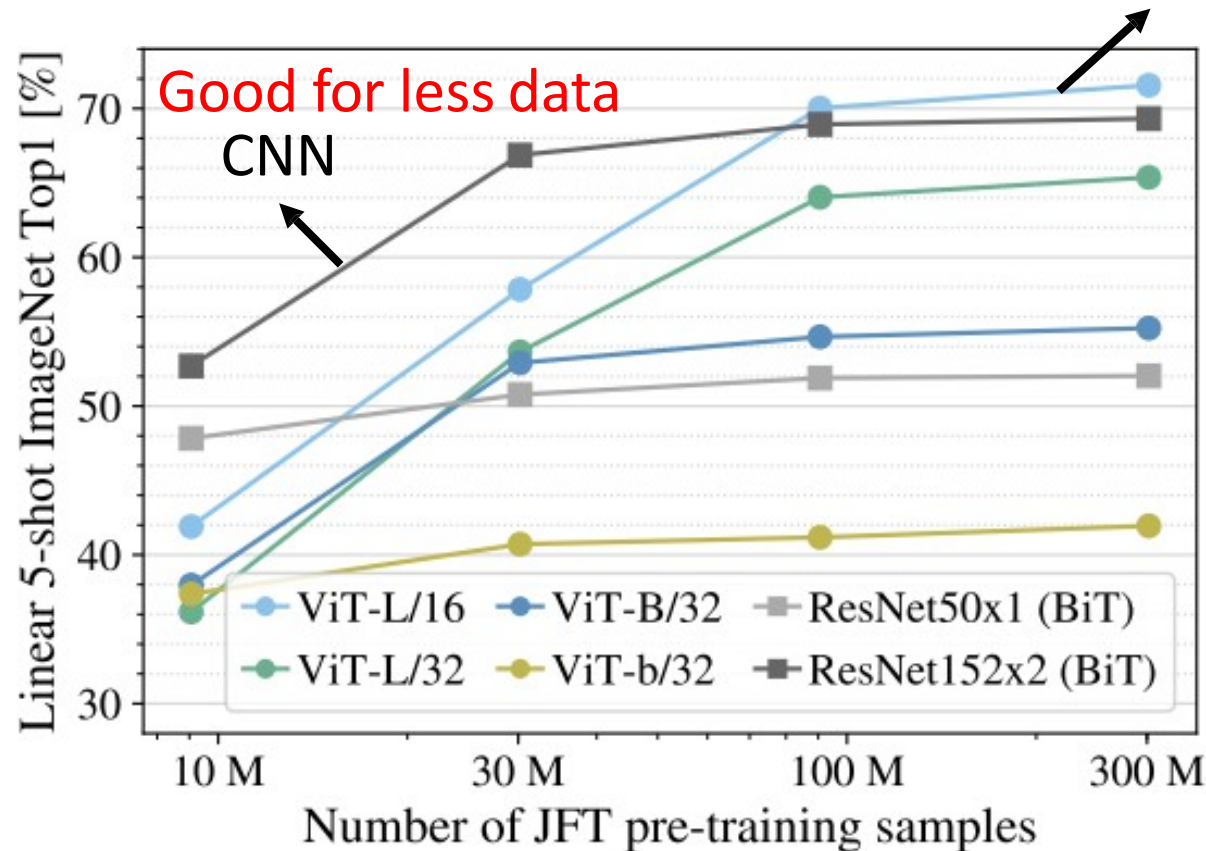
自注意力模型



Self-attention v.s. CNN

Good for more data

Self-attention



An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

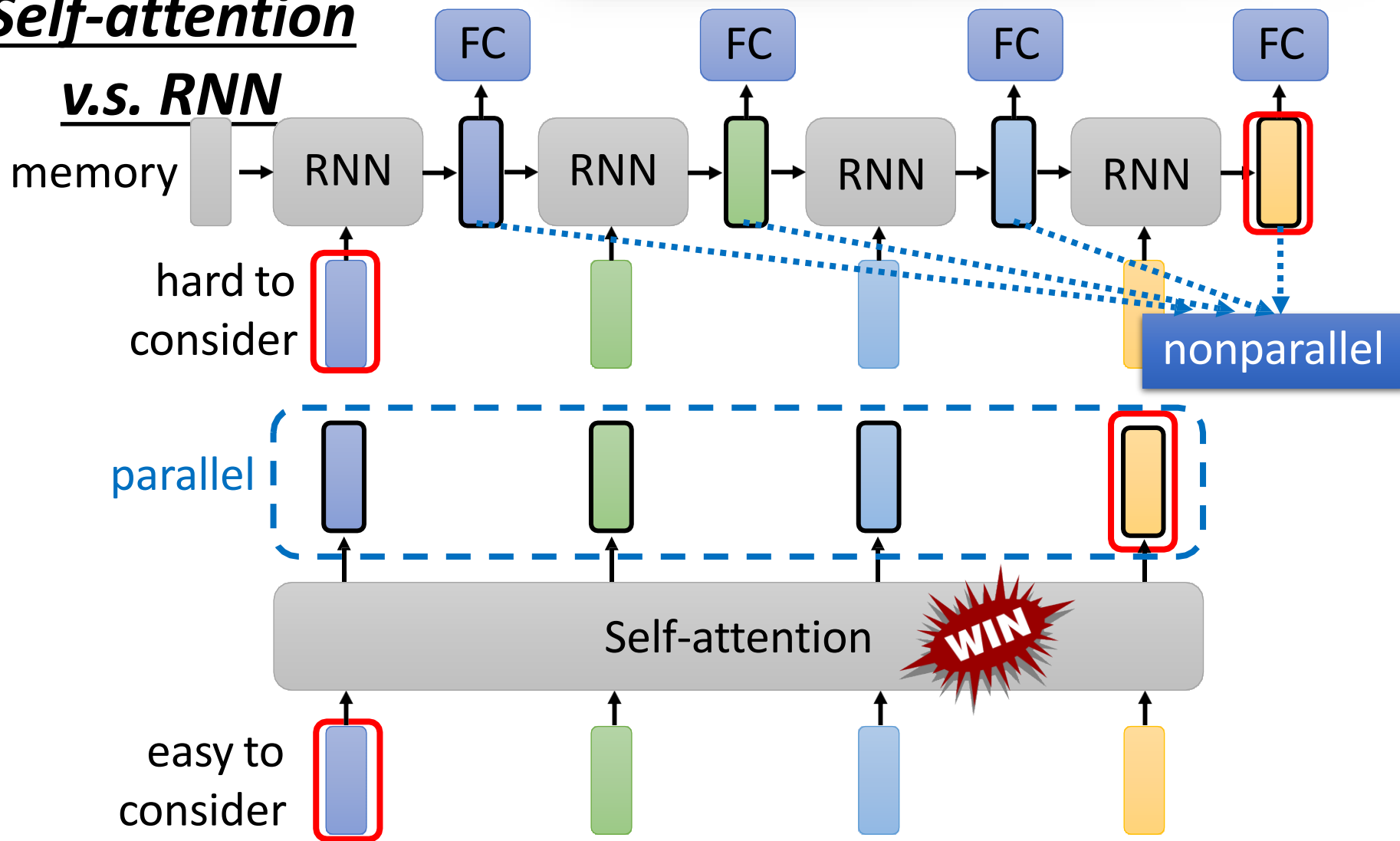
<https://arxiv.org/pdf/2010.11929.pdf>

自注意力模型

Recurrent Neural Network (RNN)

Self-attention

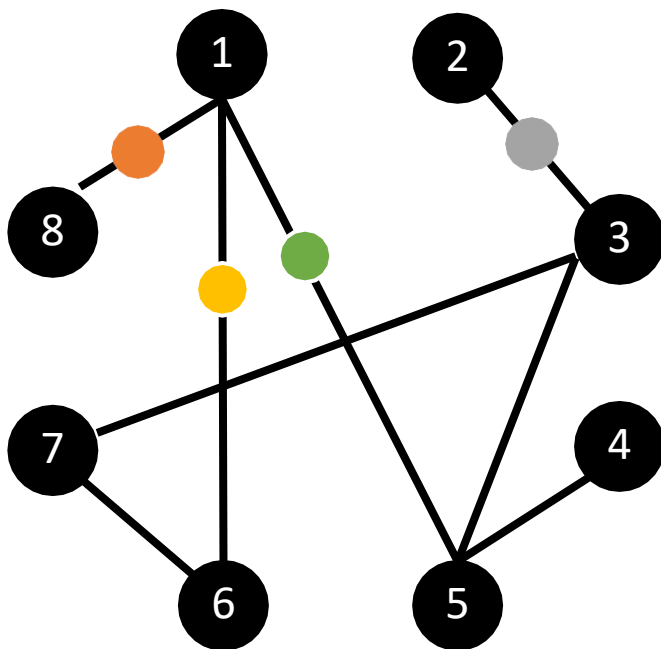
v.s. RNN



Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention

<https://arxiv.org/abs/2006.16236>

Self-attention for Graph



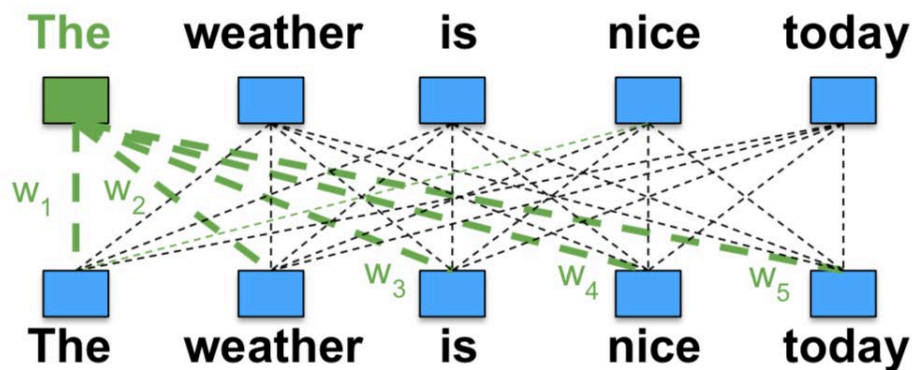
Consider **edge**: only attention to connected nodes

Attention Matrix

	1	2	3	4	5	6	7	8
1								
2								
3								
4								
5								
6								
7								
8							0	

This is one type of **Graph Neural Network (GNN)**.

自注意力示例



$$w_1, w_2, w_3, w_4, w_5 = \text{softmax} \left(\begin{bmatrix} 0.6 & 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.6 & 0.2 & 0.9 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.1 & 0.1 & 0.1 \\ 0.8 & 0.1 & 0.8 & 0.4 & 0.6 \end{bmatrix} \right)$$

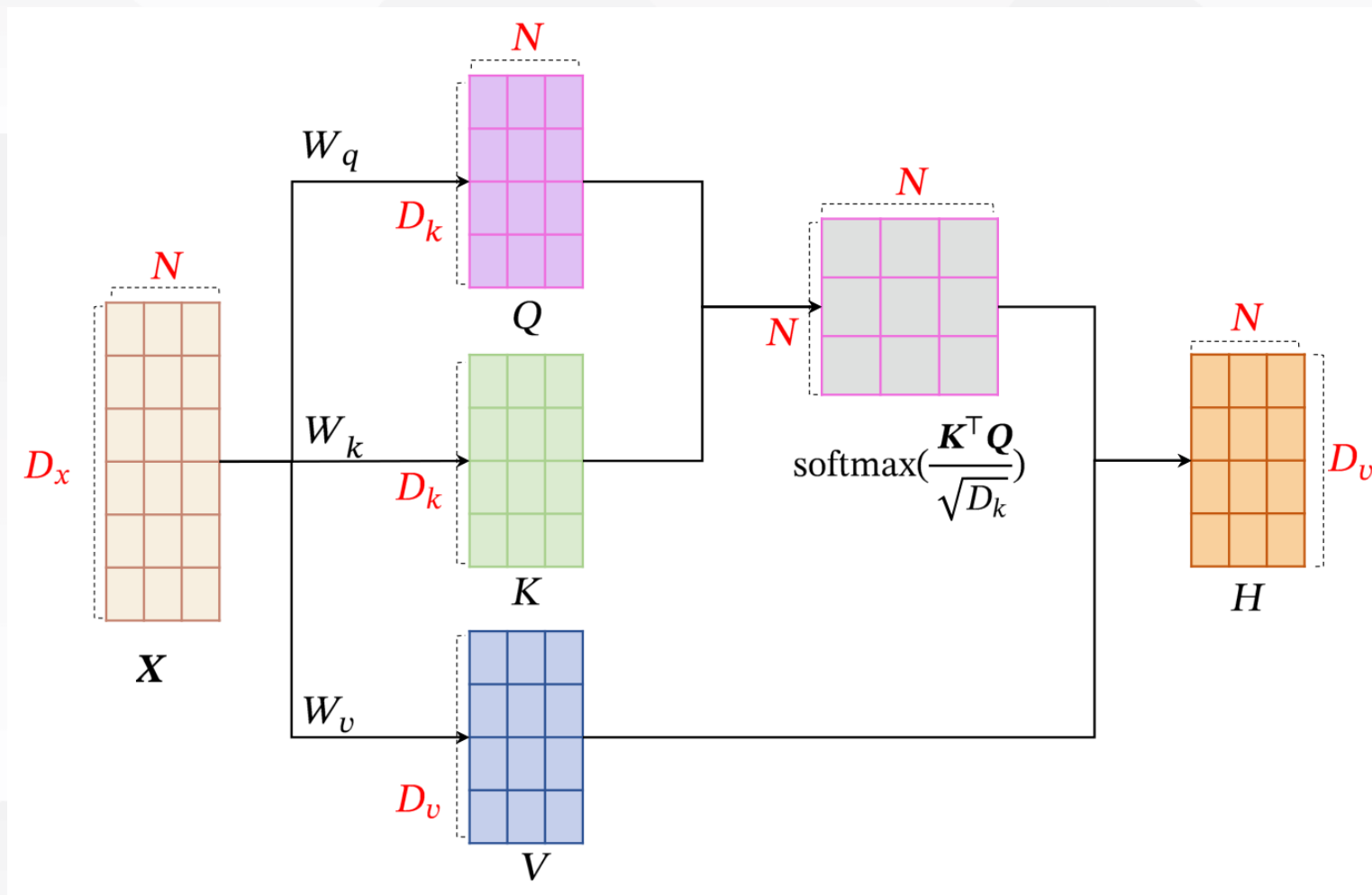
The The weather is nice today

$$\begin{bmatrix} 1.8 \\ 2.3 \\ 0.4 \end{bmatrix} = w_1 \times \begin{bmatrix} 0.6 \\ 0.2 \\ 0.8 \end{bmatrix} + w_2 \times \begin{bmatrix} 0.2 \\ 0.3 \\ 0.1 \end{bmatrix} + w_3 \times \begin{bmatrix} 0.9 \\ 0.1 \\ 0.8 \end{bmatrix} + w_4 \times \begin{bmatrix} 0.4 \\ 0.1 \\ 0.4 \end{bmatrix} + w_5 \times \begin{bmatrix} 0.4 \\ 0.1 \\ 0.6 \end{bmatrix}$$

The The weather is nice today

QKV模式

Query-Key-Value



自注意力模型



- 输入序列为 $X=[x_1, \dots, x_N] \in \mathbb{R}^{D_x \times N}$

- 首先生成三个向量序列

$$Q = W_q X \in \mathbb{R}^{D_k \times N},$$

$$K = W_k X \in \mathbb{R}^{D_k \times N},$$

$$V = W_v X \in \mathbb{R}^{D_v \times N},$$

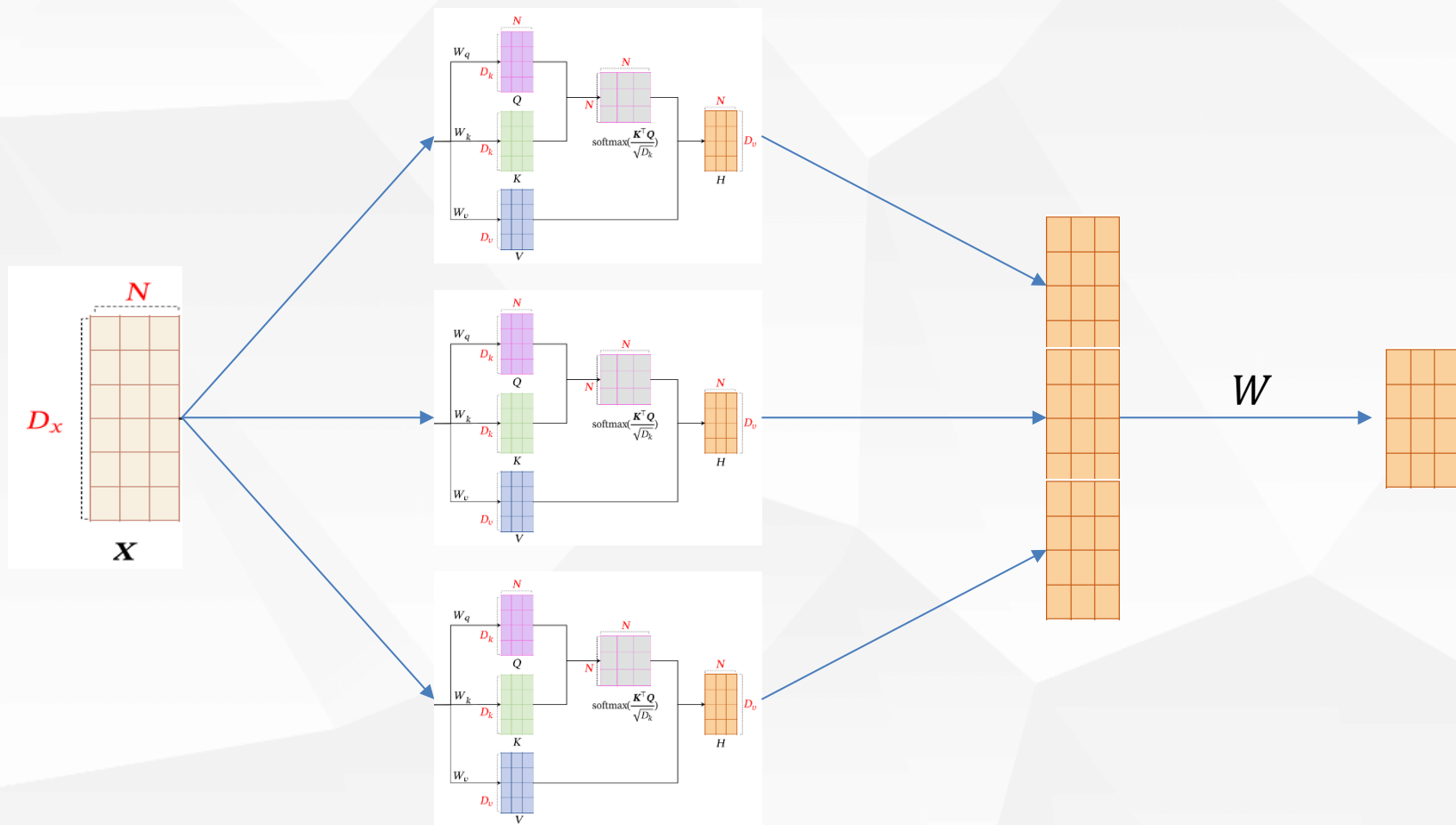
- 计算 h_n

$$h_n = \text{att}((K, V), q_n)$$

- 如果使用缩放点积来作为注意力打分函数，输出向量序列可以简写为

$$H = V \text{softmax}\left(\frac{K^T Q}{\sqrt{D_k}}\right),$$

多头自注意力模型



Transformer Encoder



- 仅仅自注意力还不够

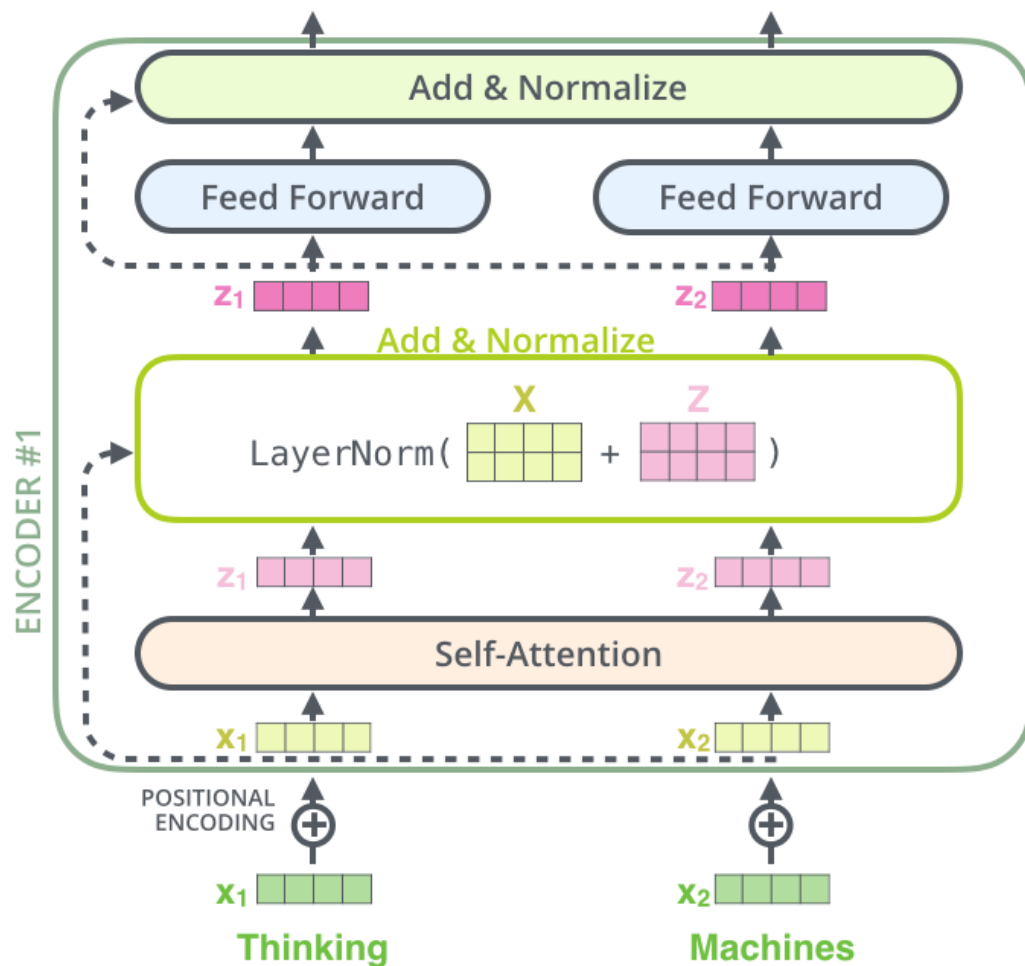
- 其它操作

位置编码

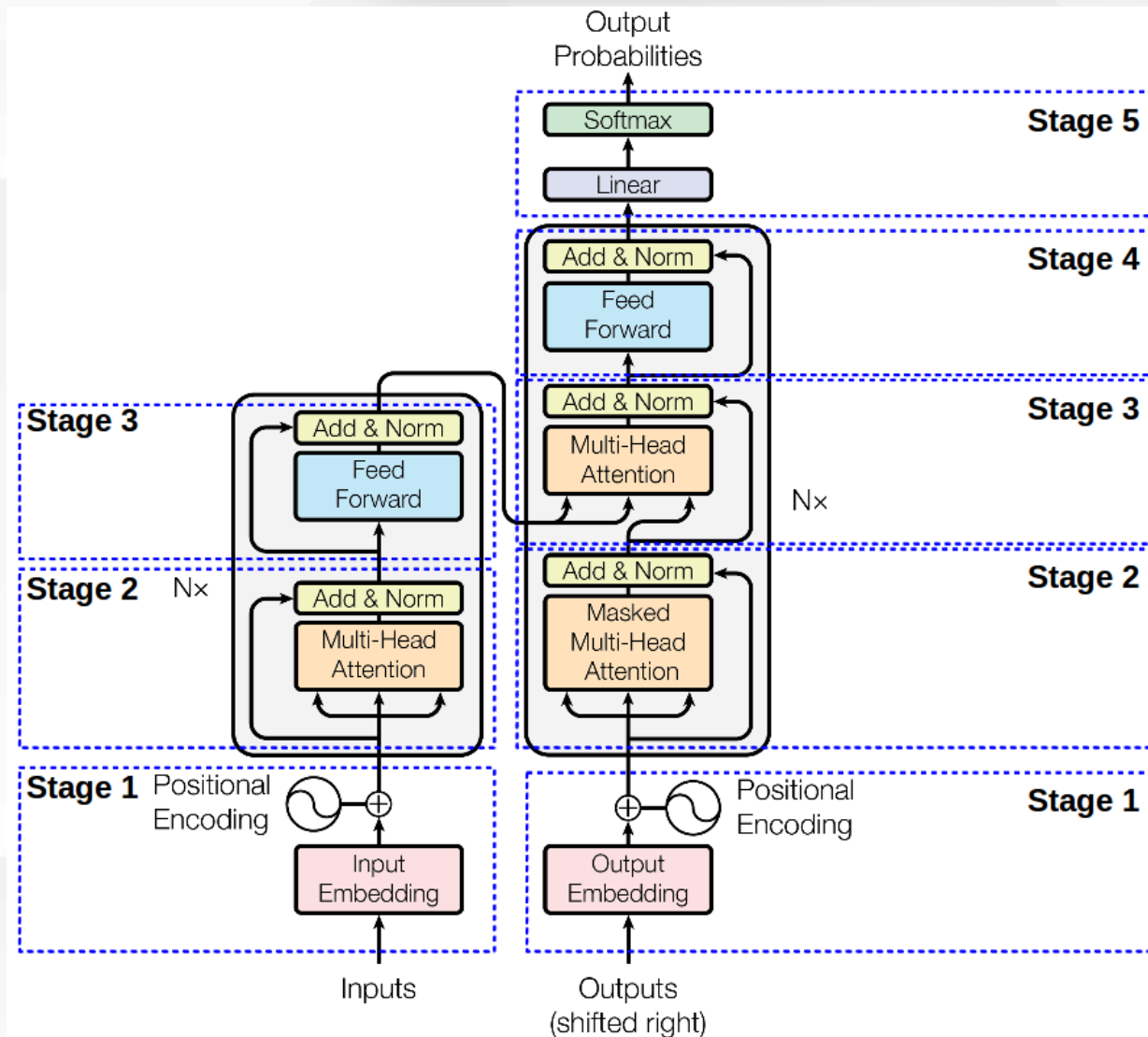
层归一化

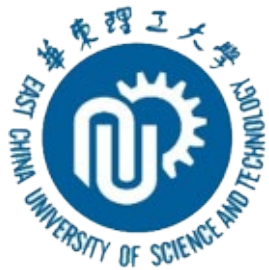
直连边

逐位的FNN

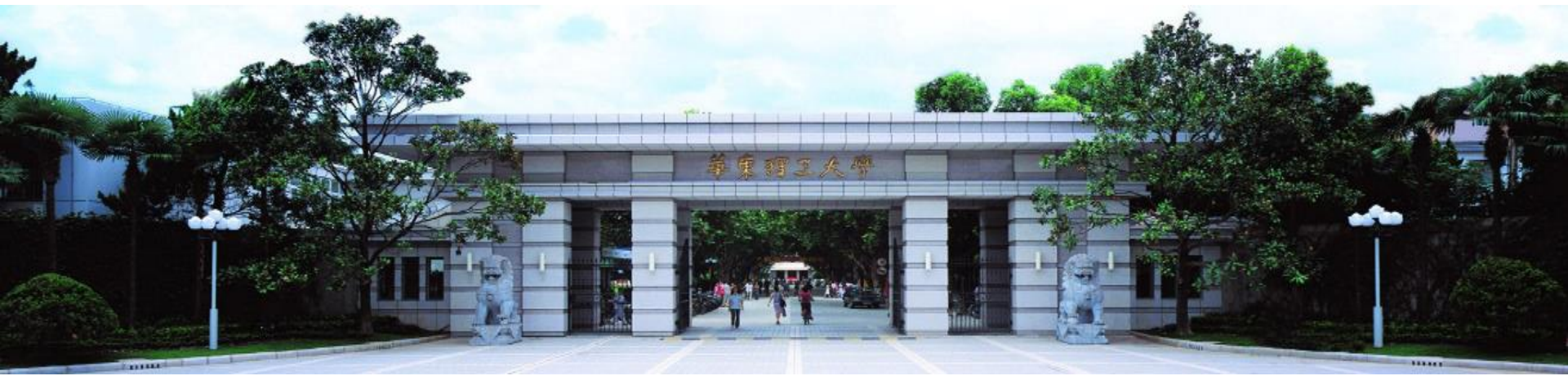


Transformer





以下了解



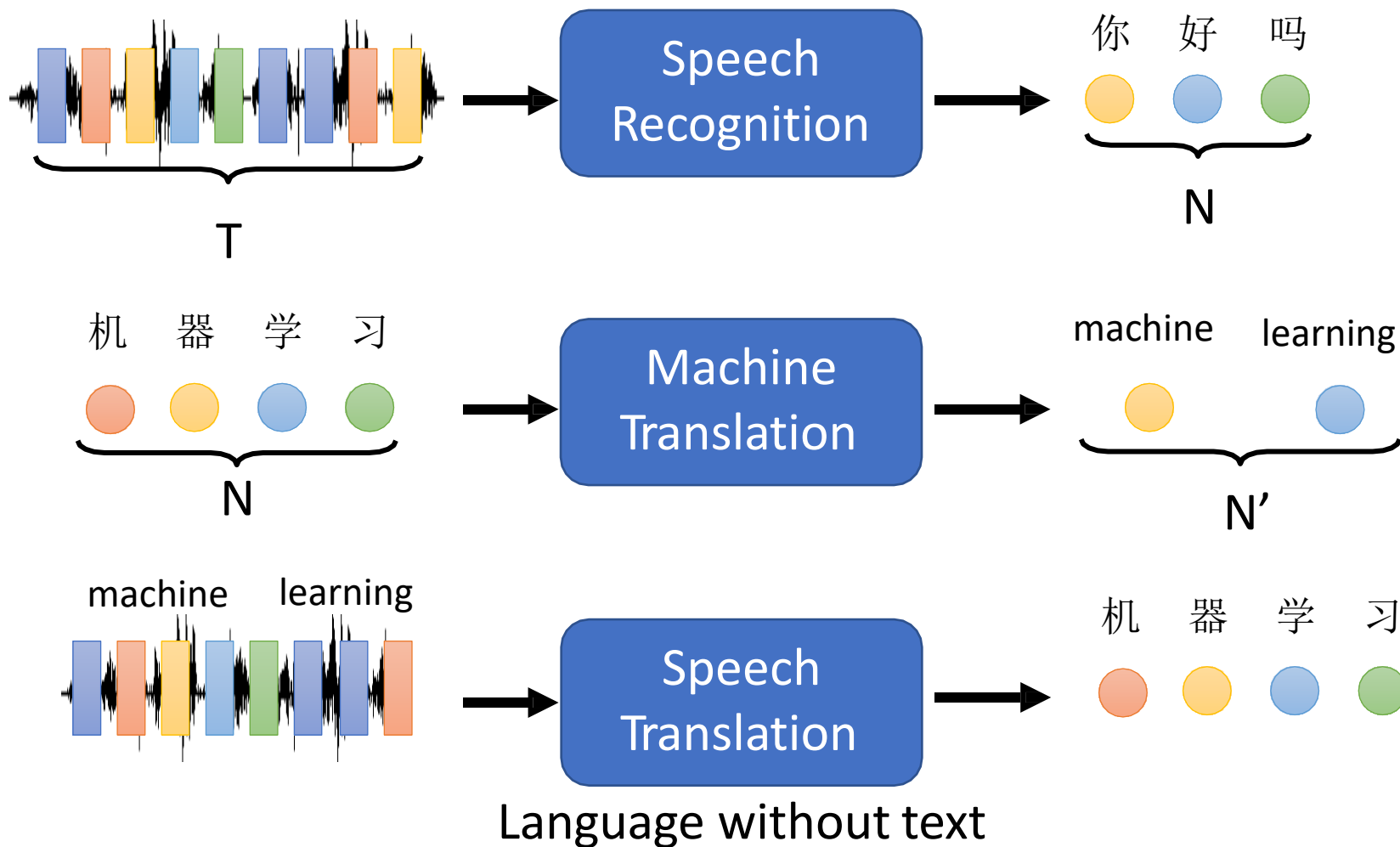
Transformer



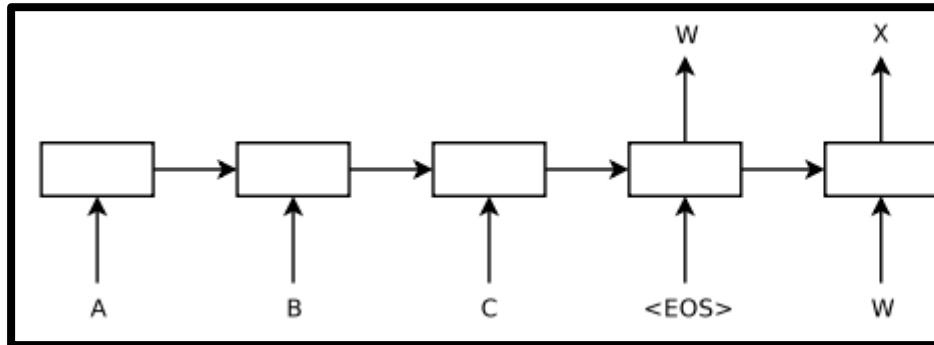
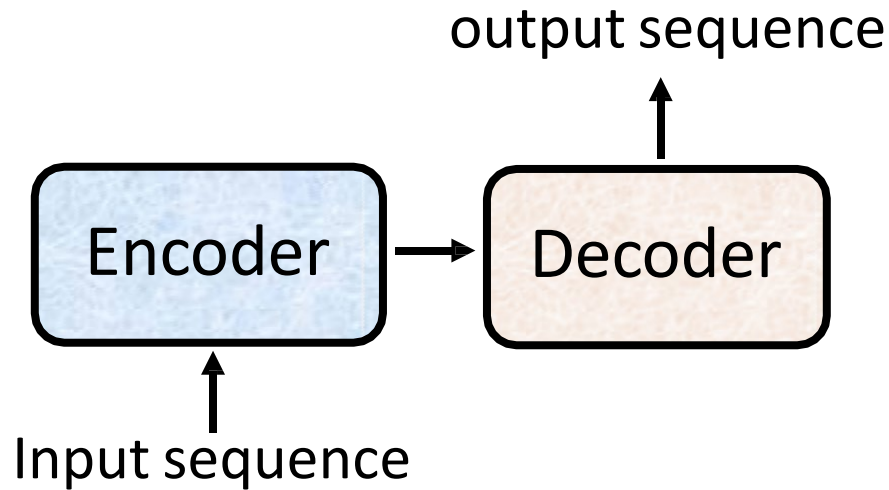
Sequence-to-sequence (Seq2seq)

Input a sequence, output a sequence

The output length is determined by model.

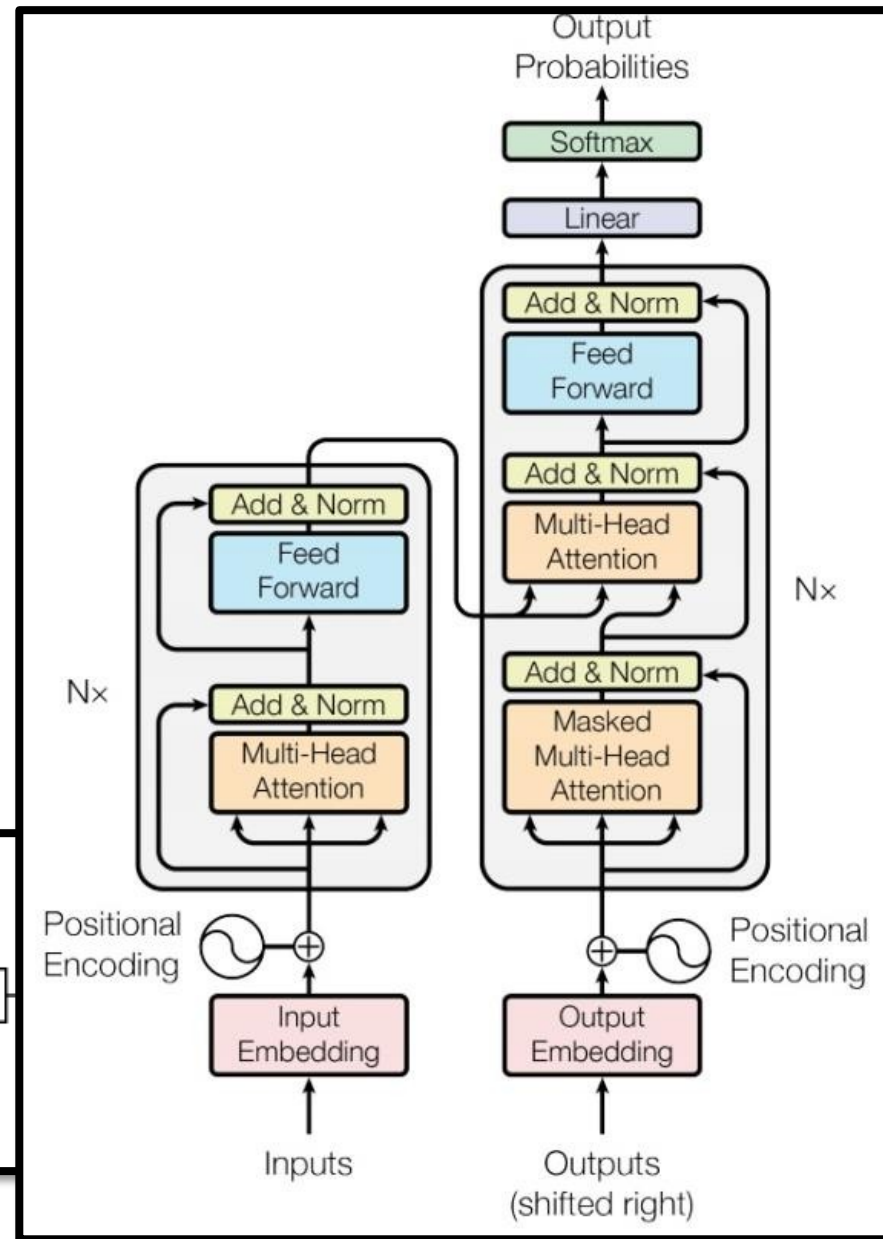


Seq2seq



Sequence to Sequence Learning with
Neural Networks

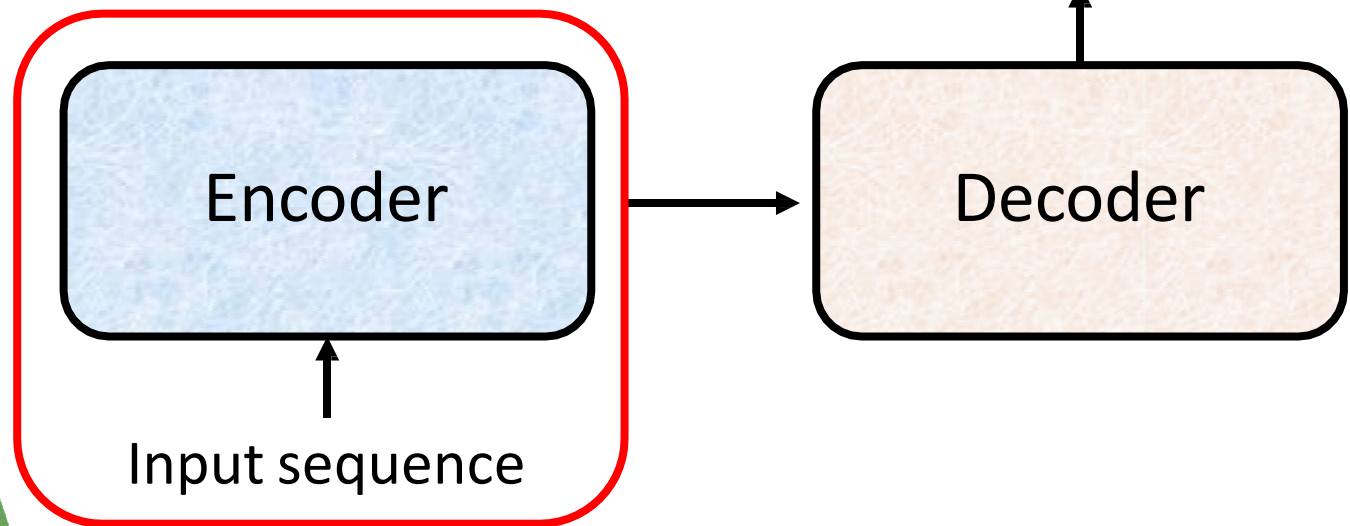
<https://arxiv.org/abs/1409.3215>



Transformer

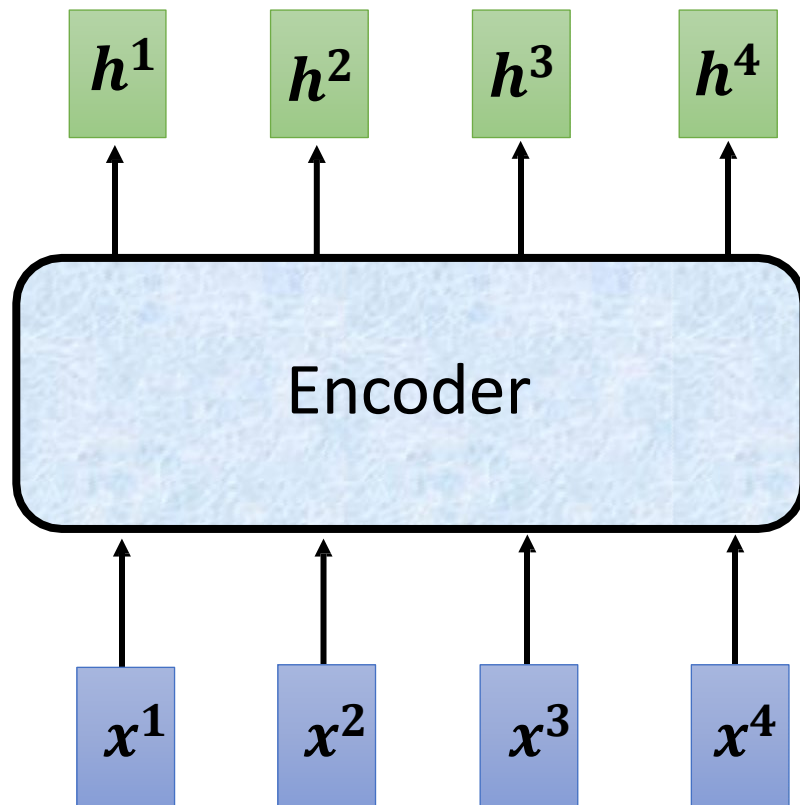
<https://arxiv.org/abs/1706.03761>₂₅

Encoder

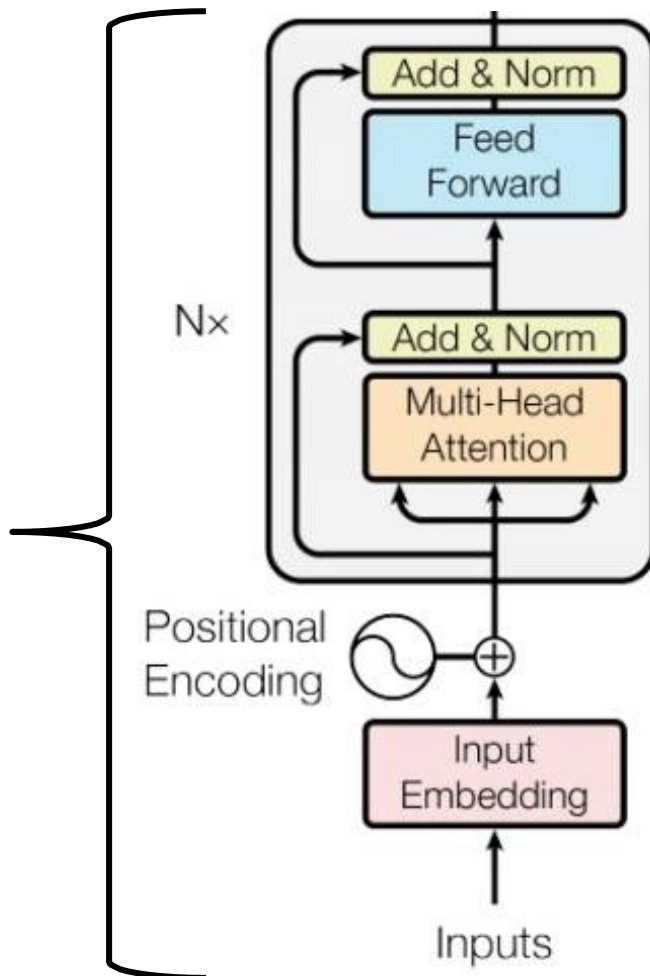


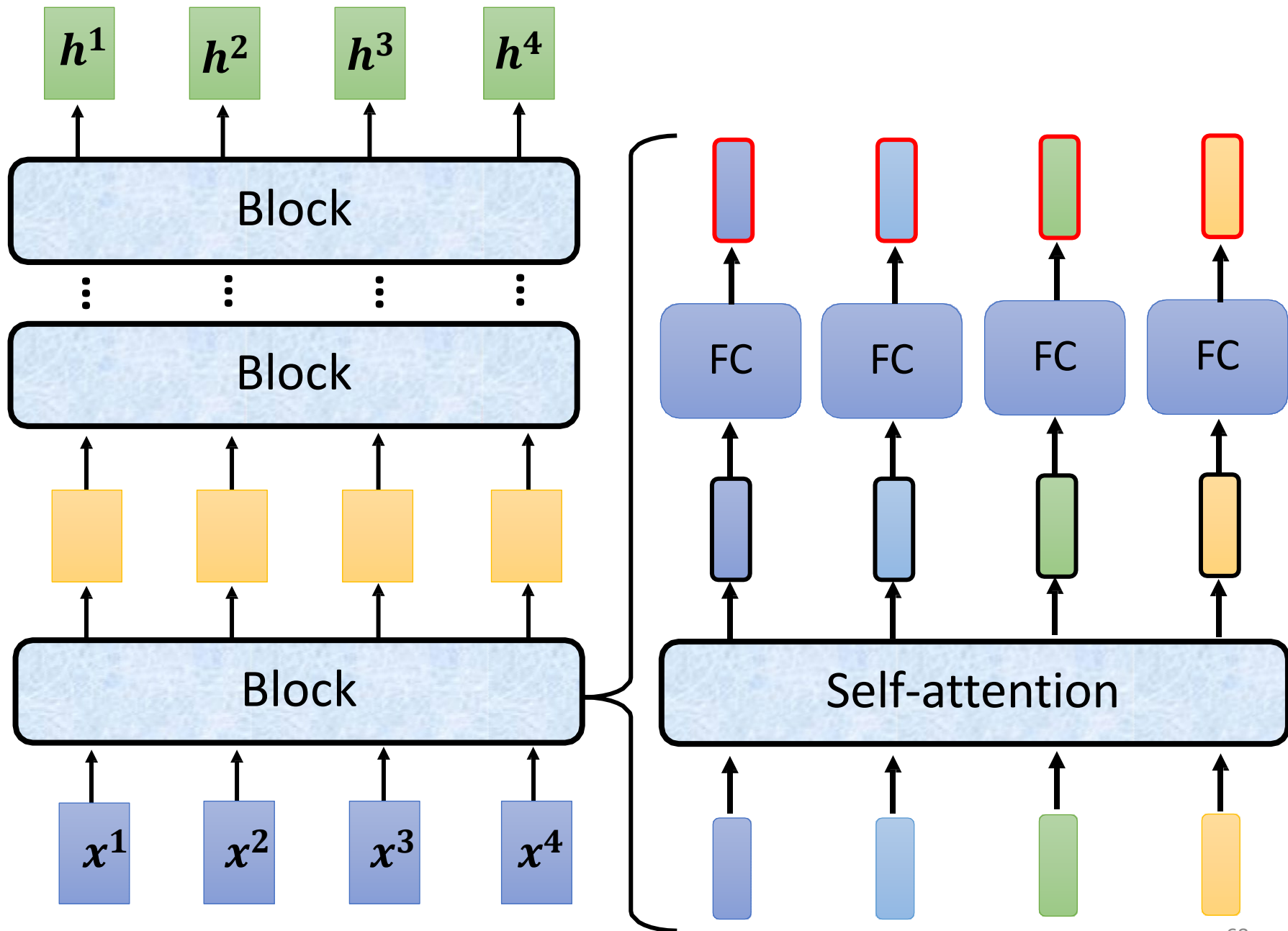
Encoder

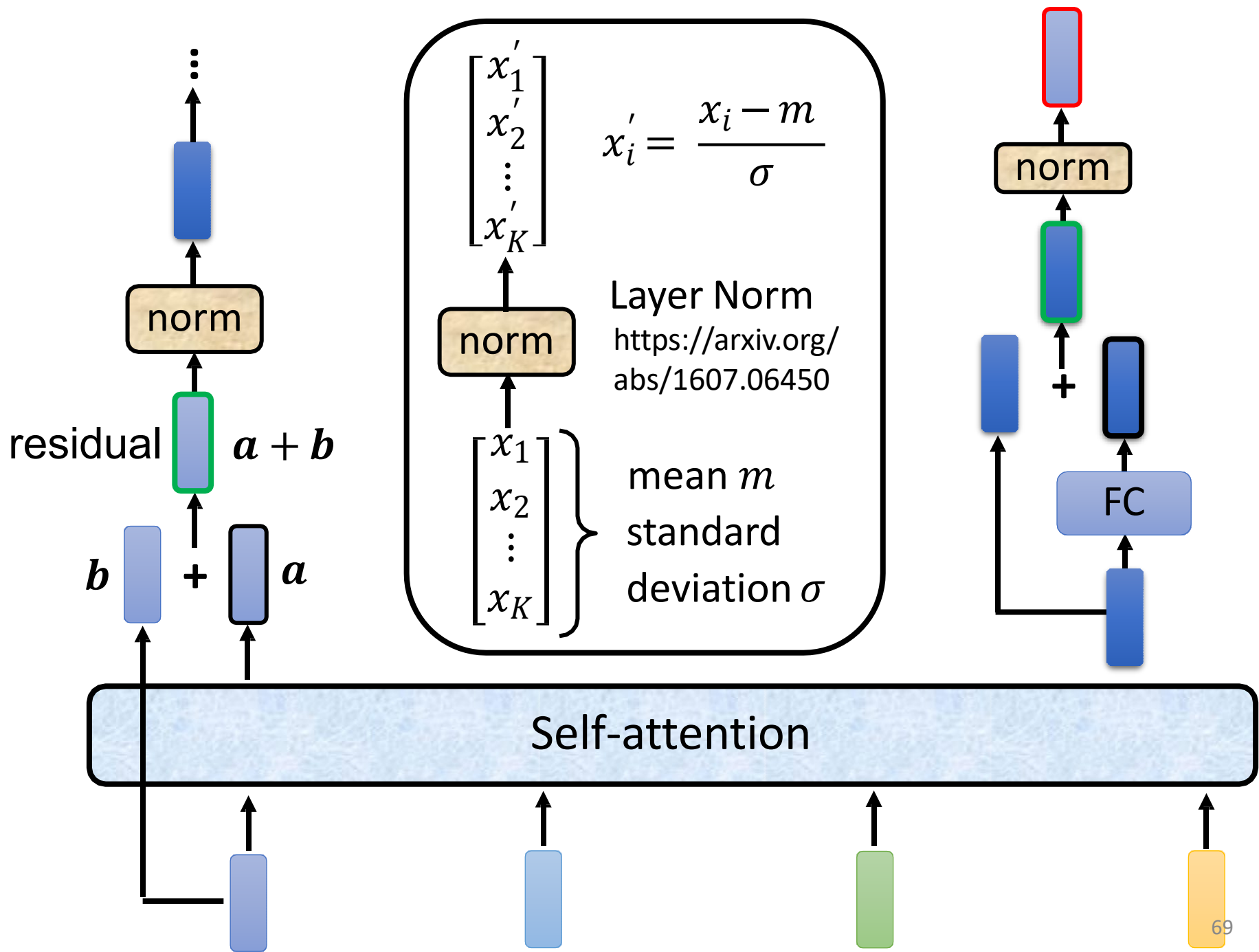
You can use **RNN** or **CNN**.



Transformer's Encoder



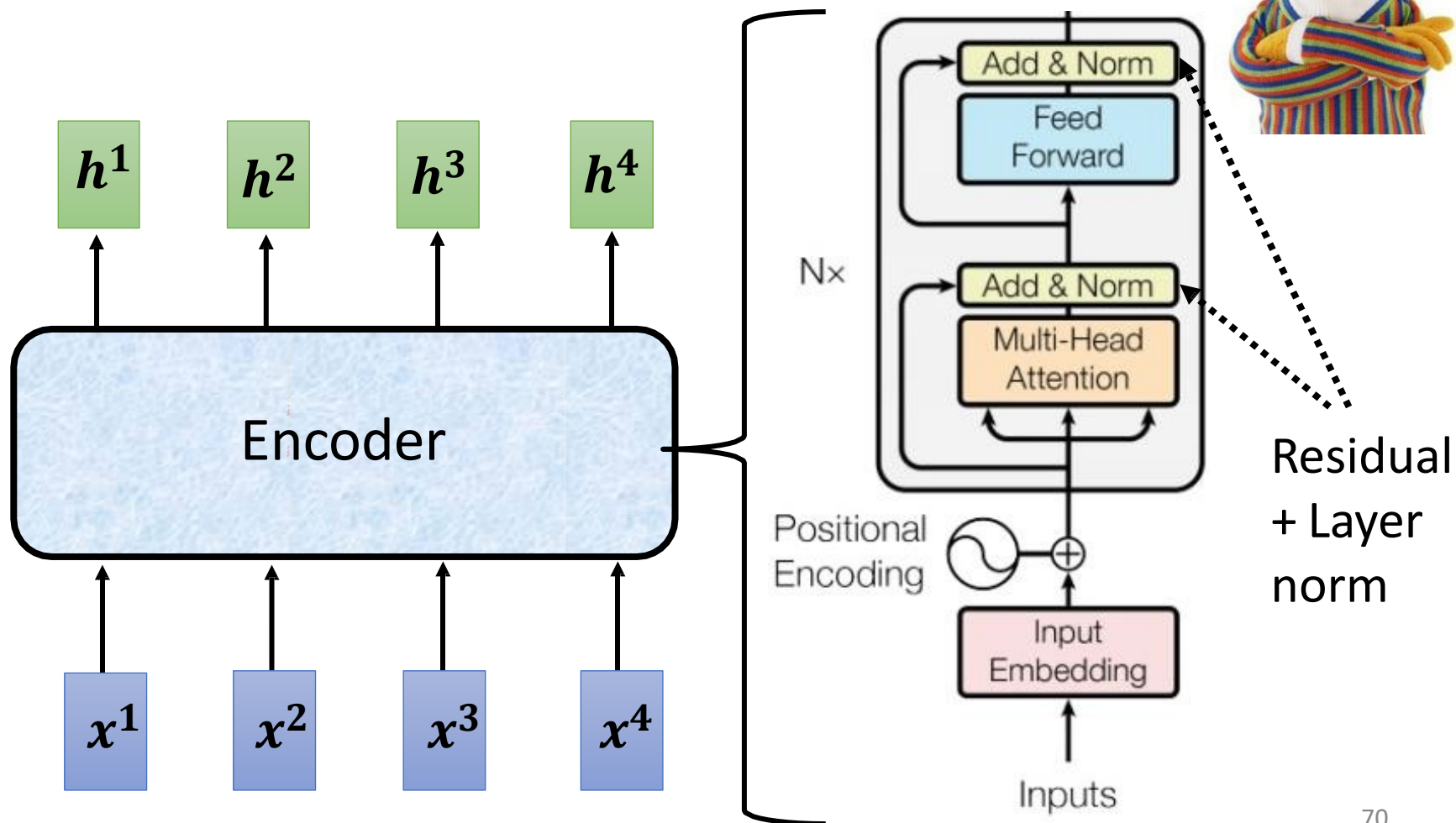




木偶动画《芝麻街》

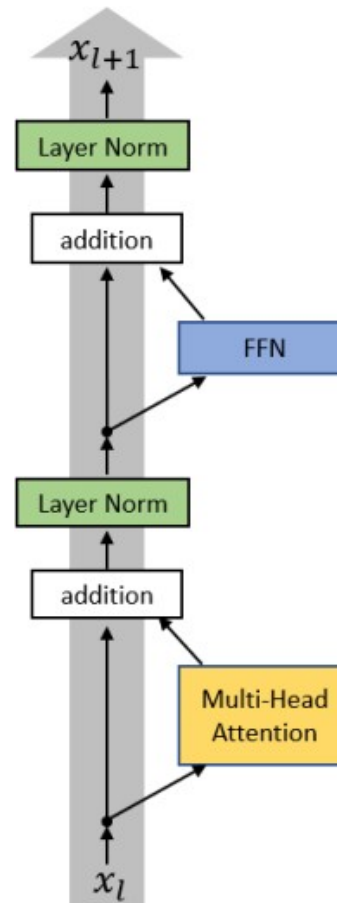
I use the **same** network architecture as transformer encoder.

BERT

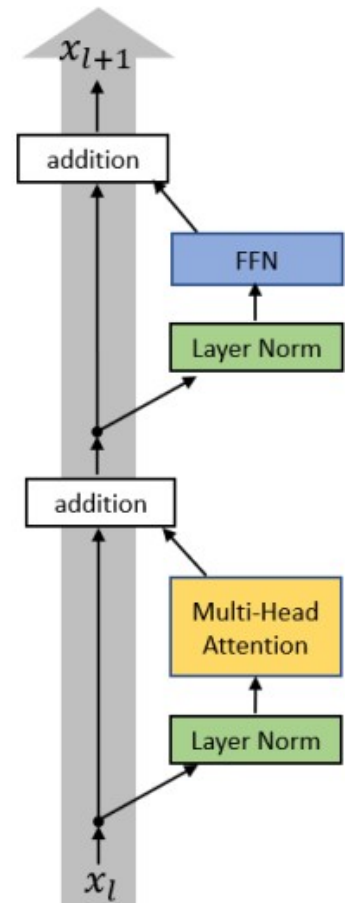


To learn more

- On Layer Normalization in the Transformer Architecture
- <https://arxiv.org/abs/2002.04745>
- PowerNorm: Rethinking Batch Normalization in Transformers
- <https://arxiv.org/abs/2003.07845>

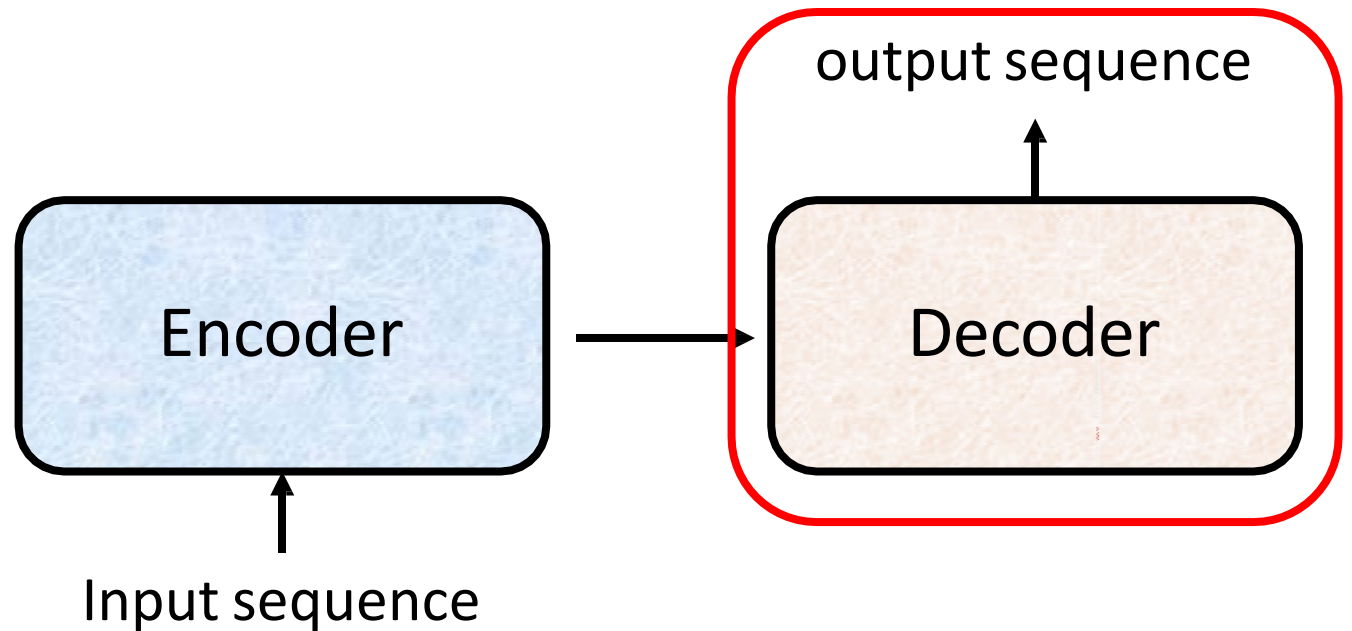


(a)



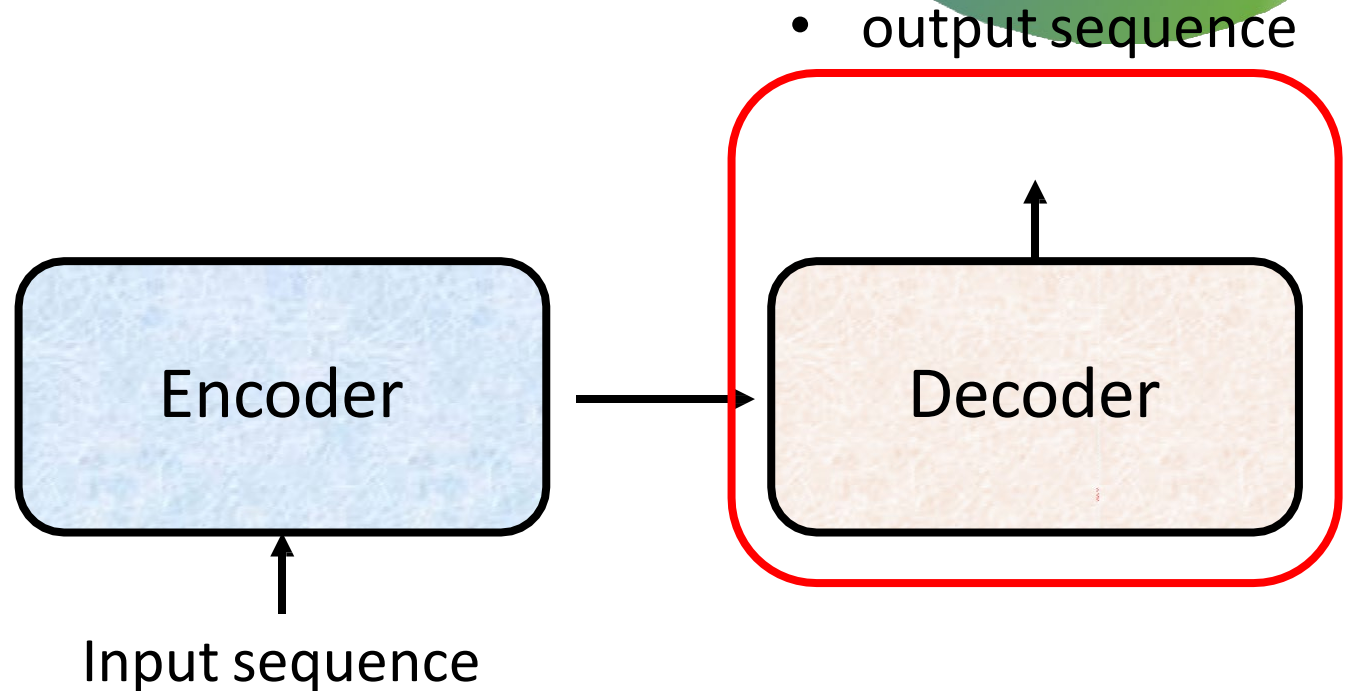
(b)

Decoder



Decoder

- – Autoregressive (AT)



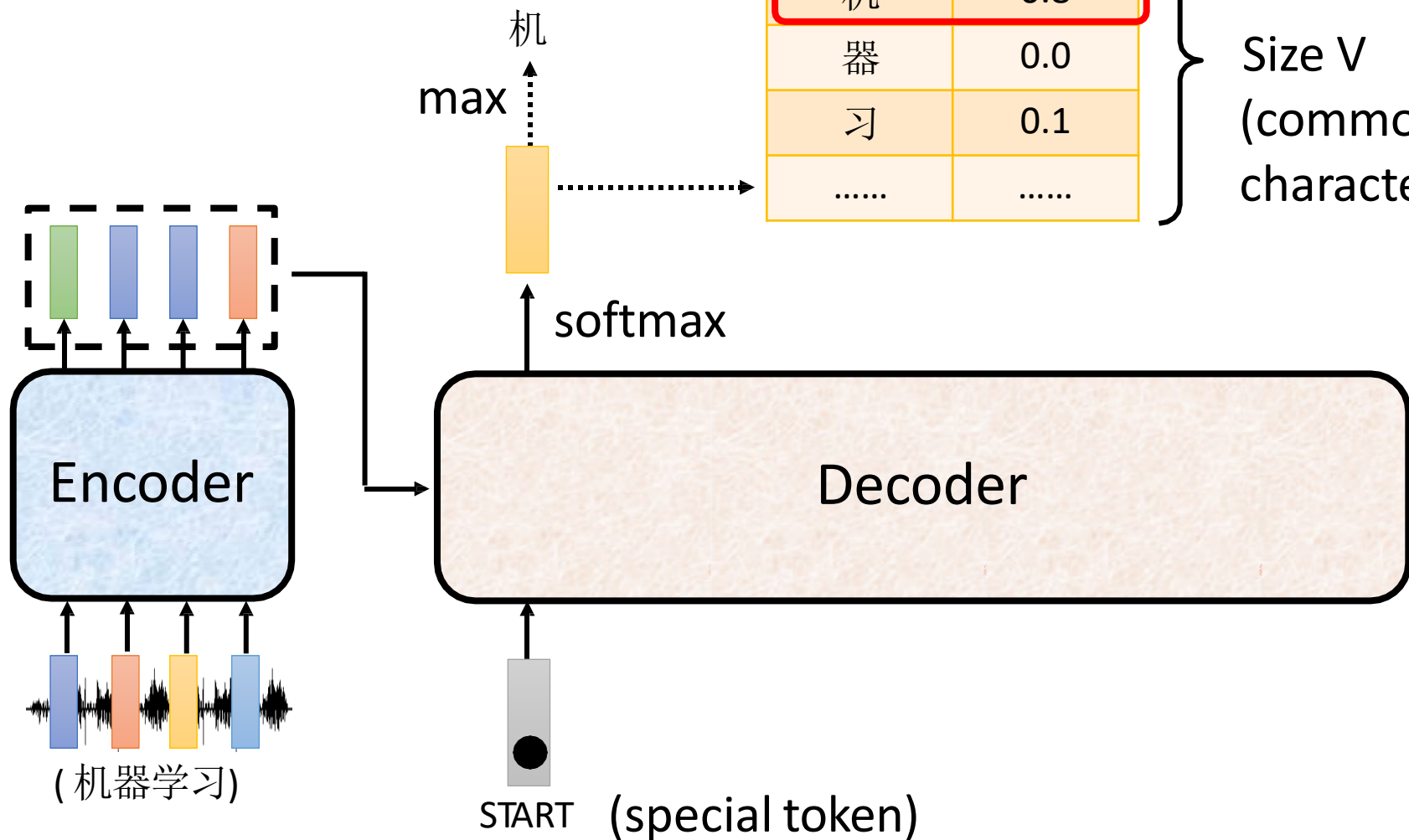
Autoregressive

(Speech Recognition as example)

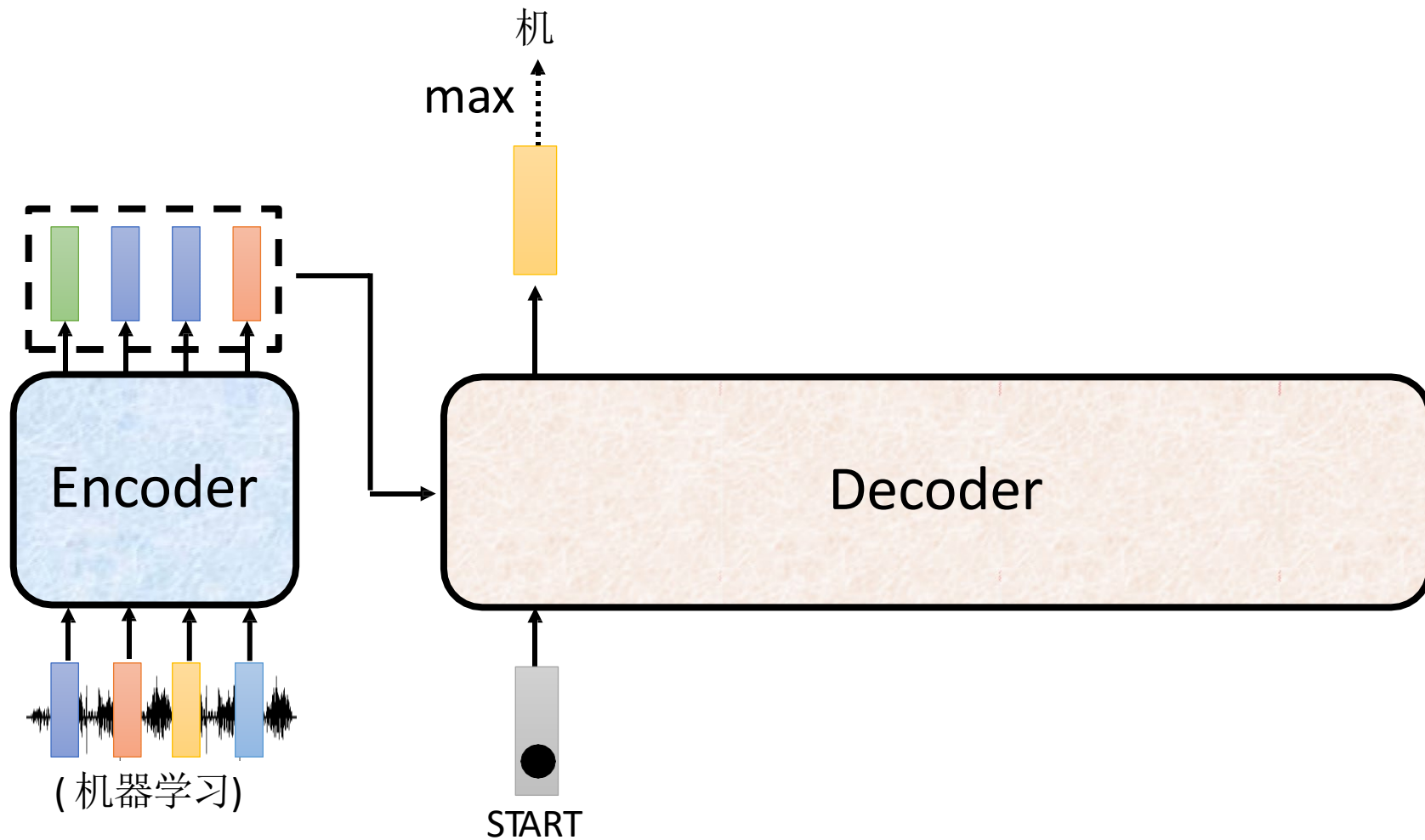
distribution

学	0.0
机	0.8
器	0.0
习	0.1
.....	

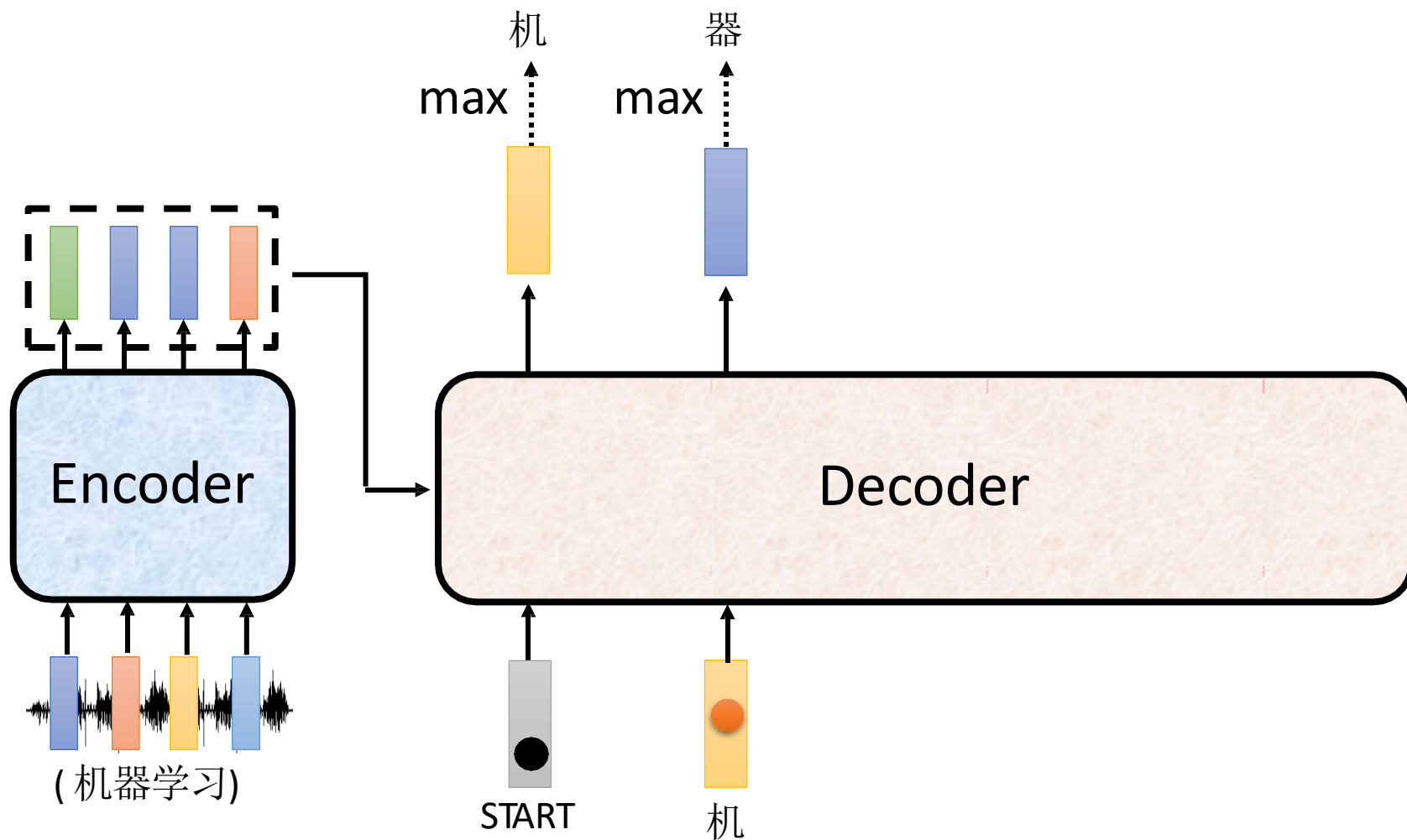
Size V
(common
characters)



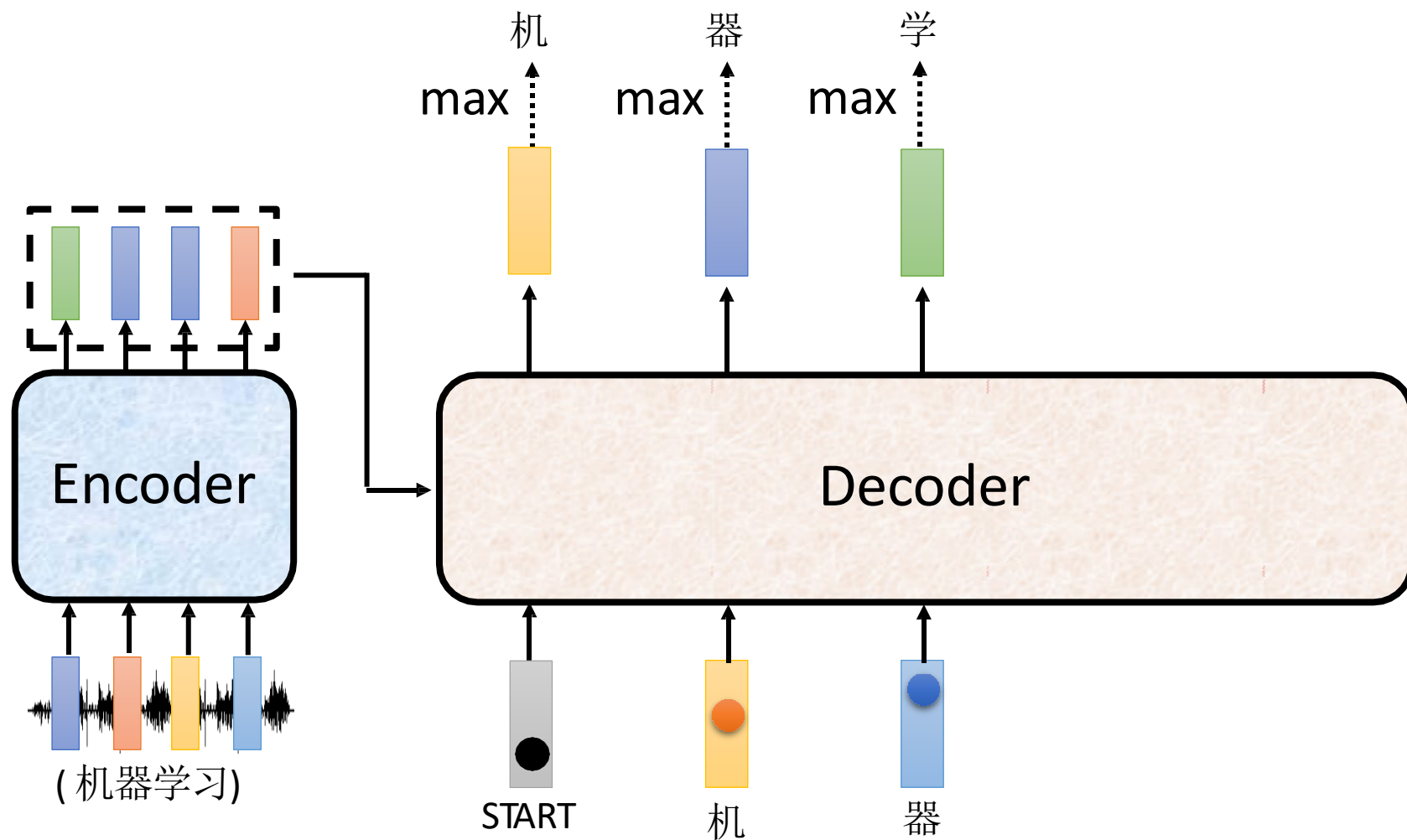
Autoregressive



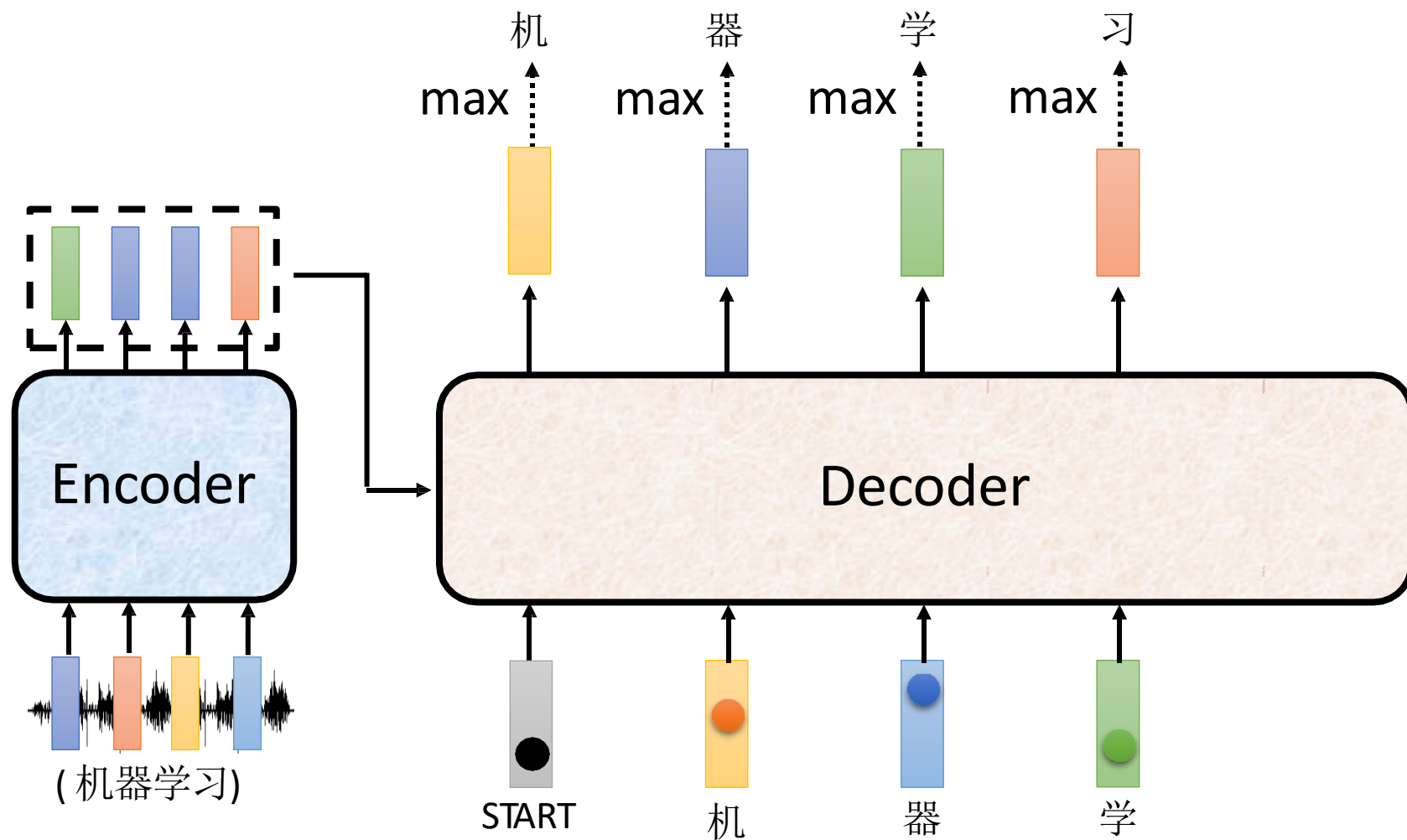
Autoregressive



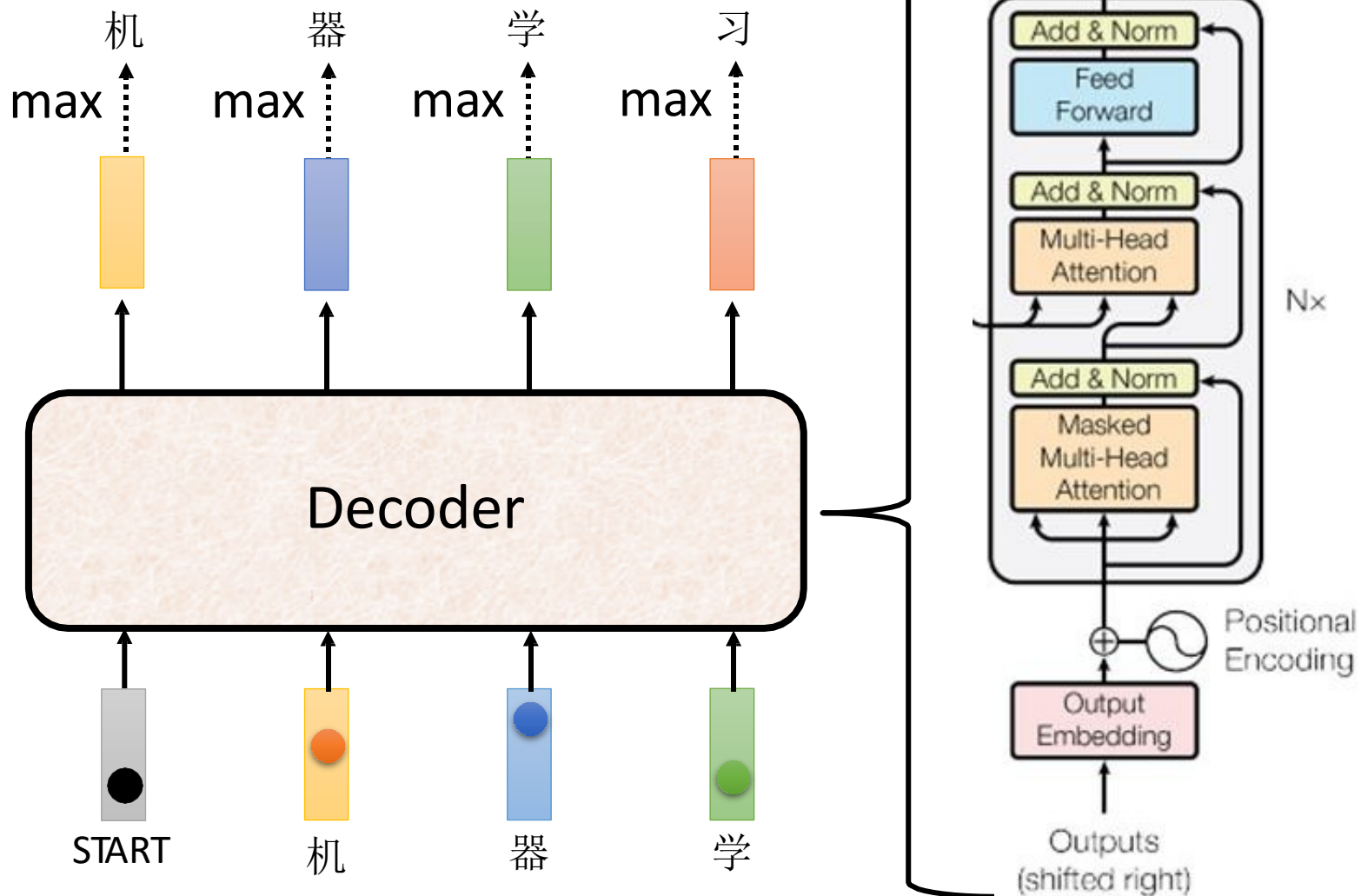
Autoregressive



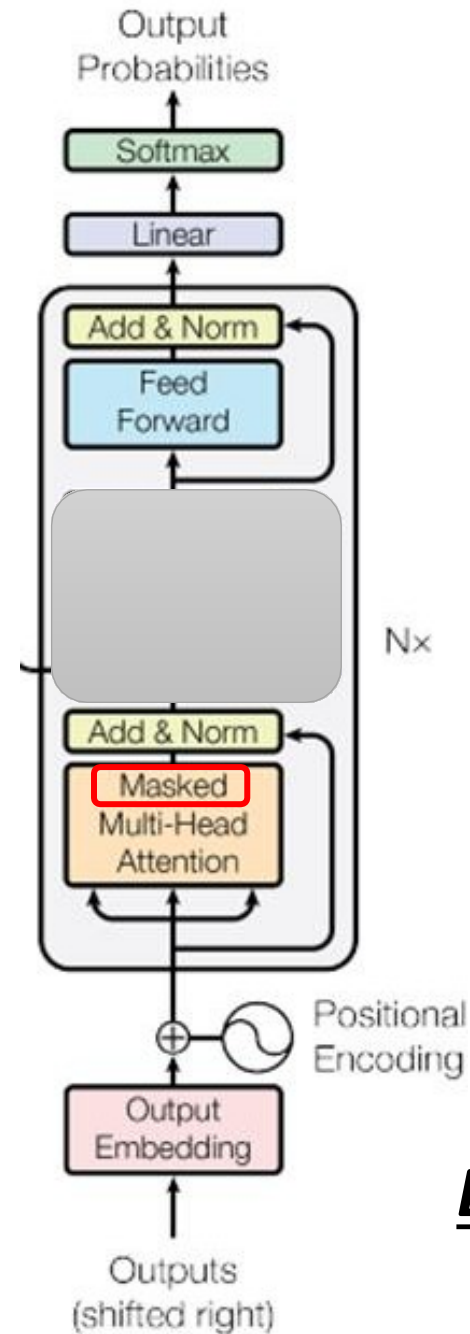
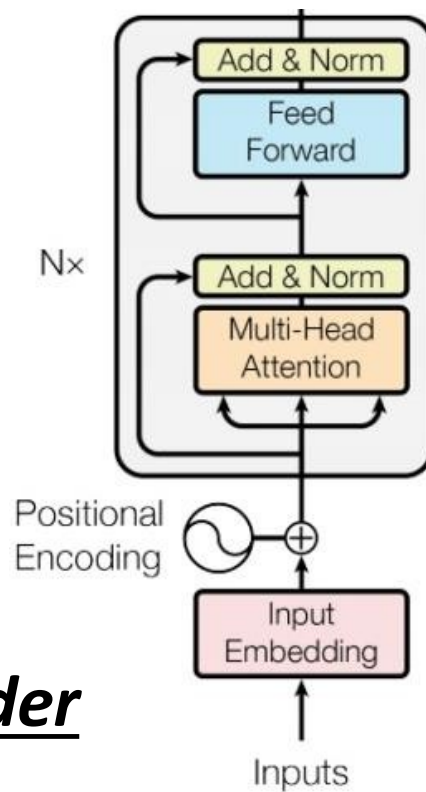
Autoregressive



ignore the input from the encoder here 😊

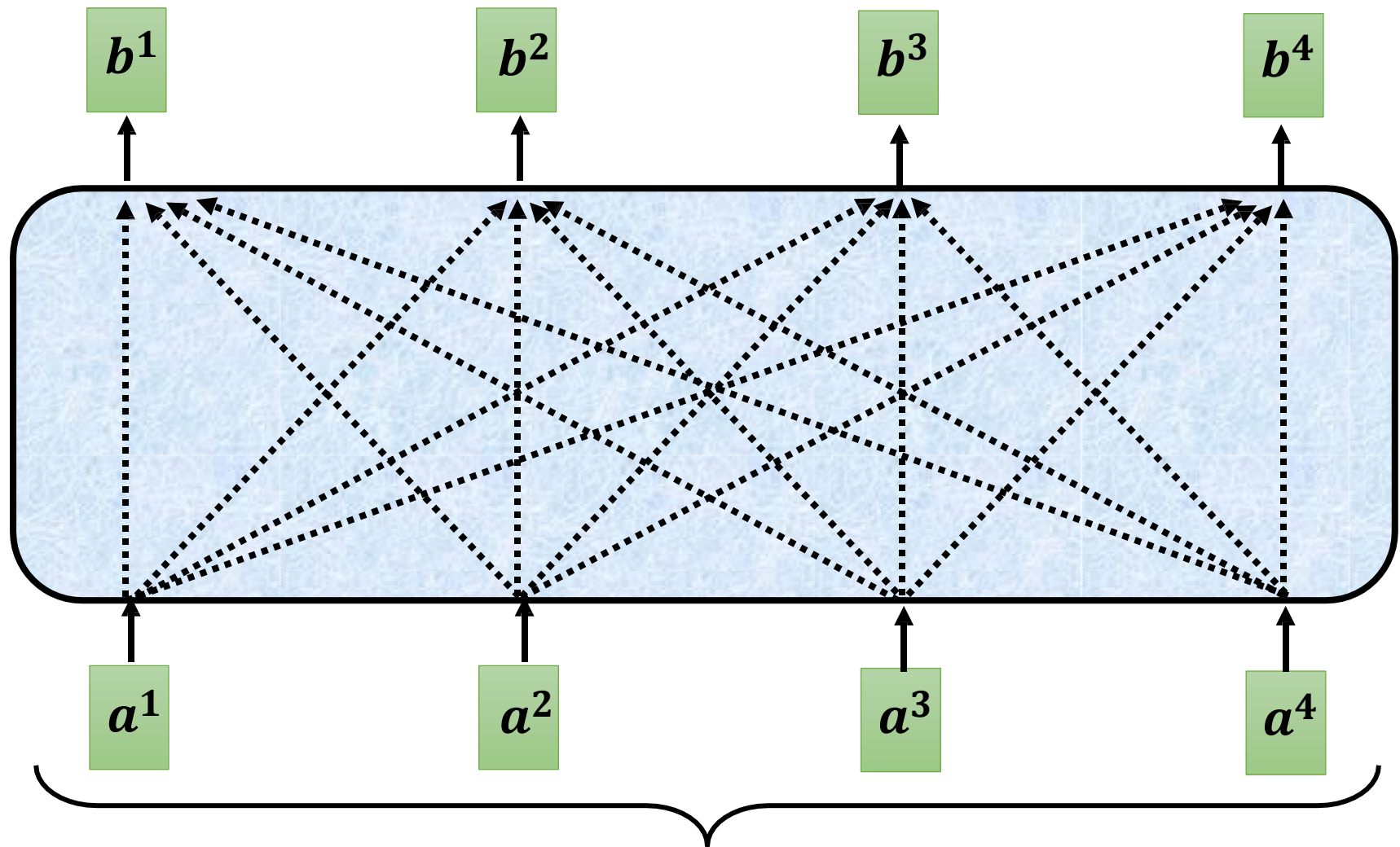


Encoder



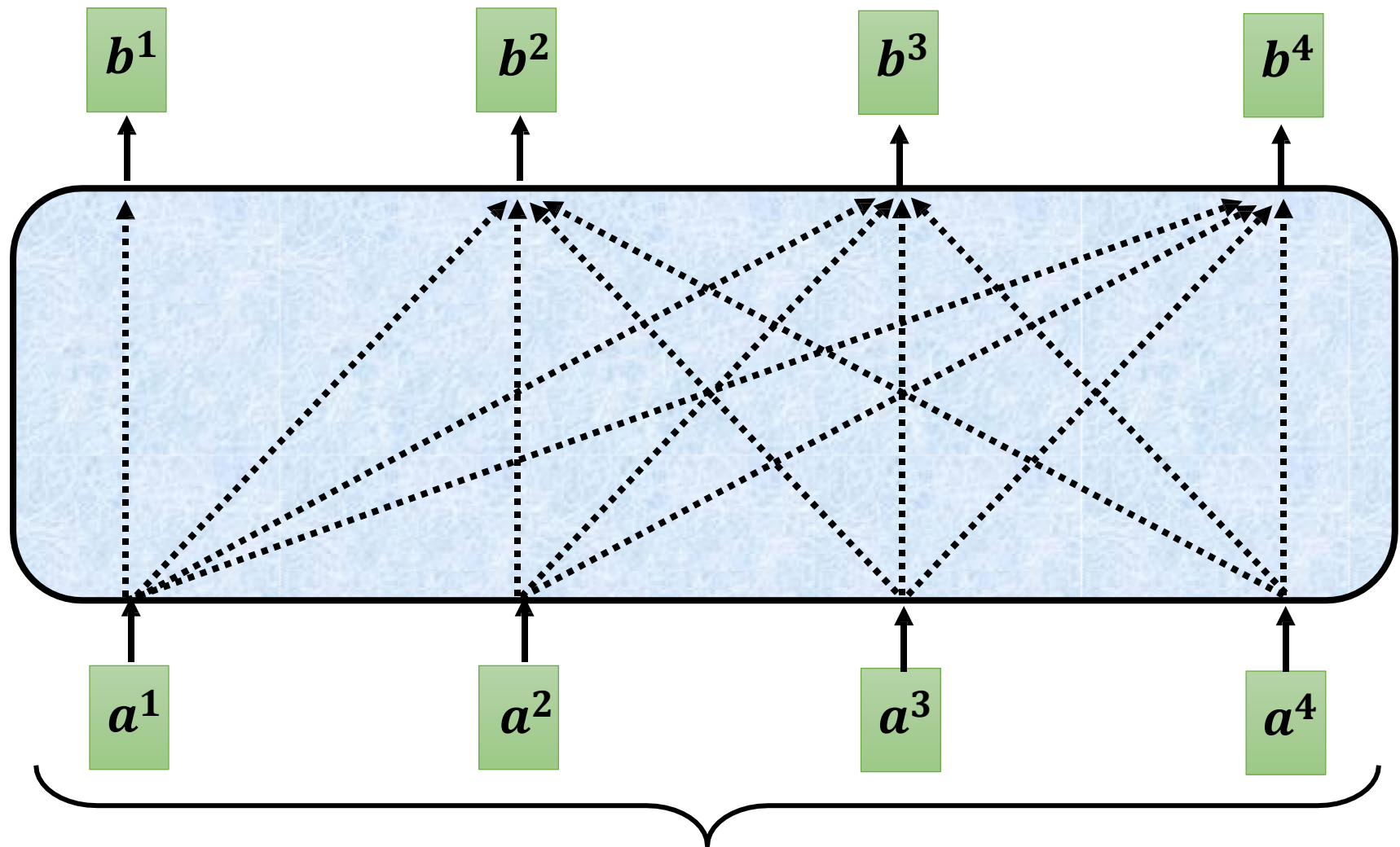
Decoder

Self-attention → Masked Self-attention (掩码的自注意力机制)



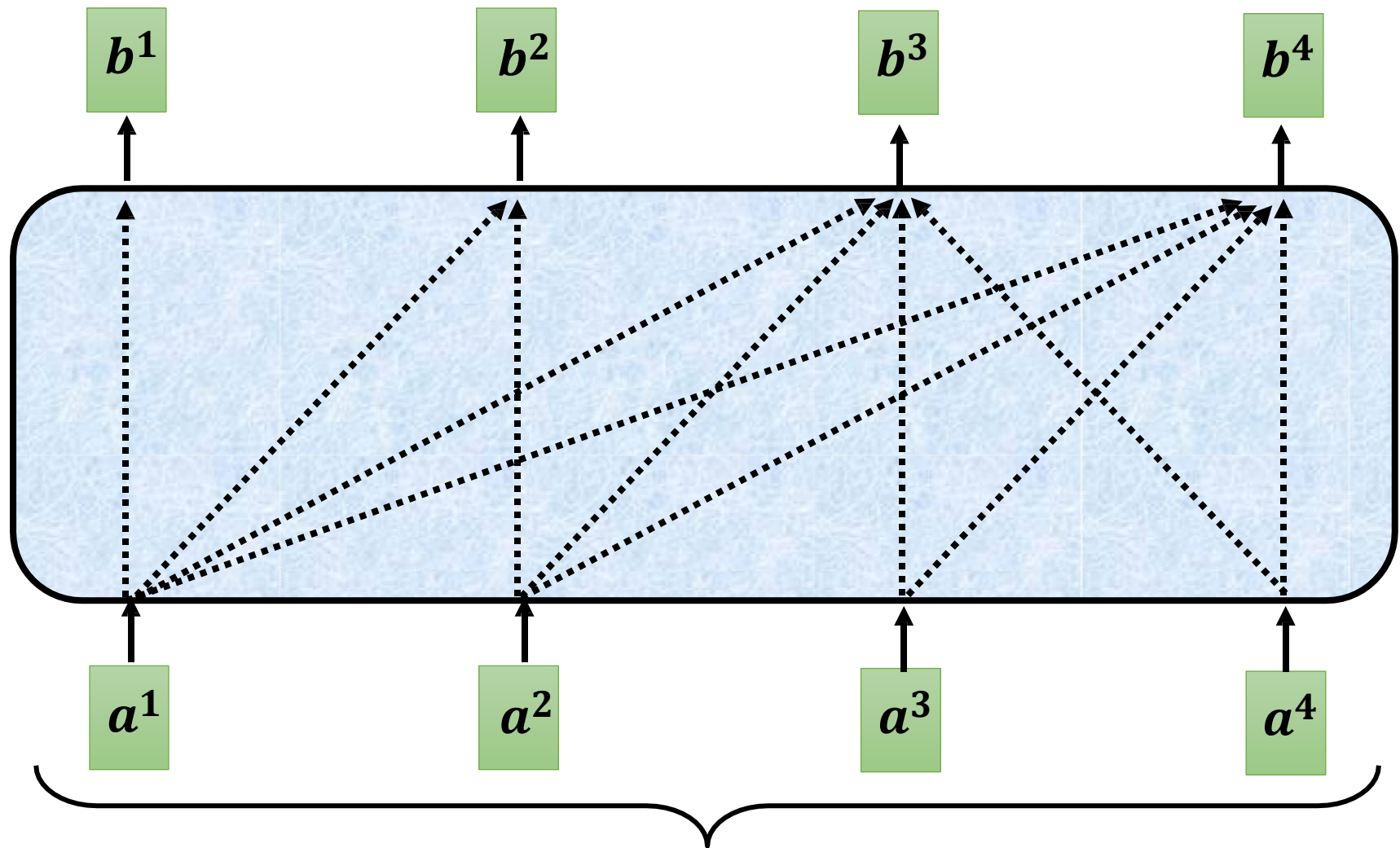
输入层或隐含层

Self-attention → Masked Self-attention



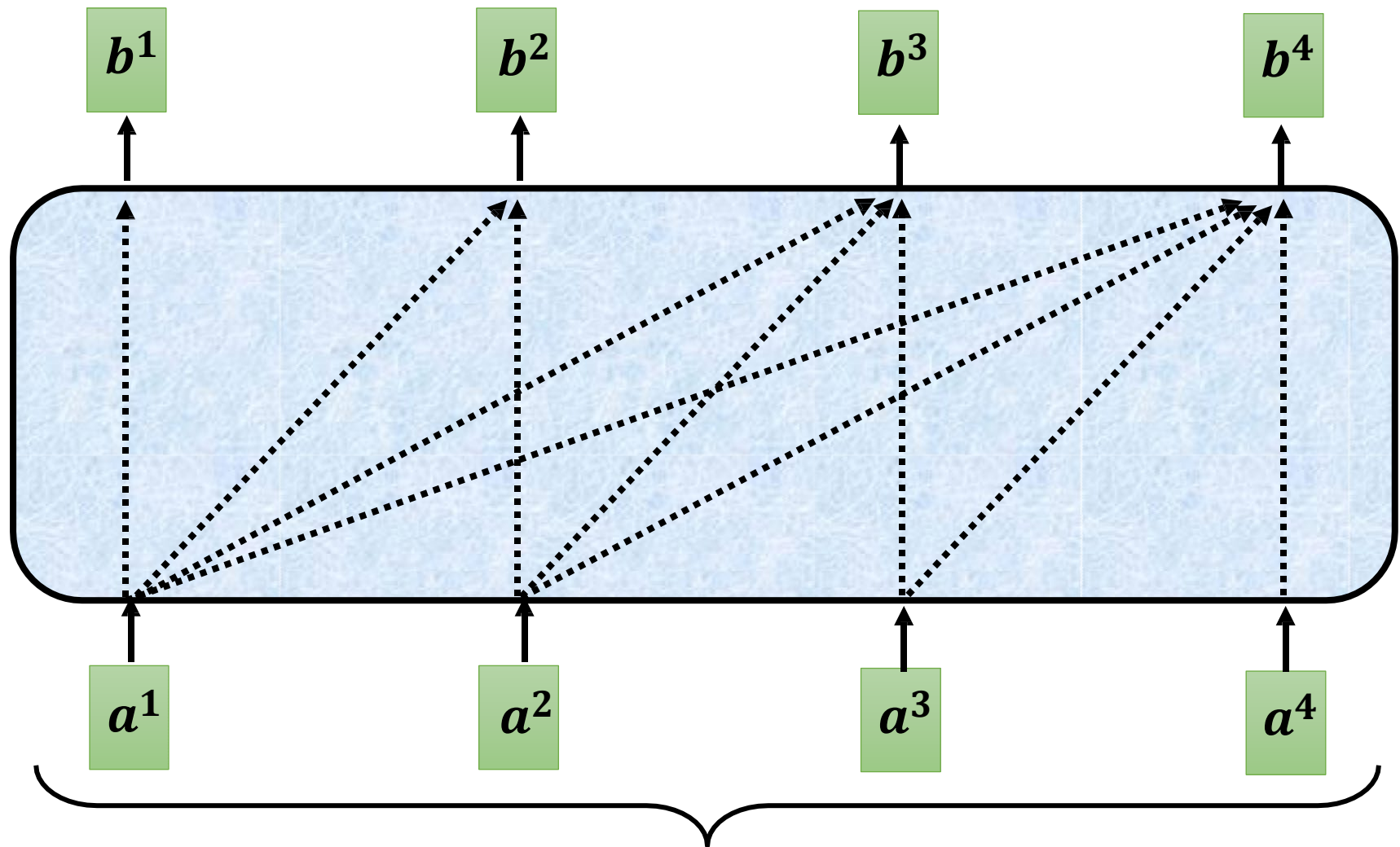
Can be either **input** or a **hidden layer**

Self-attention → Masked Self-attention



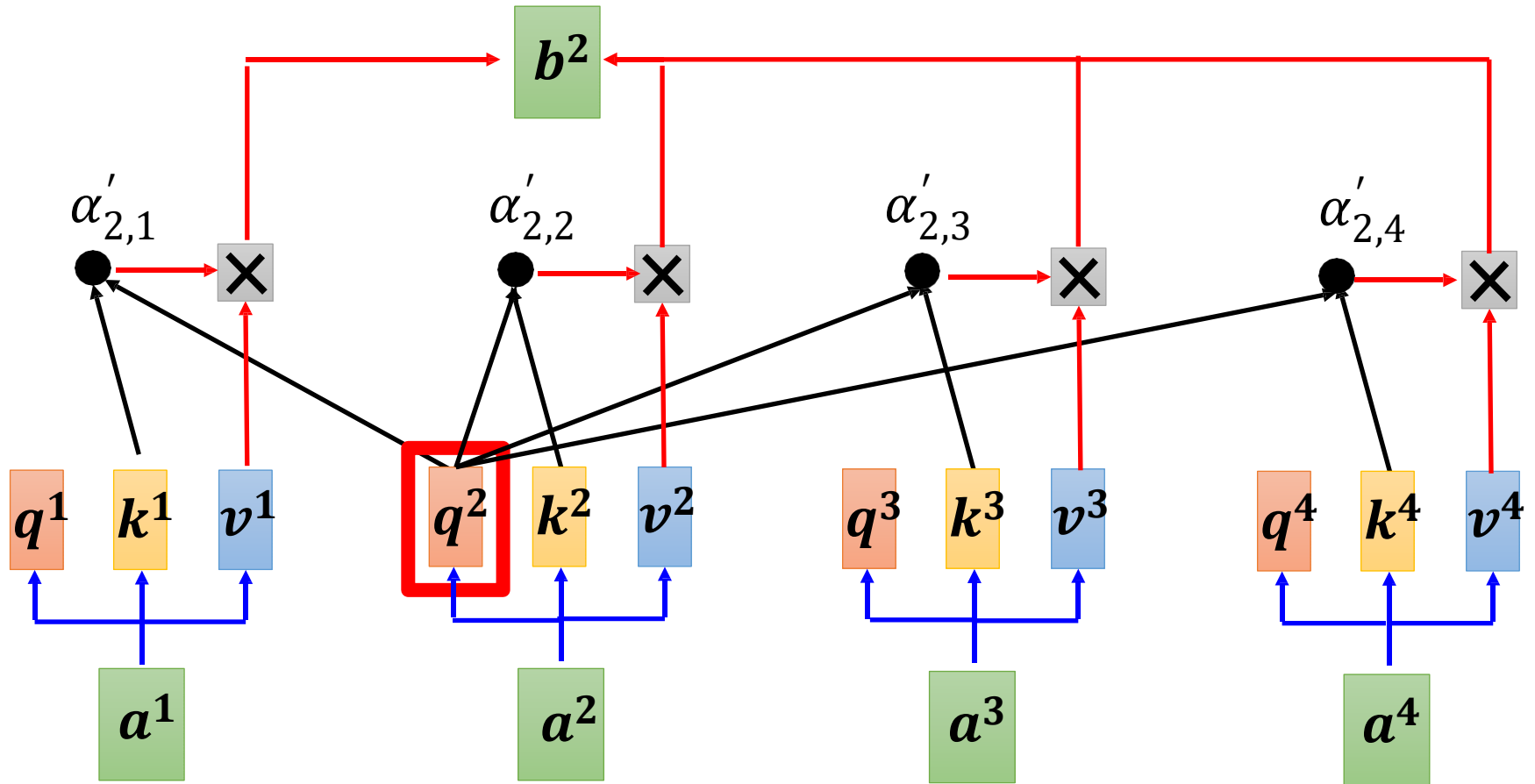
Can be either **input** or a **hidden layer**

Self-attention → Masked Self-attention



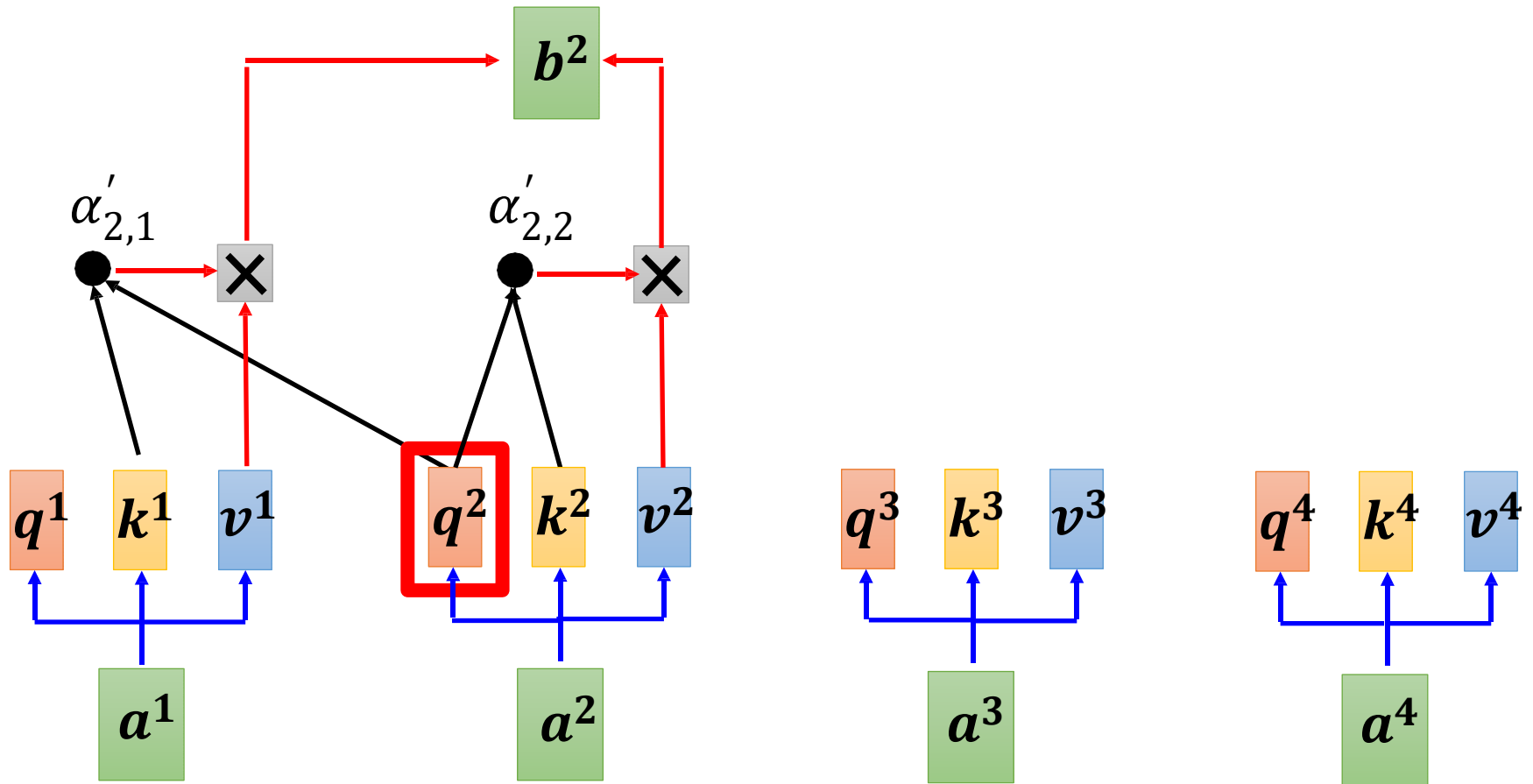
Can be either **input** or a **hidden layer**

Self-attention $\rightarrow$ Masked Self-attention



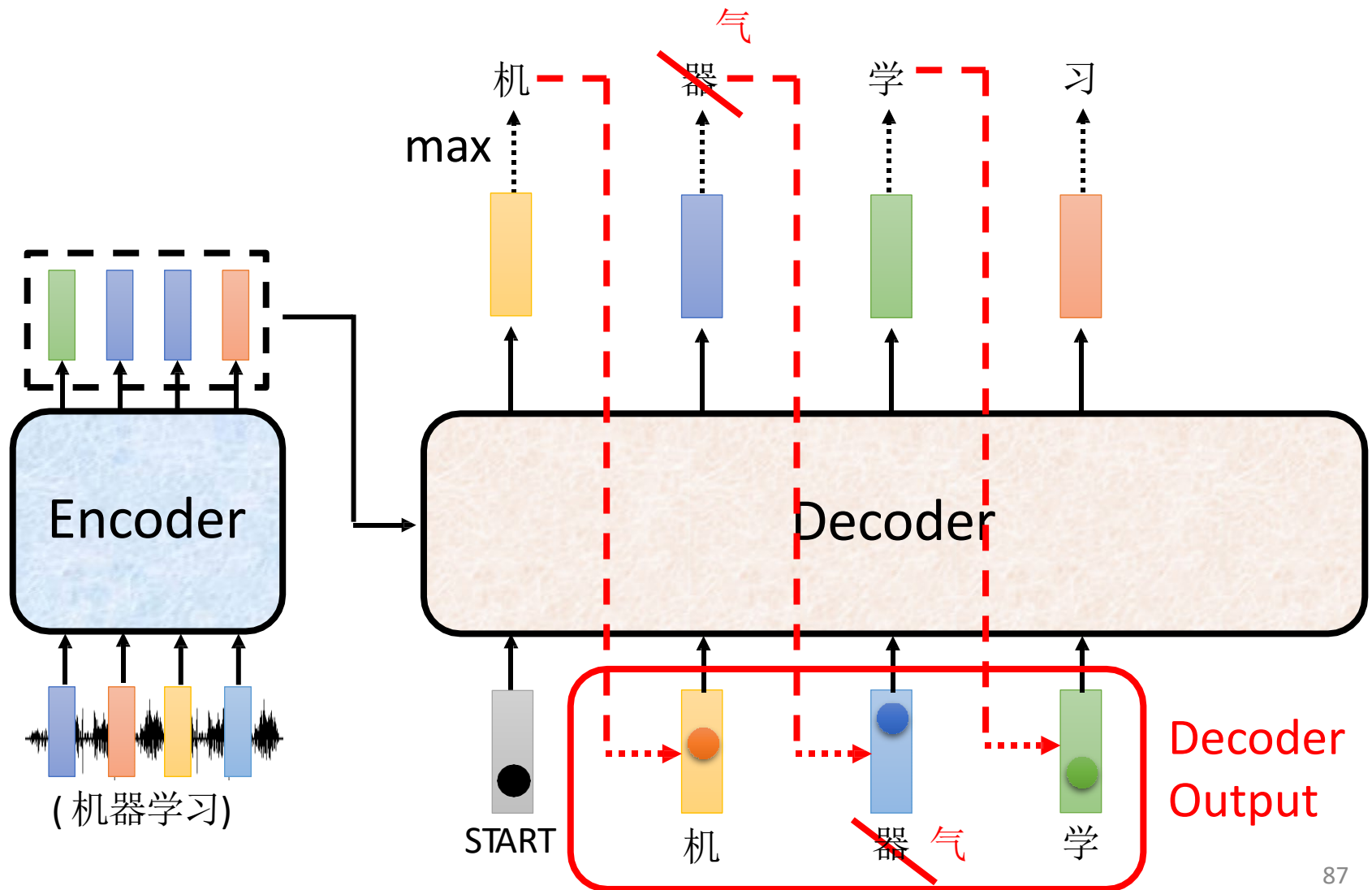
Why masked?

Self-attention $\rightarrow$ Masked Self-attention



Why masked?

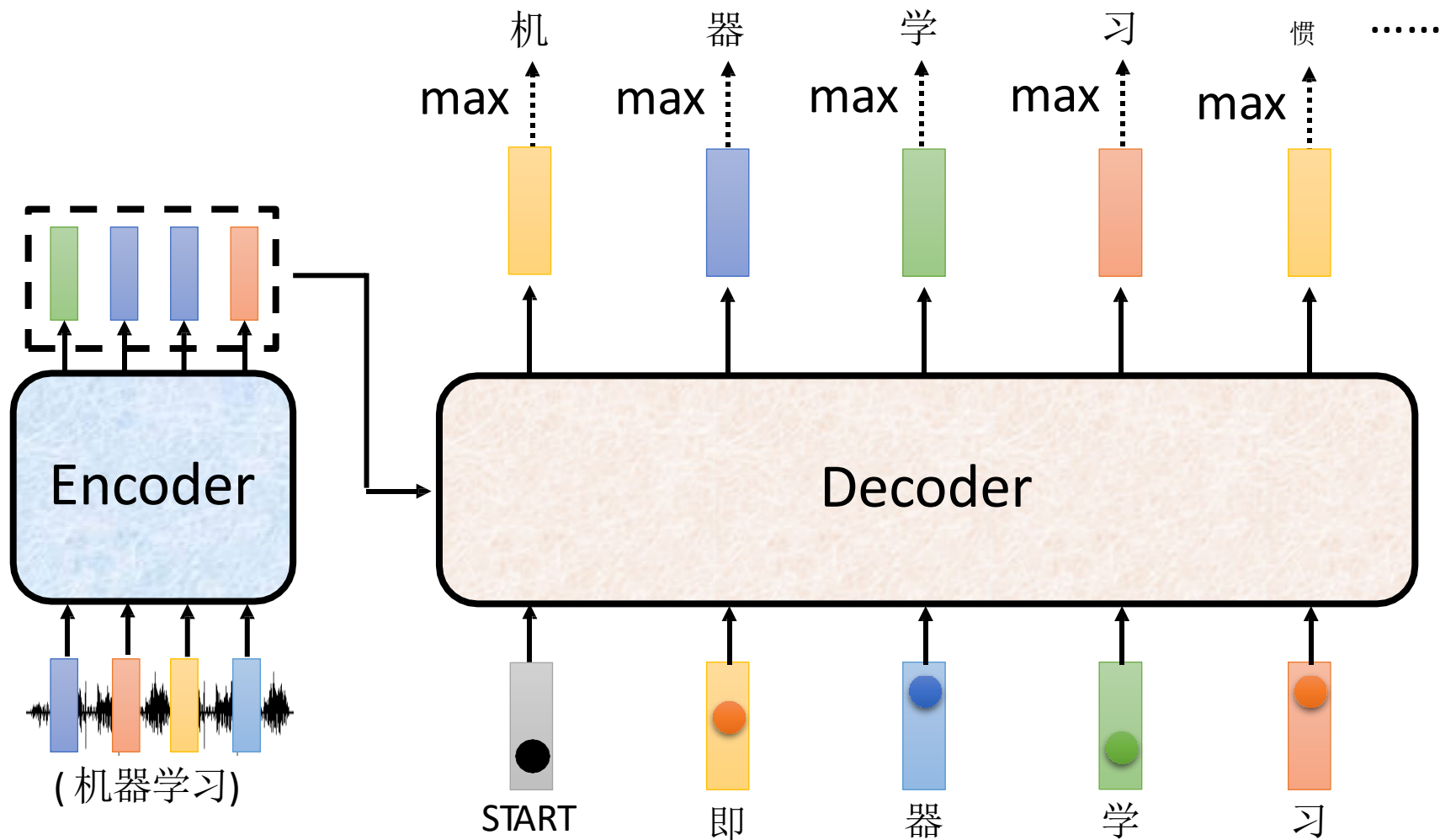
Autoregressive (自回归)



Autoregressive

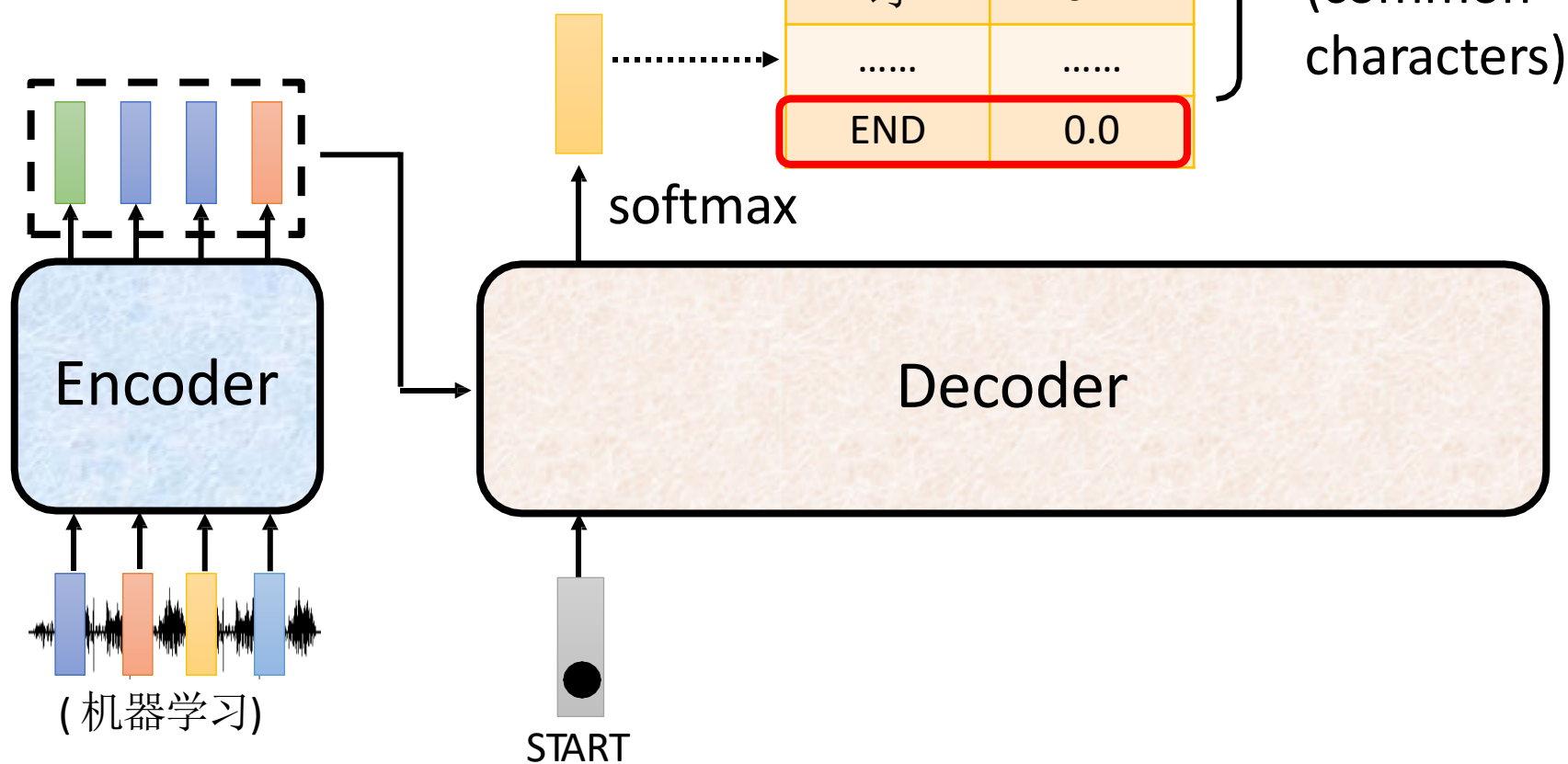
输出的长度是未知的。

Never stop!



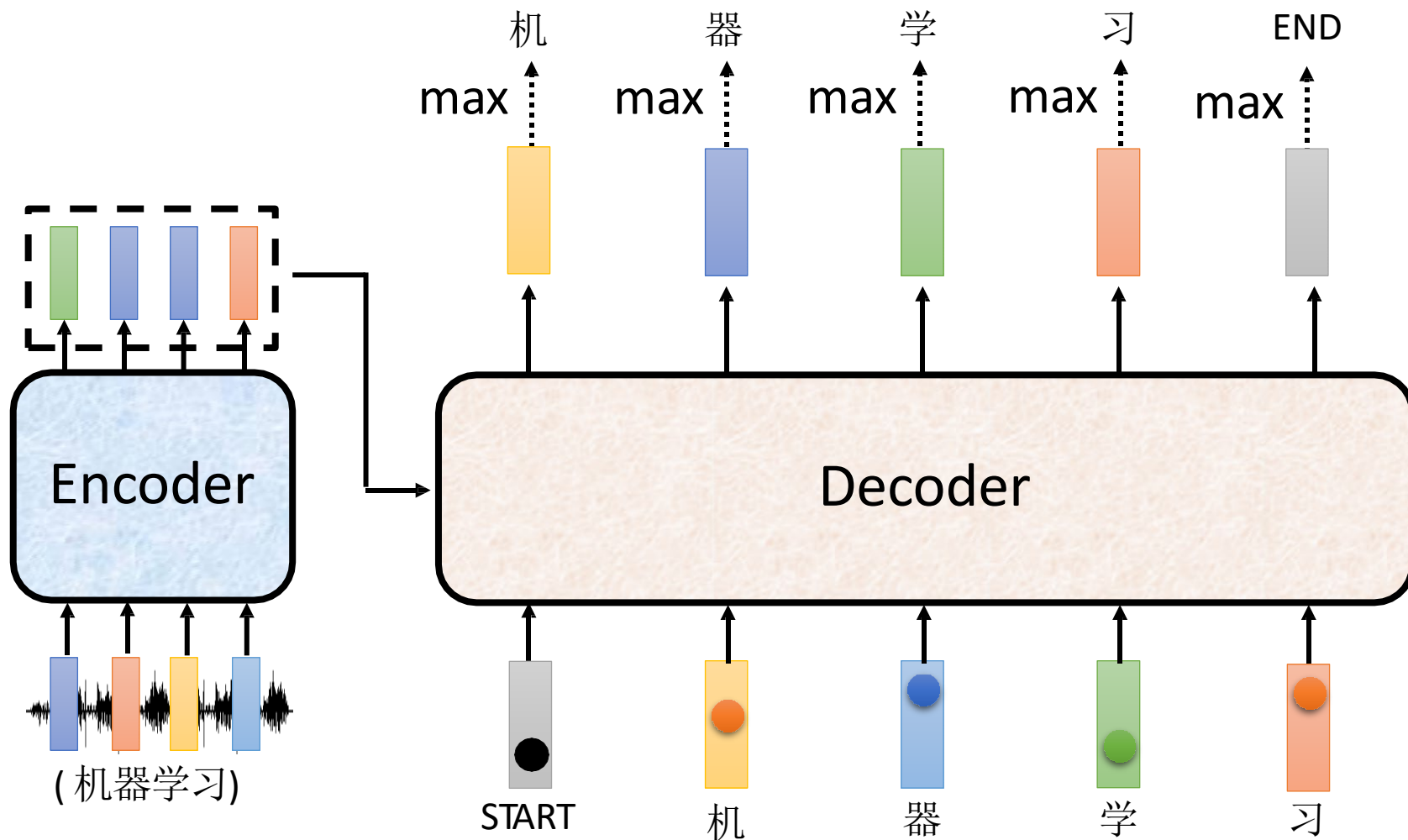
Adding "Stop Token"

停止符

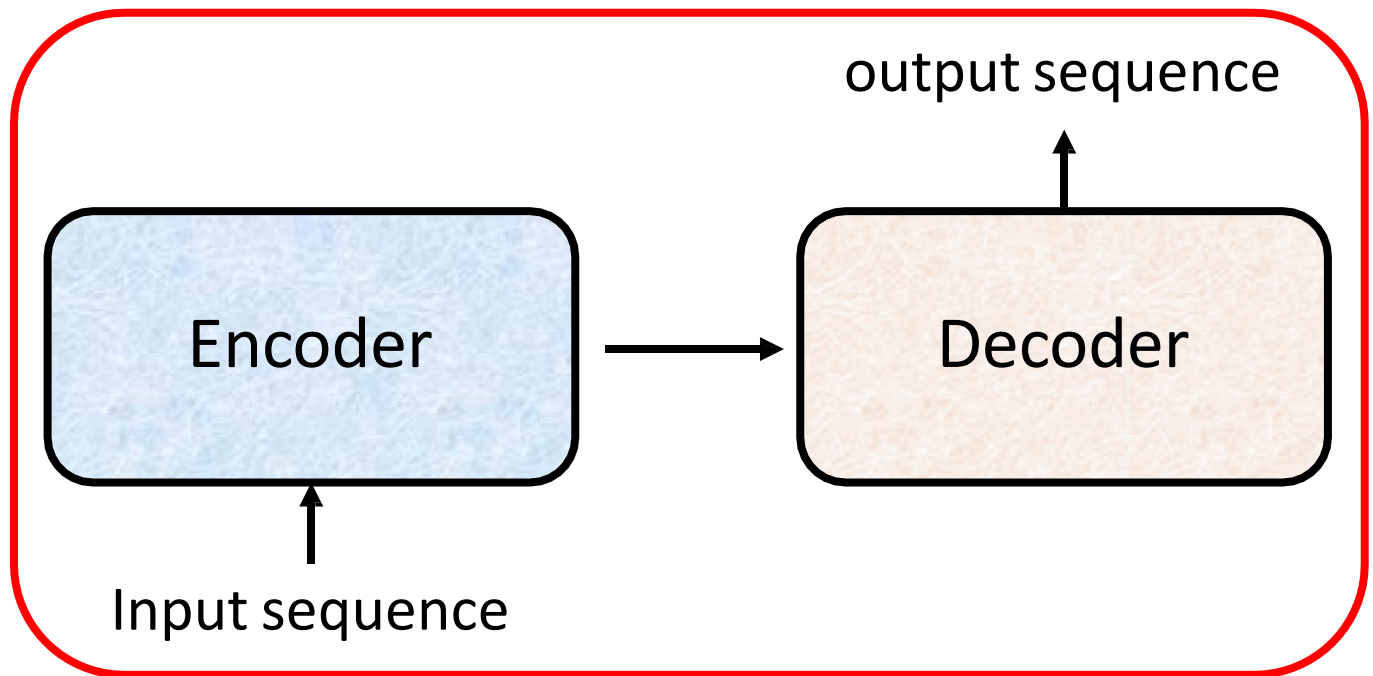


Autoregressive

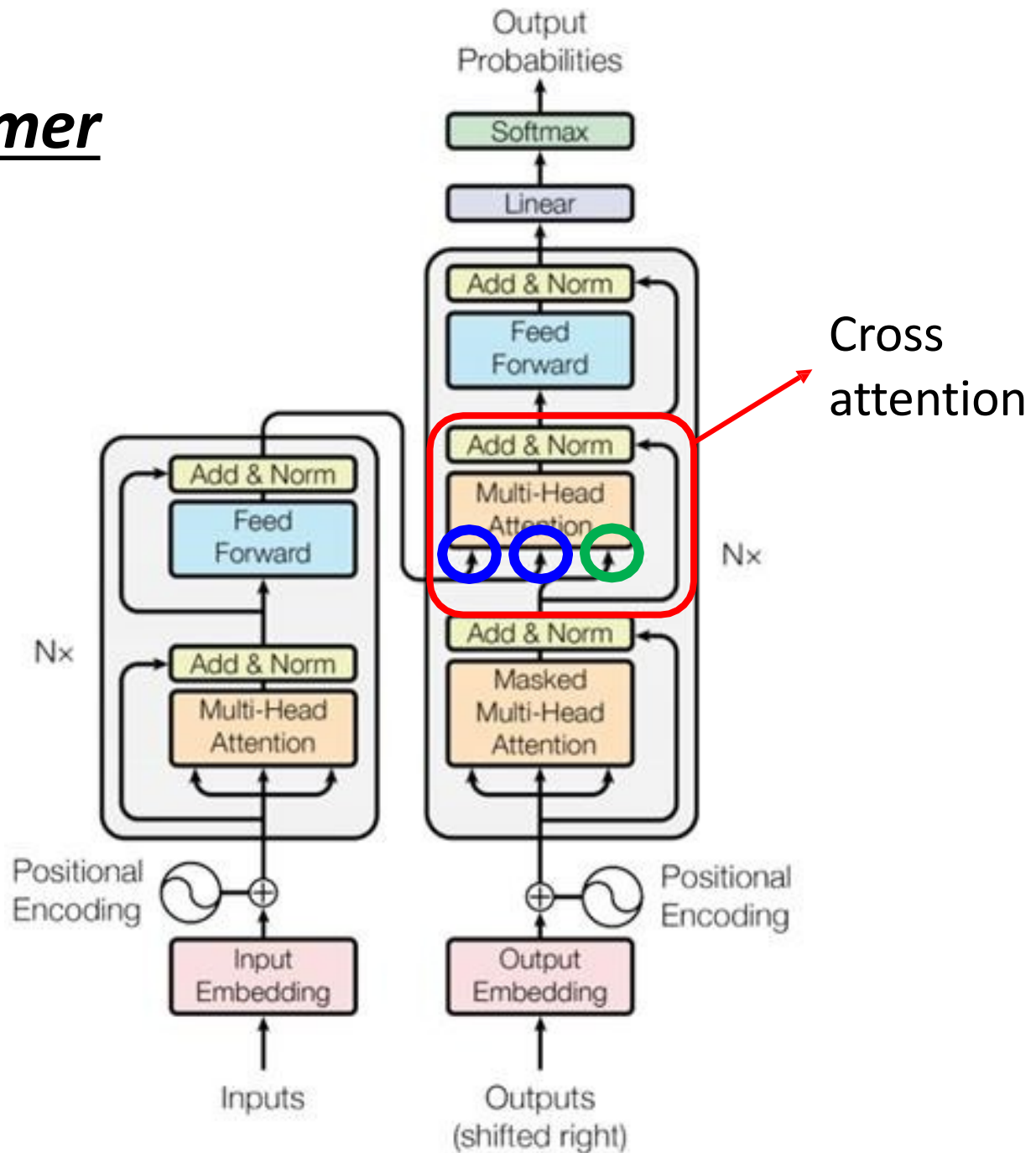
Stop at here!

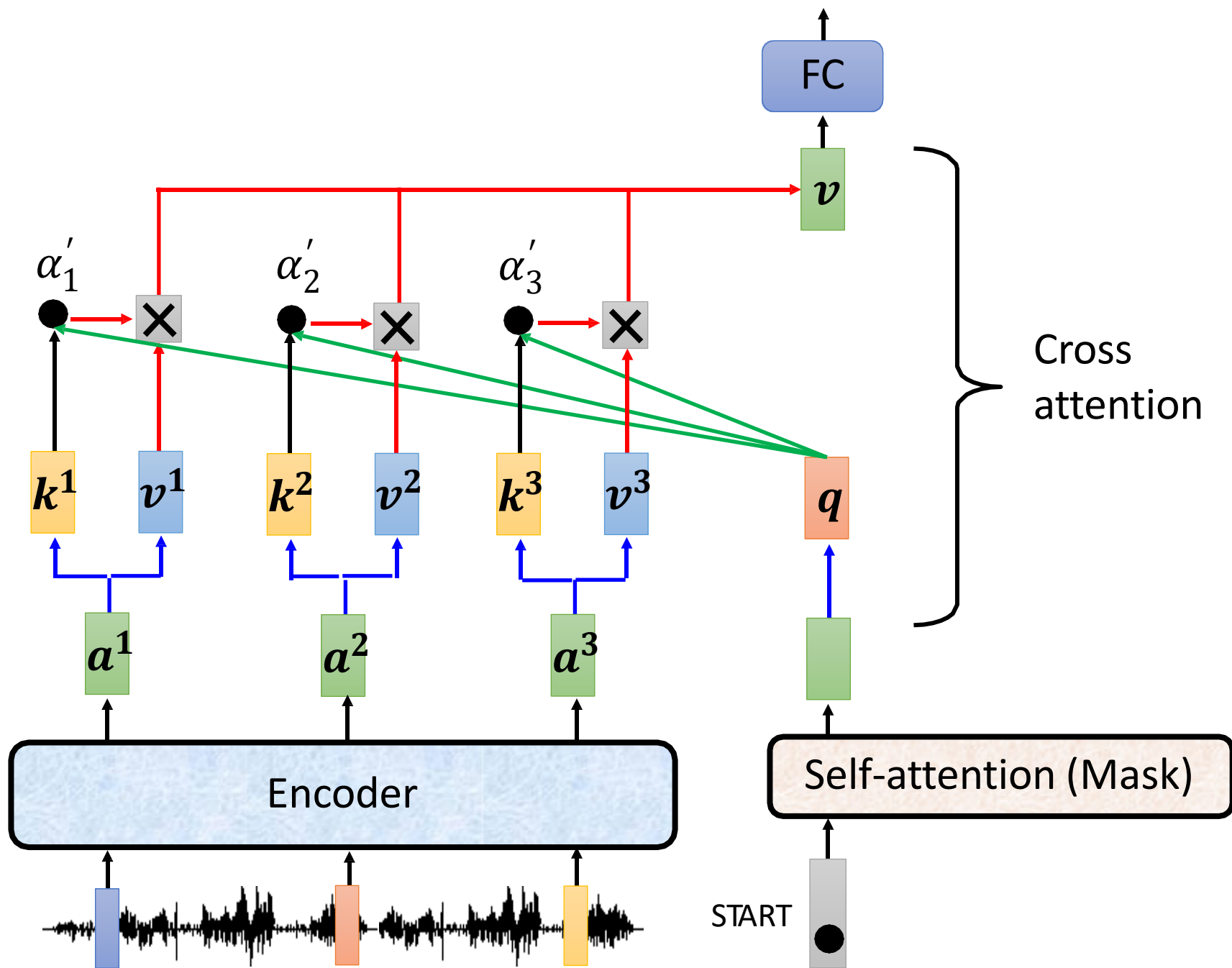


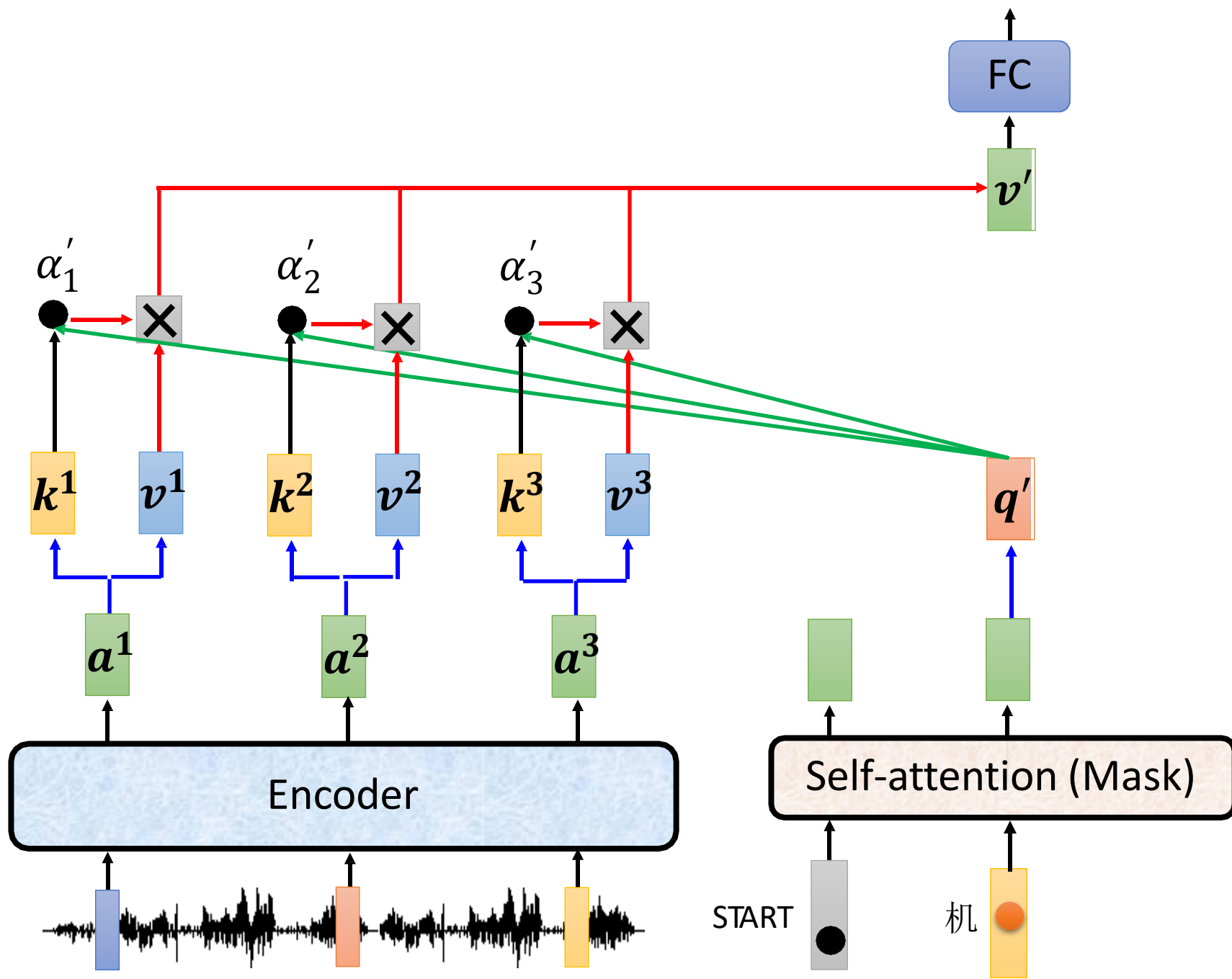
Encoder-Decoder



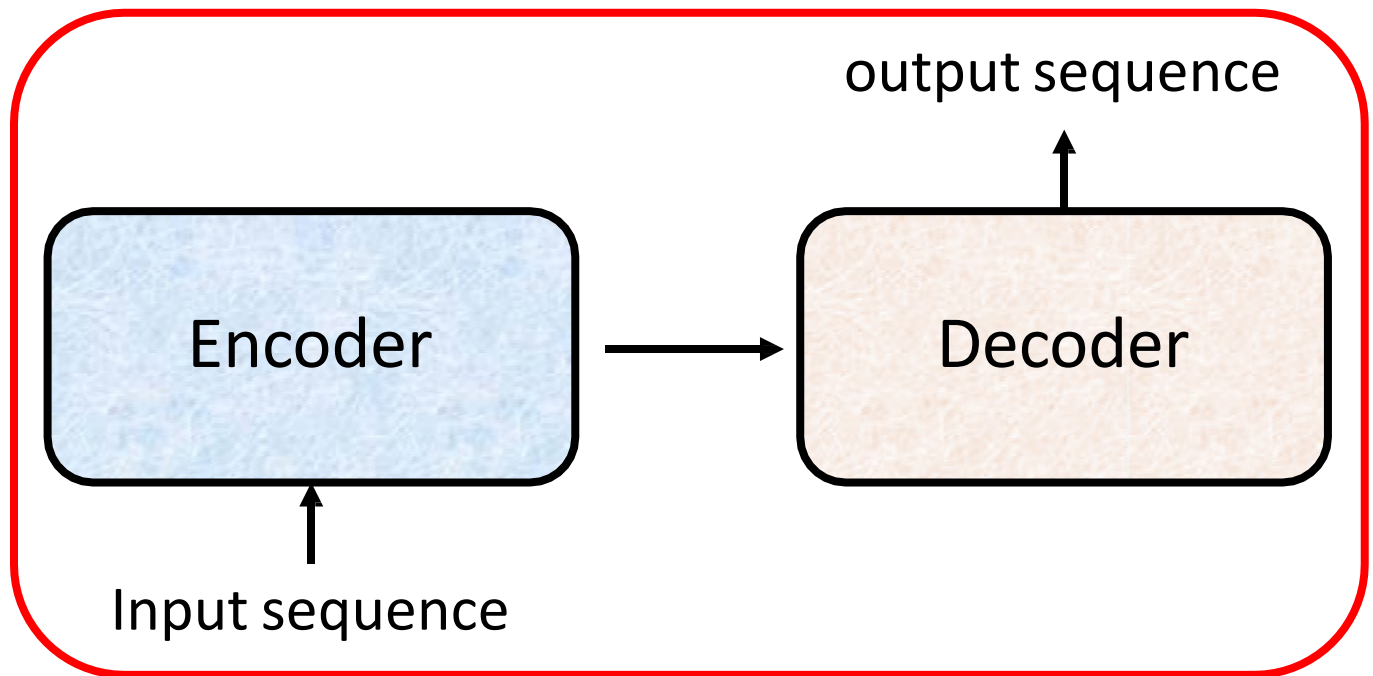
Transformer

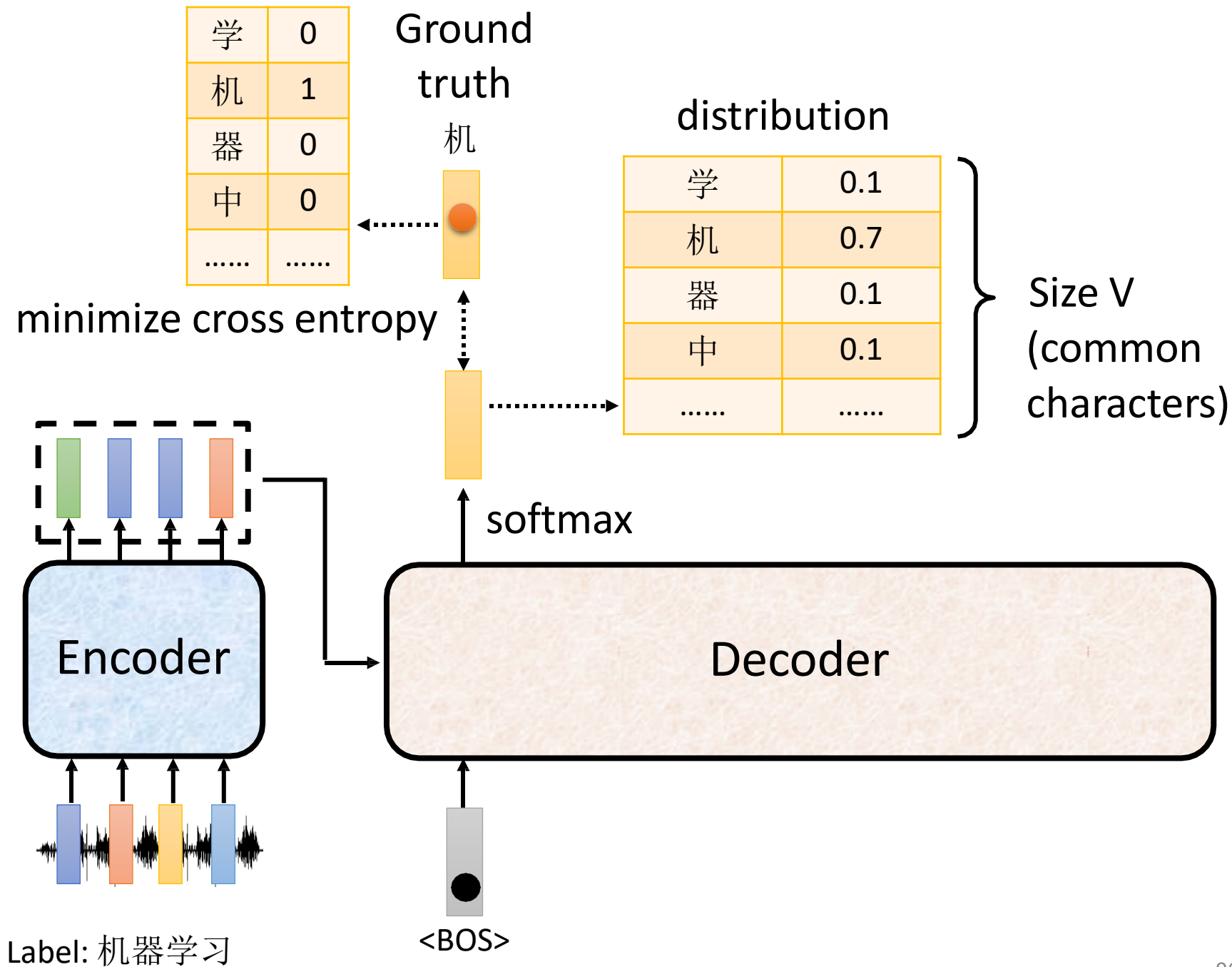




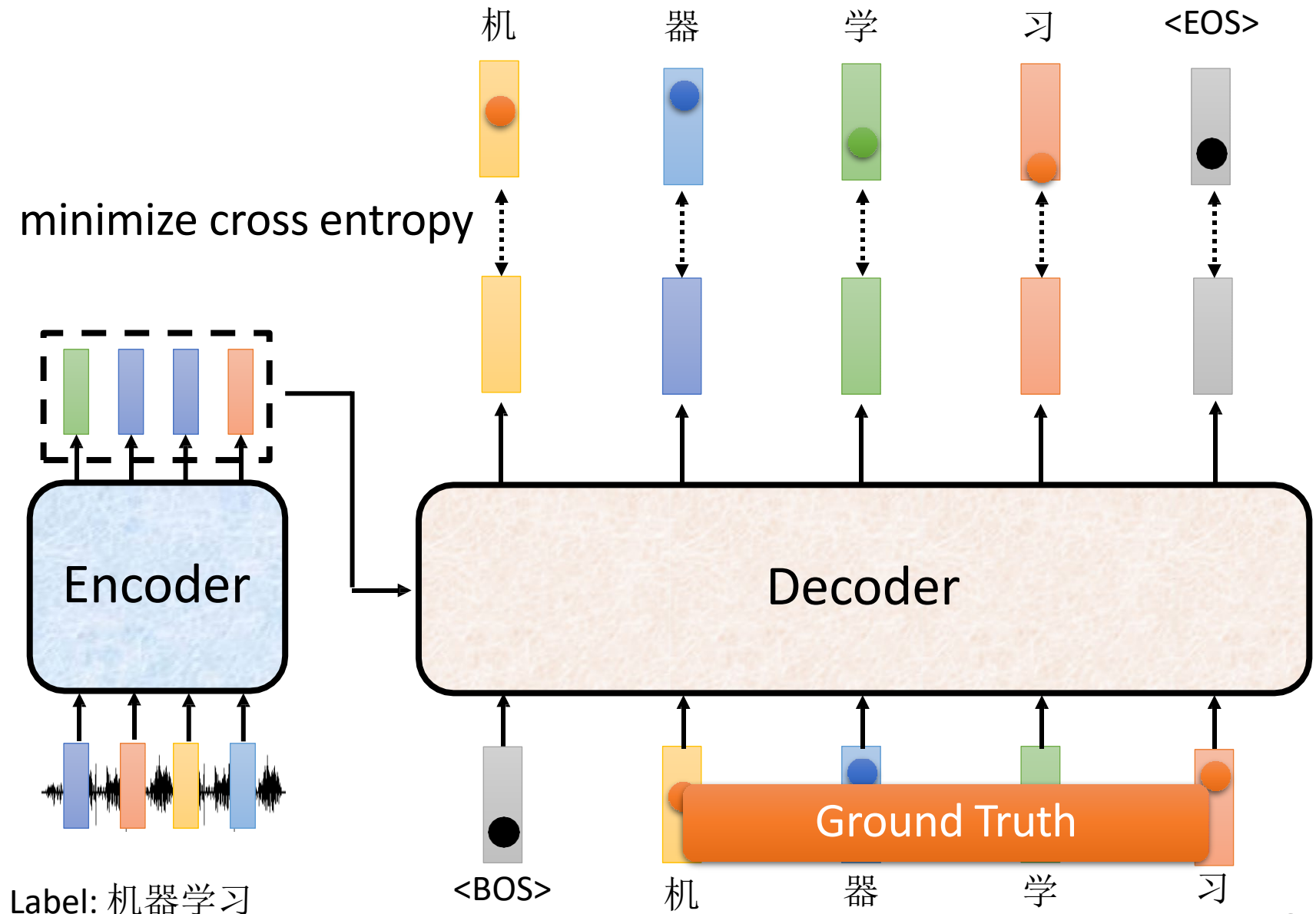


Training

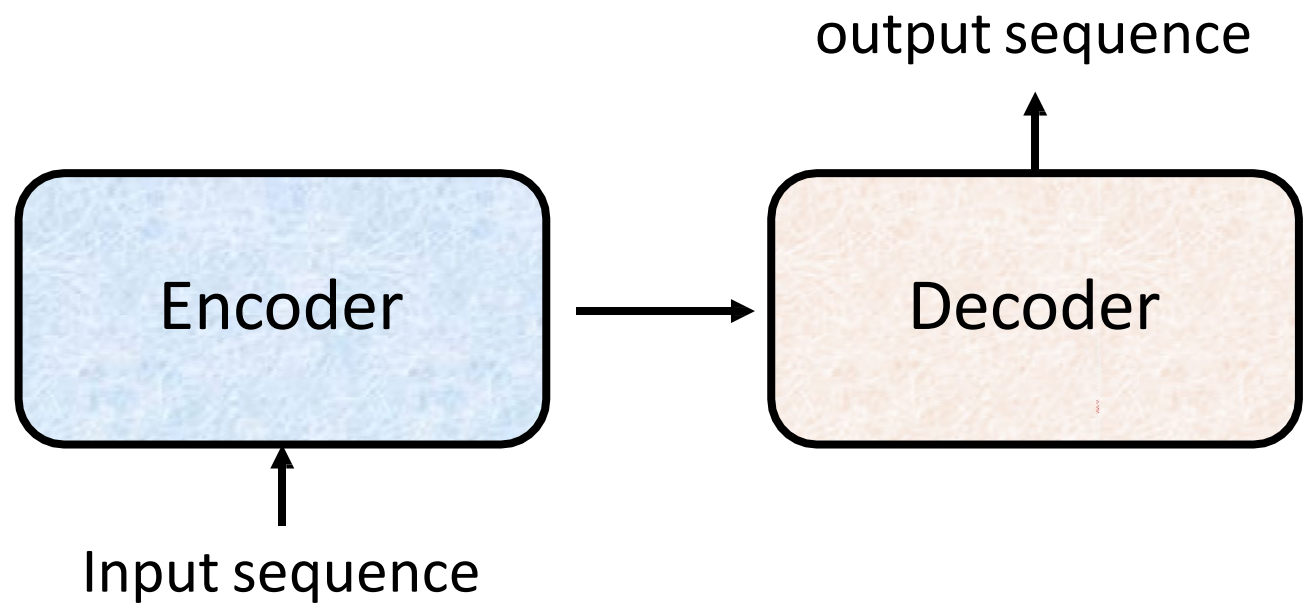




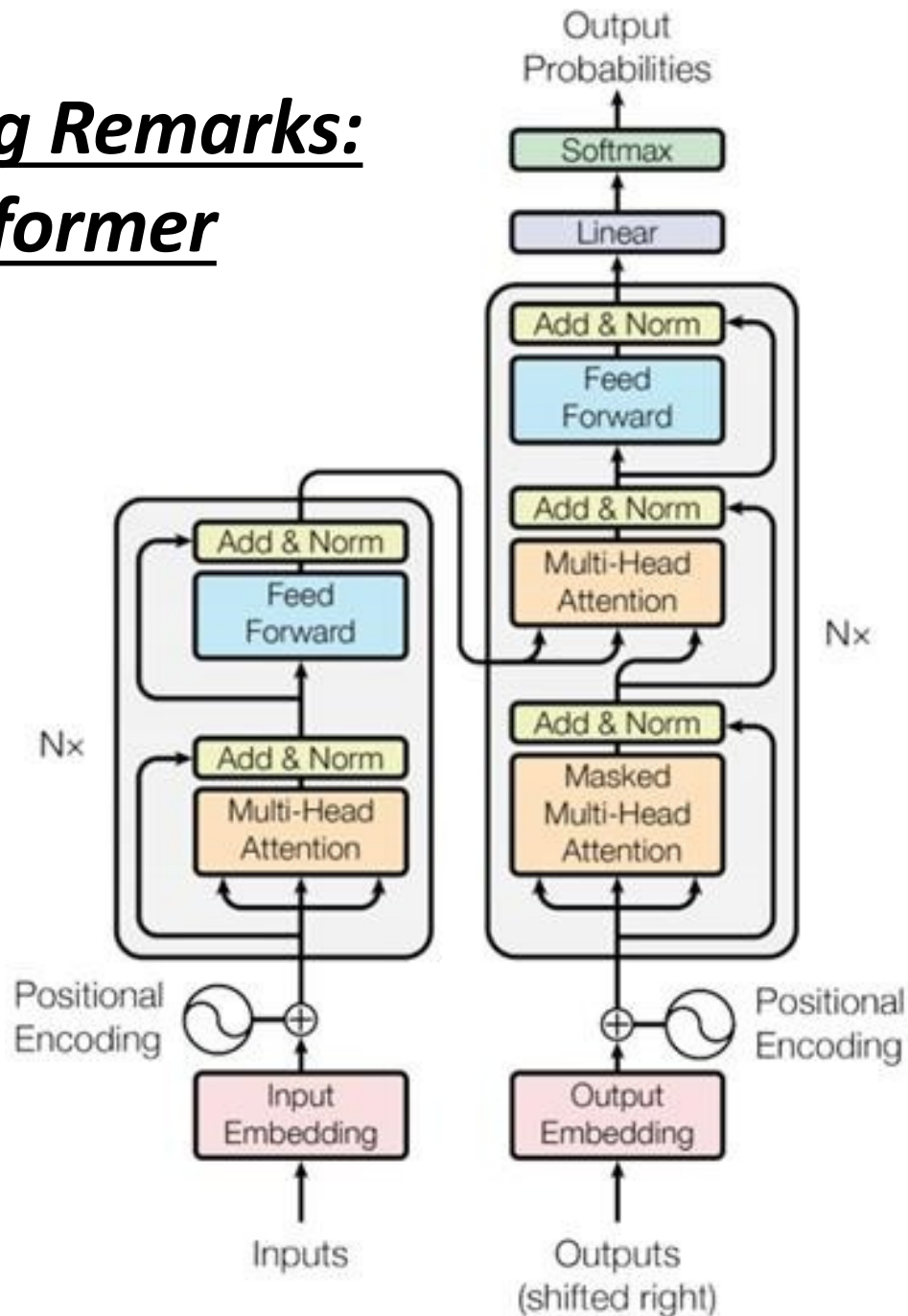
Teacher Forcing: using the ground truth as input.



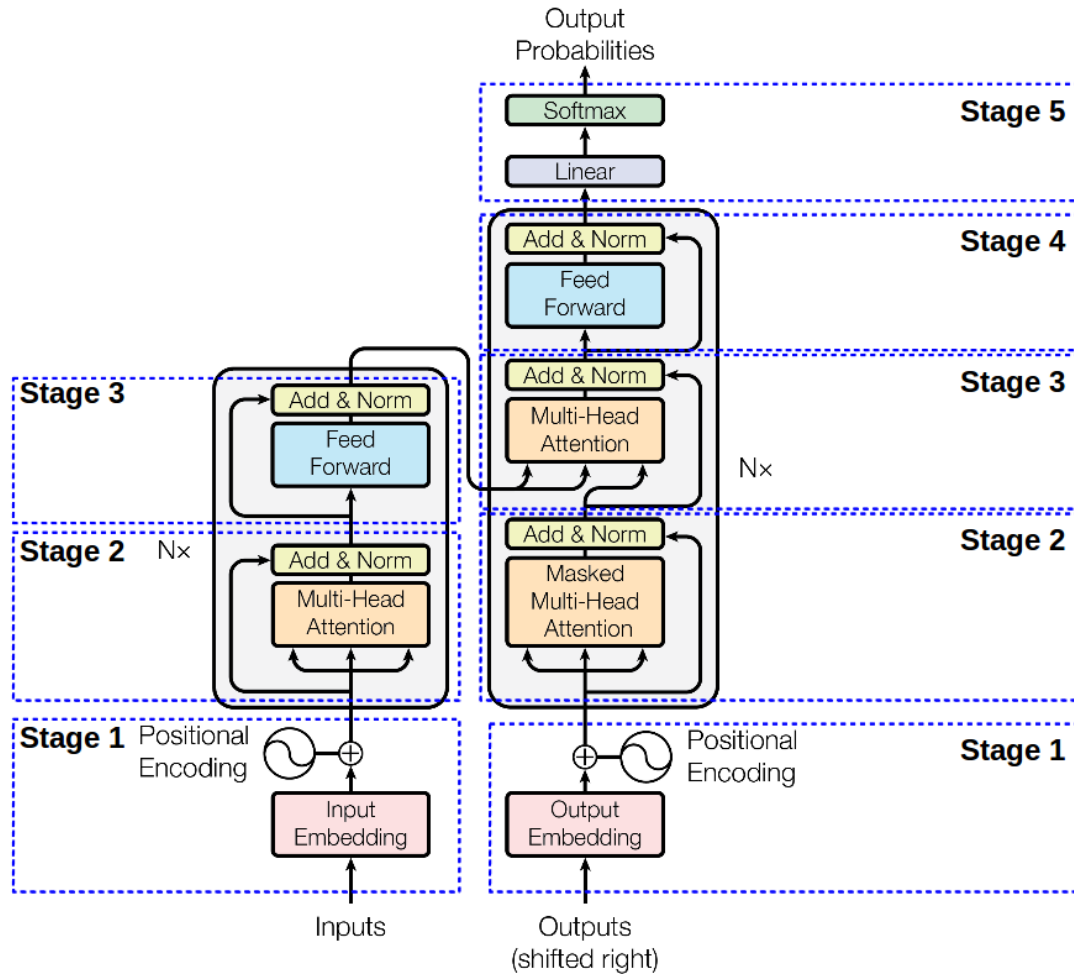
Tips



Concluding Remarks: Transformer



Transformer



Transformer



大脑中的记忆



记忆：外界信息在人脑中的内部表示

特点：联想记忆（Associative Memory）

记忆
过程

工作记忆（短期记忆）

情景记忆

结构记忆（长期记忆）

不严格的类比



记忆周期	计算机	人脑	神经网络
短期	寄存器	短期记忆	状态（神经元活性）
中期	内存	工作记忆	外部记忆
长期	外存	长期记忆	可学习参数
存储方式	随机寻址	内容寻址	内容寻址为主

记忆容量实验

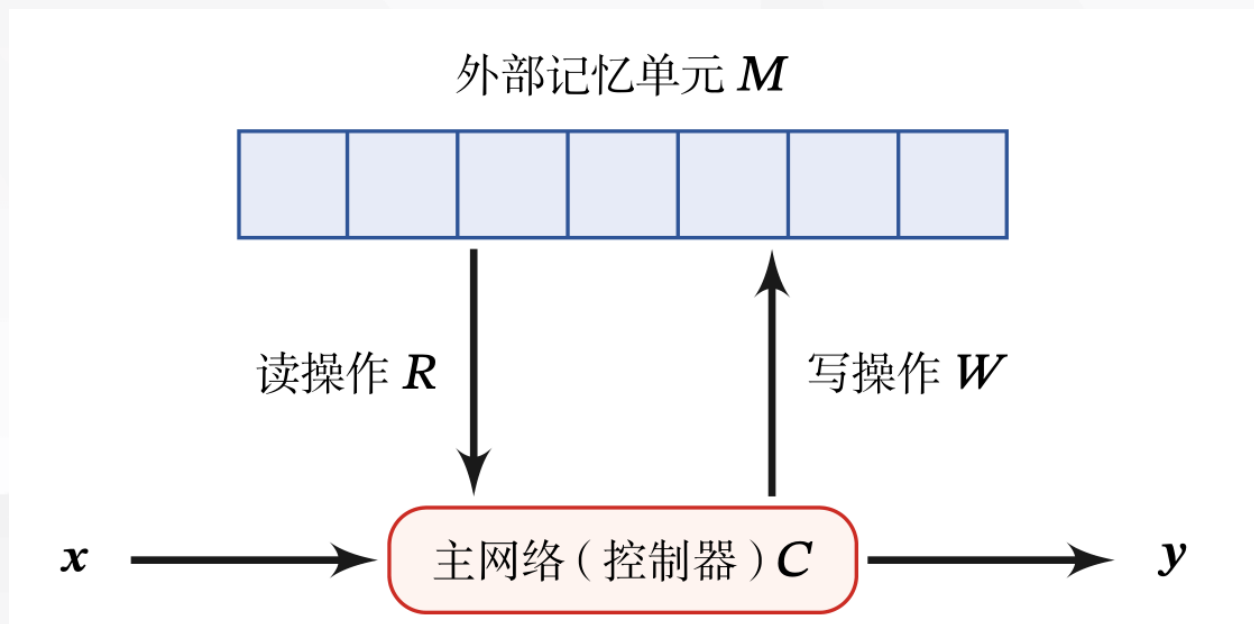


请尽力记住所有的字母



记忆增强神经网络 (Memory Augmented Neural Network)

- 主网络
- 外部记忆
- 读写操作



外部记忆



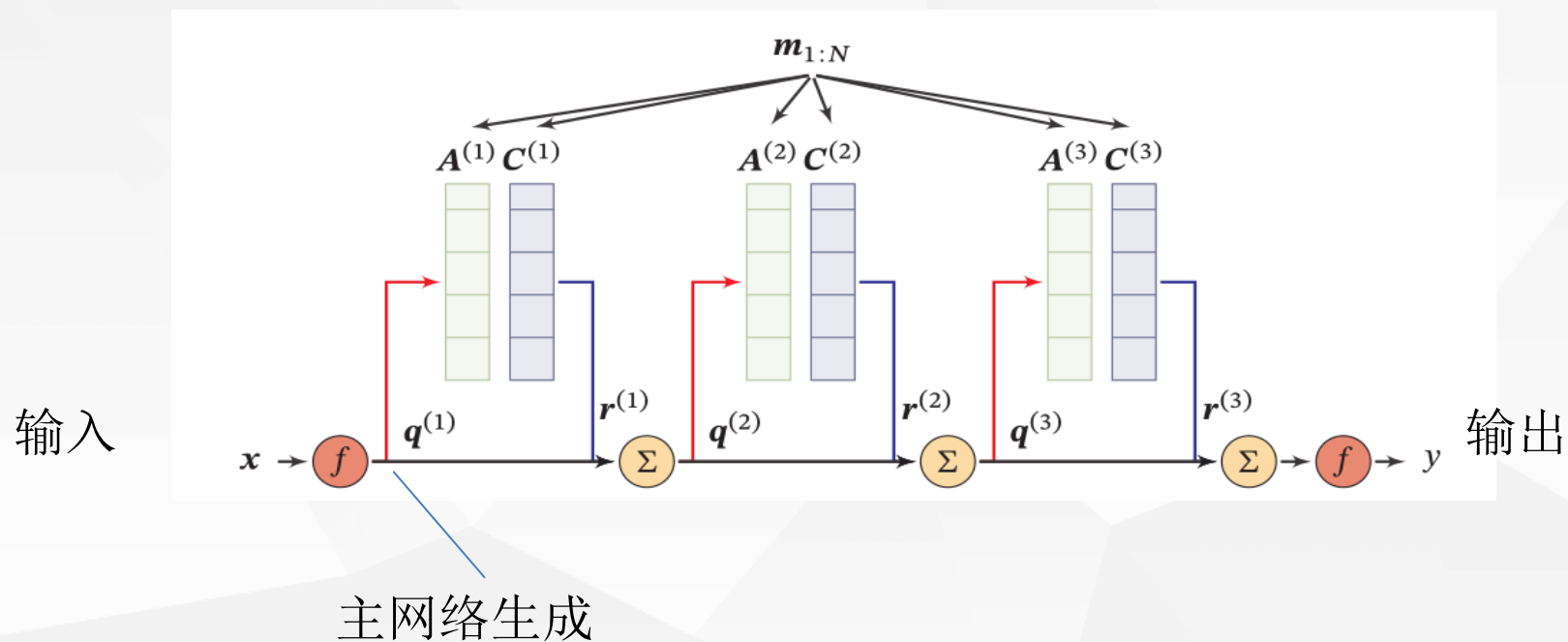
- 外部记忆定义为矩阵 $M \in \mathbb{R}^{d \times k}$
k是记忆片段的数量，d是每个记忆片段的大小
- 外部记忆类型
只读：Memory Network、RNNN中的 h_t
可读写：NTM
- 如何读写？

注意力机制

记忆网络



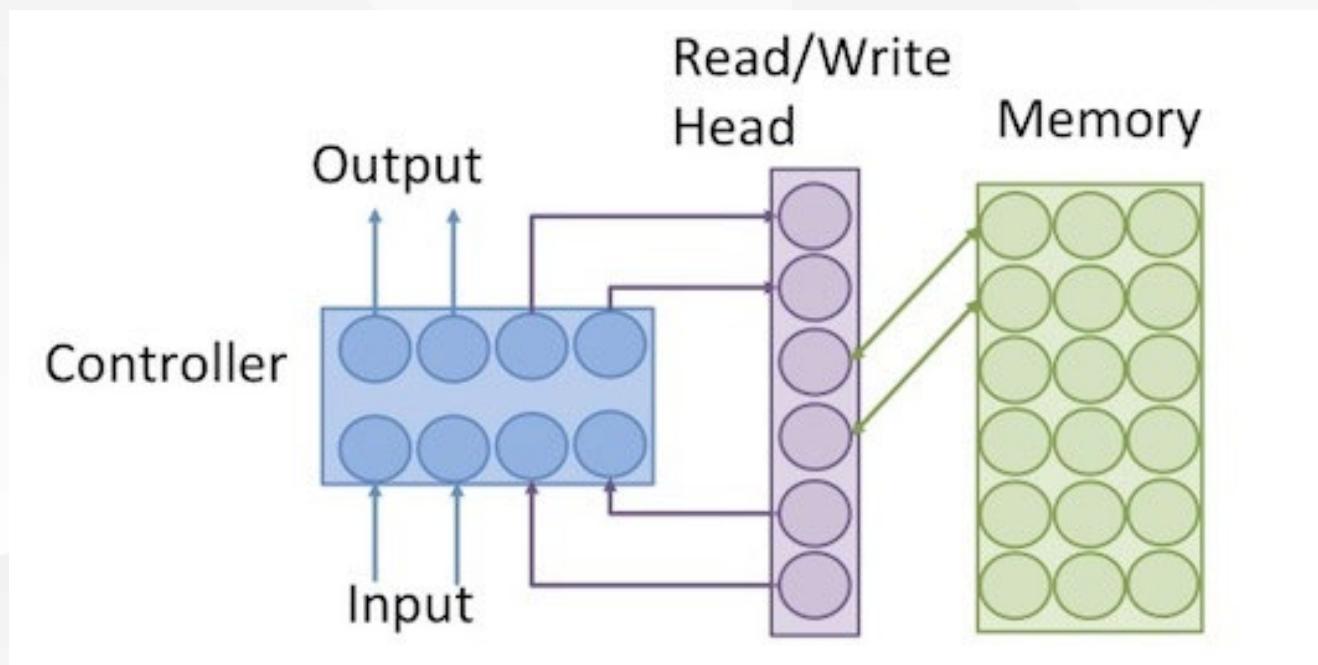
外部记忆



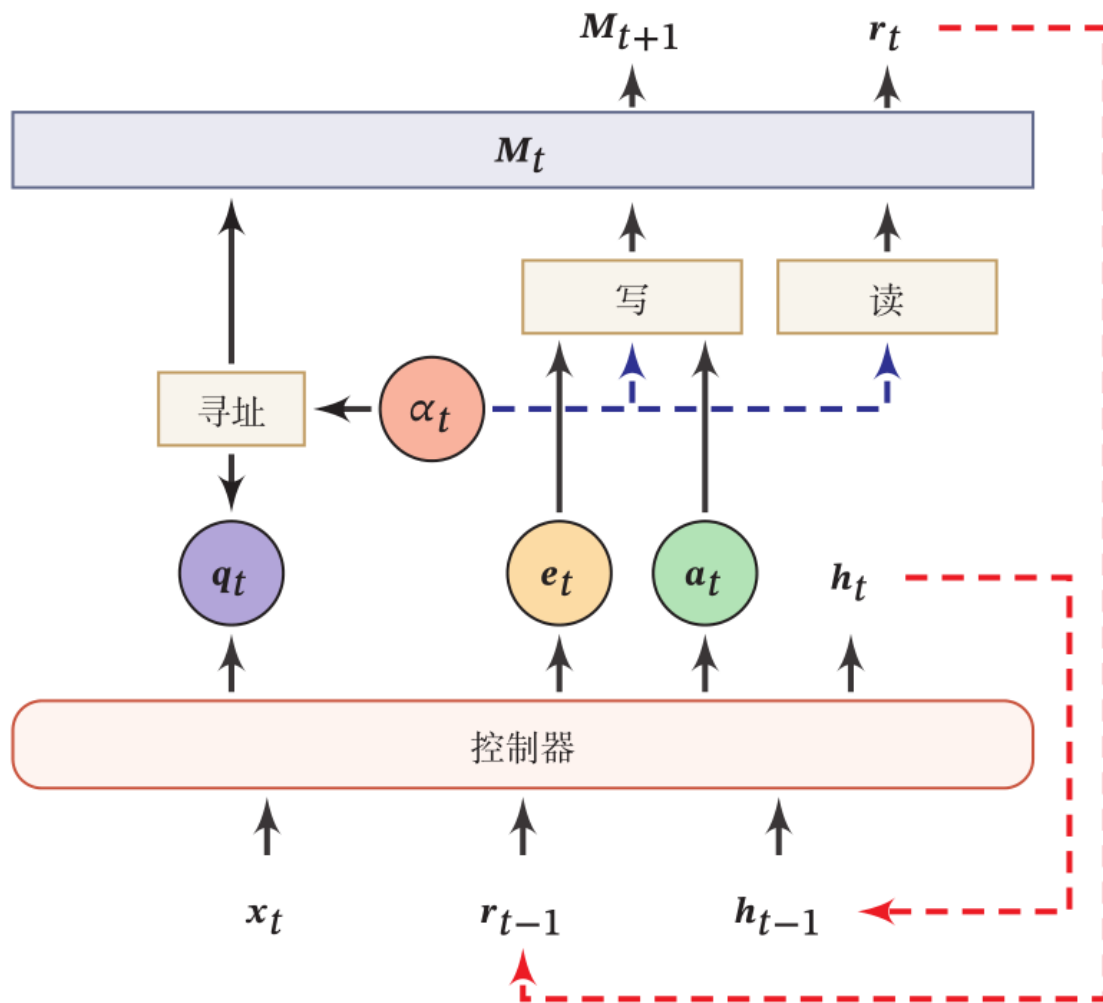
神经图灵机



- 组件：控制器 部记忆 读写操作
- 整个架构可微分



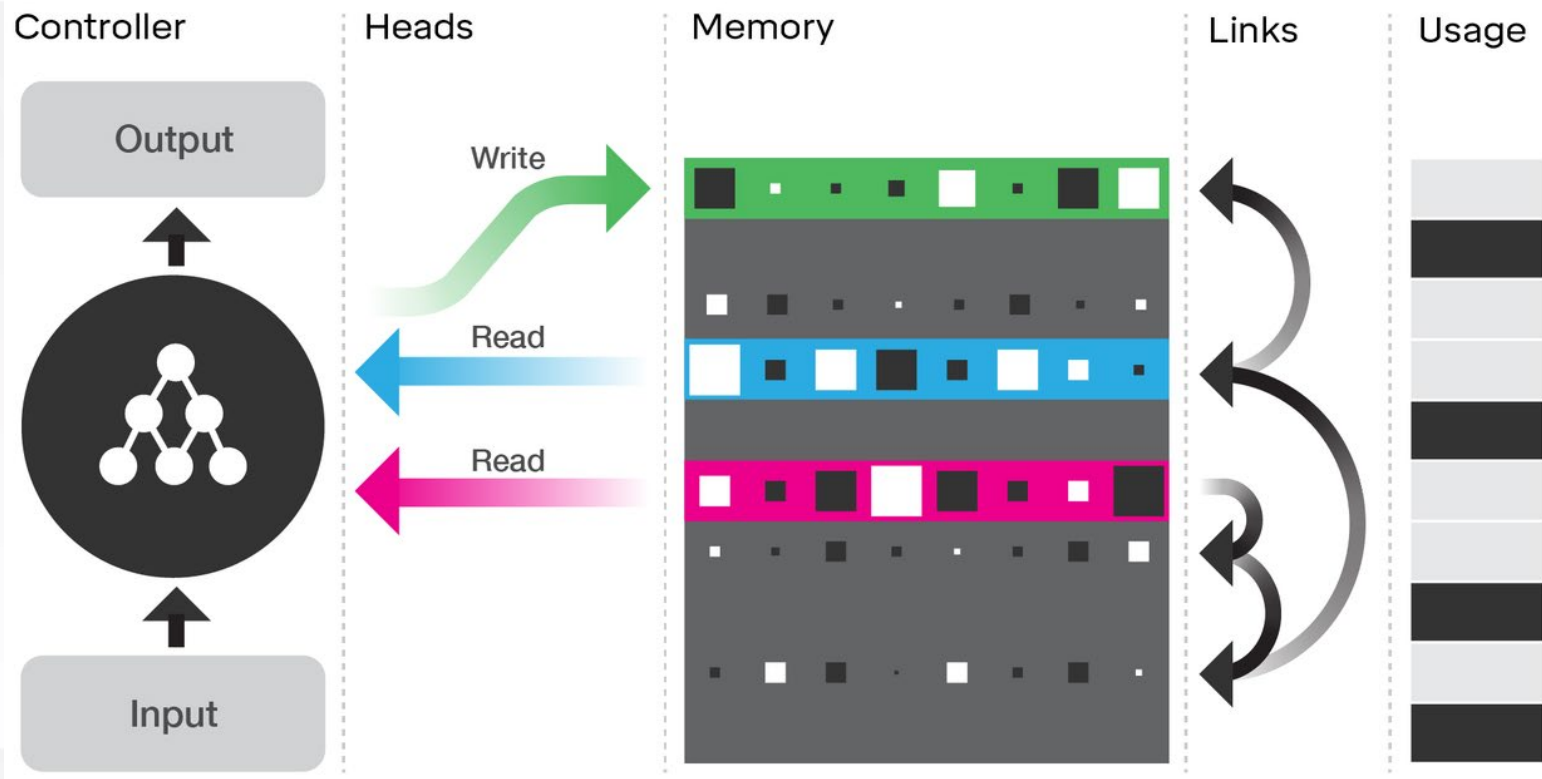
神经图灵机示例



Differentiable neural computers



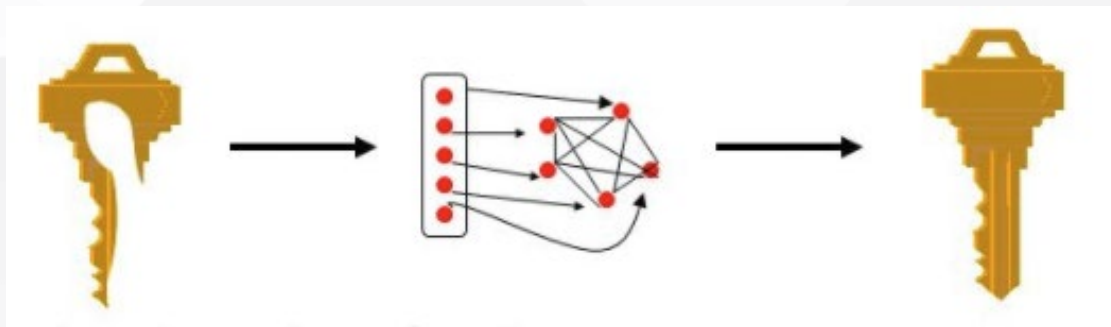
Illustration of the DNC architecture



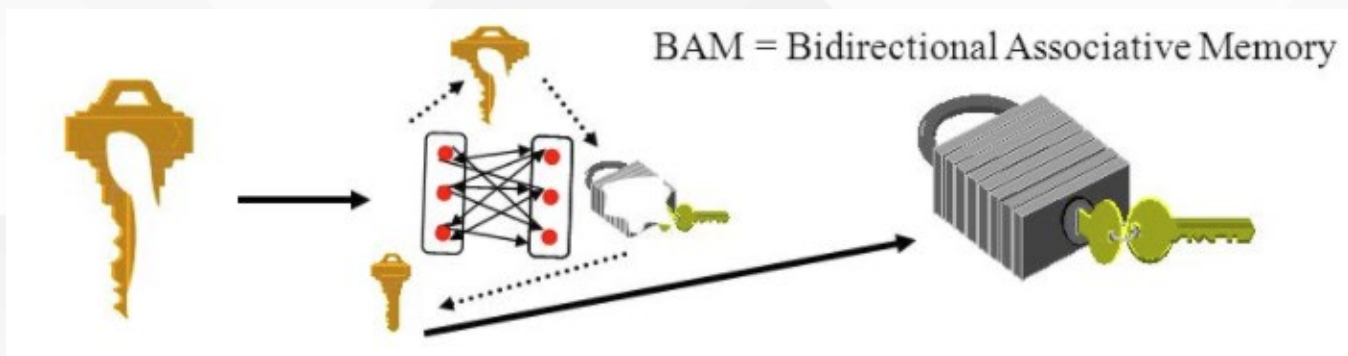
联想记忆



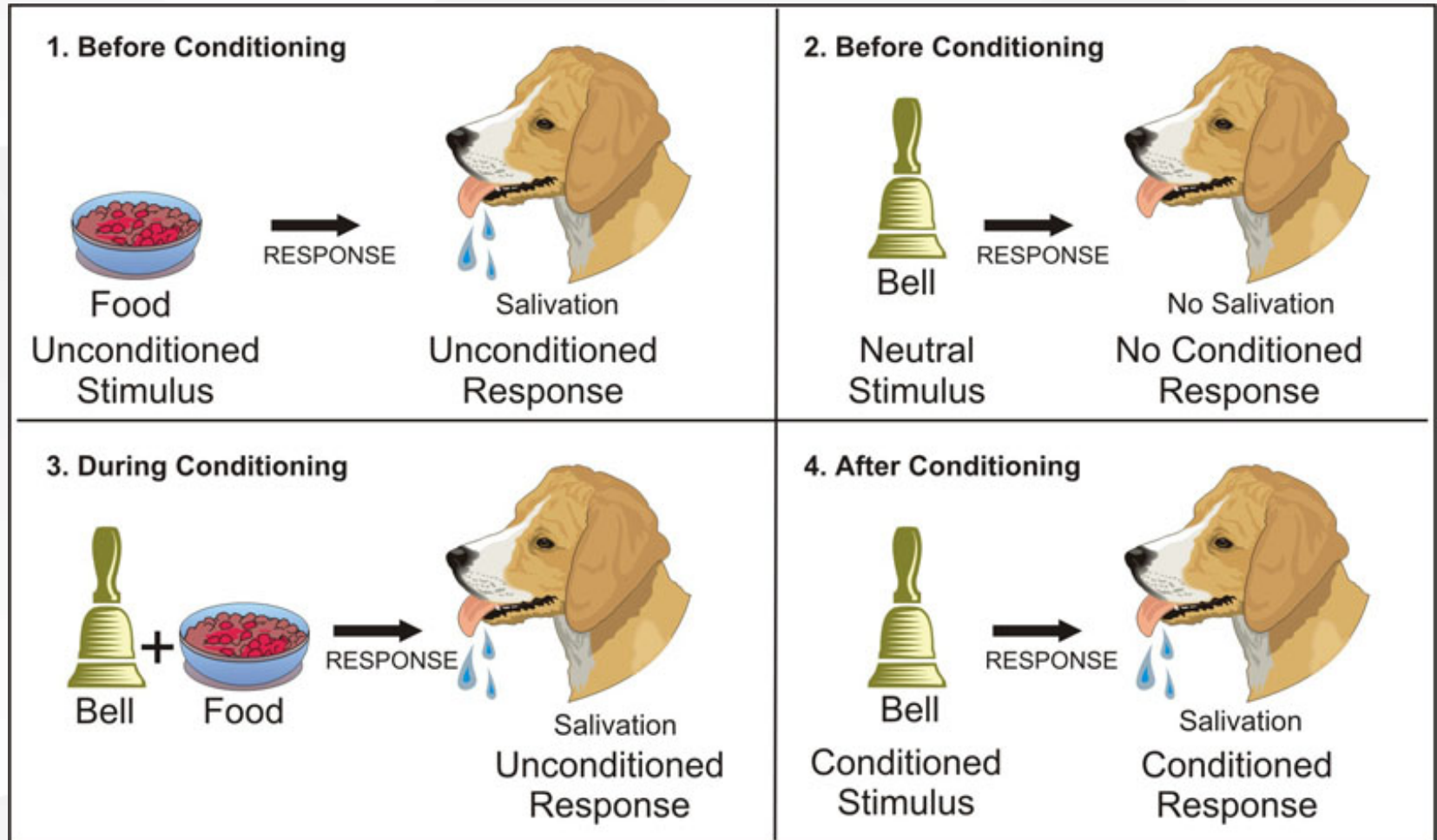
- 自联想



- 异联想



如何实现联想记忆？



Classical Conditioning

Hopfield网络



神经网络作为一种记忆的存储和检索模型。

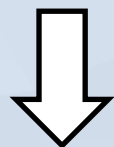
J. J. Hopfield, "Neural networks and physical systems with emergent collective computational abilities", Proceedings of the National Academy of Sciences of the USA, vol. 79 no. 8 pp. 2554-2558, April 1982.

更新规则:

$$w_{ii} = 0 \quad \forall i \in [1, m]$$

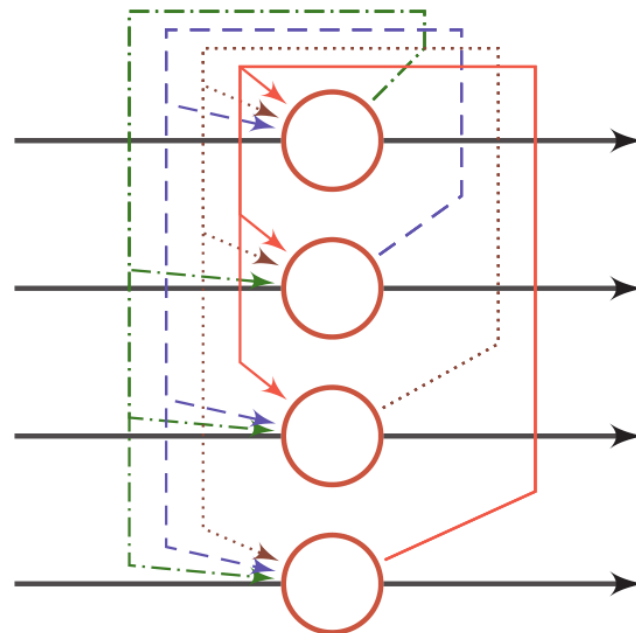
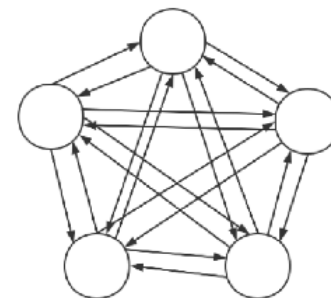
$$w_{ij} = w_{ji} \quad \forall i, j \in [1, m].$$

$$s_i = \begin{cases} +1 & \text{if } \sum_{j=1}^m w_{ij} s_j + b_i \geq 0, \\ -1 & \text{otherwise,} \end{cases}$$



同步更新

$$\mathbf{s}_t = f(W\mathbf{s}_{t-1} + \mathbf{b})$$



能量函数



- 在Hopfield网络中，我们给每个不同的网络状态定义一个标量属性，称为“能量”。

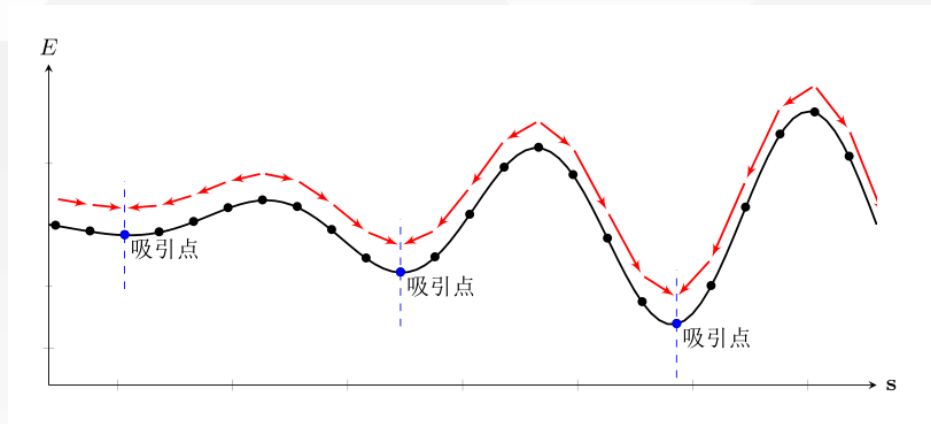
$$\begin{aligned} E &= -\frac{1}{2} \sum_{i,j} w_{ij} s_i s_j - \sum_i b_i s_i \\ &= -\frac{1}{2} \mathbf{s}^T \mathbf{W} \mathbf{s} - \mathbf{b}^T \mathbf{s}. \end{aligned}$$

- Hopfield网络是稳定的，即能量函数经过多次迭代后会达到收敛状态。权重对称是一个重要特征，因为它保证了能量函数在神经元激活时单调递减，而不对称的权重可能导致周期性震荡或者混乱。

检索过程（联想记忆）



- 给定一个外部输入，网络经过演化，会达到某个稳定状态



- 检索过程

每个吸引点 u 都对应一个“管辖”区域。如果输入向量 x 落入这个区域，网络最终会收敛到 u 。

吸引点可以看作是网络中存储的信息。

存储过程



信息存储是指将一组向量 $X=[x_1, \dots, x_N]$ 存储在网络中的过程

- 存储过程主要是调整神经元之间的连接权重，因此可以看作是一种学习过程。
- 神经元 i 和 j 之间的连接权重

$$w_{ij} = \frac{1}{N} \sum_{n=1}^N x_i^{(n)} x_j^{(n)},$$

- 如果两个神经元经常同时激活，则它们之间的连接加强；如果经常不同时激活，则连接消失。
- 类似于人脑的Hebbian法则

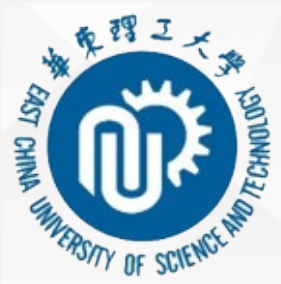
使用联想记忆增加网络容量



- 既然联想记忆具有存储和检索功能，我们可以利用联想记忆来增加网络容量。
- 和结构化的外部记忆相比，联想记忆具有更好的生物学解释性。

Associative long short-term memory

Using fast weights to attend to the recent past



谢谢！

THANK YOU FOR LISTENING

